9

Electrodiagnostic Medicine

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SUMMARY PEARLS

- In electrodiagnostic testing, be aware of the referral goals, electrodiagnosis laboratory standards for adults and children, specific type of instrumentation used, and common pitfalls.
- Electrodiagnostic testing is an extension of the physical examination.
- For needle electromyography, be aware of waveform definitions and clinical caveats.
- Be able to differentiate electrodiagnostic findings between the focal clinical conditions of peripheral entrapments, radiculopathies, and plexopathies.
- Be aware of the specific electrodiagnostic findings in generalized clinical conditions, including polyneuropathies, myopathies, neuromuscular junction disorders, and motor neuron disease.

Introduction

Electrodiagnostic (EDX) medicine and testing involve the application of anatomic and physiologic knowledge in the evaluation of peripheral nervous system pathophysiologic changes to establish diagnosis and prognosis and to guide treatment decisions. EDX testing is an extension of the history and physical examination tailored to the clinical scenario. A wide range of tests are available, and it is incumbent on the electrodiagnostic medicine consultant (EMC) to select those relevant to the clinical circumstances. The testing procedure may change as data are acquired and new findings interpreted. High-quality consultations are rendered by physicians with experience and technical competence coupled with an understanding of the peripheral nervous system. This chapter provides an overview of EDX tests, the extent of such testing for different clinical scenarios based on guidelines, and the types of abnormalities expected with different neuromuscular disorders.

Clinical Assessment: History and Physical Examination

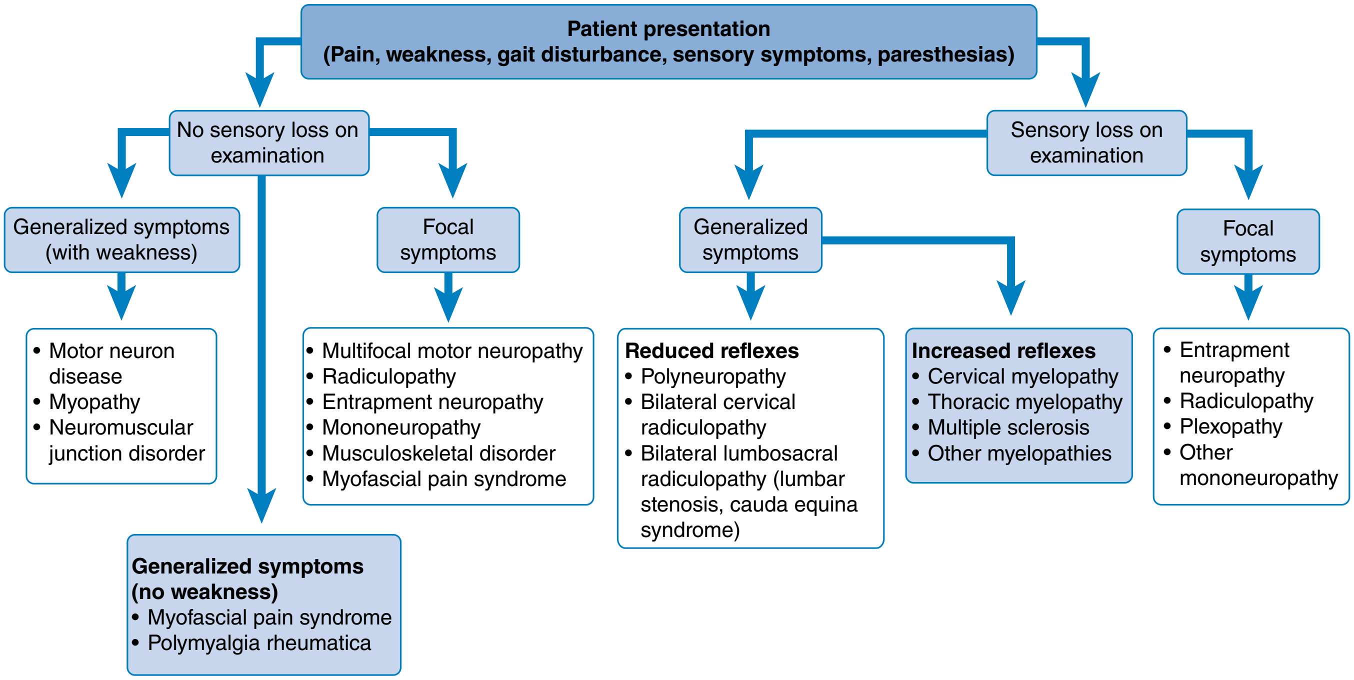
History

A directed history establishes a conceptual framework for subsequent testing. The history should include the referral reason and presenting symptom timing, characteristics, and distribution (localized or diffuse). Pain and numbness are indicative of sensory system involvement. Nocturnal hand numbness is commonly associated with carpal tunnel syndrome. Motor weakness without sensory symptoms indicates primarily motor system involvement: motor axons, neuromuscular junction, or muscles. For patients with more generalized symptoms, inquiries about other neurologic complaints, such as dyspnea, bladder urgency, dysphagia, dysphonia, and visual changes, are indicated. History in patients with generalized, insidious onset of unexplained symptoms should include inquiries about diabetes, alcohol use, and a family history of potential inherited neurologic disorders. Established cervical or lumbar spine conditions, such as disc herniation, spinal stenosis, or prior surgical interventions, should be noted. Activities that alter or relieve symptoms often help differentiate musculoskeletal conditions from entrapments to plexopathies, and radiculopathies. A brief recording of the person's work activities is often helpful to identify risk for overuse syndromes.

Important past medical history includes any history of malignancy, chemotherapy, or radiation because these conditions and treatments can have peripheral neurologic implications. Previous surgeries for entrapment neuropathies, such as ulnar transposition or carpal tunnel release, should be ascertained. For instance, a transposed ulnar nerve complicates nerve conduction calculations for the segment across the elbow because measurement is less precise, and stimulus sites above and below the elbow can be more difficult to find. Whether a person is thrombocytopenic, has bleeding disorders, or is receiving anticoagulants is important information to avoid bleeding complications with needle electromyography (EMG) testing.

Physical Examination

Strength, sensation, and reflex testing are the most important aspects of the physical examination when establishing an EDX testing plan. Motor weakness is often seen in neuromuscular disorders. Subtle motor weakness can be elucidated by side-to-side comparison and examination of smaller distal muscles, such as extensor hallucis longus and foot or hand intrinsic muscles.



• **Fig. 9.1** An algorithmic approach to using physical examination and symptom information to tailor the electrodiagnostic evaluation.

Reduced reflexes indicate peripheral nervous system problems, whereas increased reflexes are suggestive of central nervous system disorders, particularly when seen in combination with a Babinski sign, spasticity, and increased tone. Cranial nerve examination is helpful in many clinical situations. An important caveat is that in acute spinal cord injury, there can be an initial reduction in reflexes because of spinal shock.

An algorithmic approach to using physical examination and symptom information to tailor the EDX evaluation is shown in Fig. 9.1. In this approach, the patient's sensory complaints and physical examination signs of sensory loss and weakness create a conceptual framework for approaching these problems. Sensory loss with sensory symptoms is a key finding on examination for someone presenting with generalized subjective symptoms and provides a branch point in the algorithmic approach to tailoring the EDX study. For the purposes of this discussion, generalized findings are defined as being present in two or more limbs. Although helpful, there are many exceptions to this taxonomy. The EMC must refocus the diagnostic effort and prioritize testing choices because data are acquired dynamically during the testing process to obtain a patient-specific EDX diagnosis.

Myopathies, neuromuscular junction disorders, motor neuron disease, and multifocal motor neuropathy are all characterized by sensory system preservation. In contrast, polyneuropathies, radiculopathies, myelopathies, and central nervous system disorders frequently result in reduced sensation. In persons with no sensory loss or weakness on examination and preserved reflexes, the EMC should maintain a heightened suspicion for myofascial pain syndrome, fibromyalgia, polymyalgia rheumatica, or other musculoskeletal disorders.[15] For patients with suspected cervical or lumbosacral radiculopathies, the presence of a lost reflex, clear sensory loss, or motor weakness makes the probability of a positive EDX study many times greater.[116,117,187] The presence of a musculoskeletal disorder should not curtail EDX testing when indicated because they frequently coexist with entrapment neuropathies and radiculopathies.[27,28]

Purpose of Electrodiagnostic Testing

EDX testing, as an extension of the history and physical examination, provides objective neurophysiologic data to narrow the clinical differential diagnoses, establishing an EDX physiologic diagnosis to assist in management decisions (see eAppendix 9A for physiologic basis for EDX testing). EDX testing excludes conditions in the differential diagnosis and alters the referring diagnostic impression 42% of the time.[86] EDX testing can also suggest severity or extent of the disorder beyond the clinical symptoms and provide potential information about prognosis. There also is utility in solidifying a diagnosis. For example, an unequivocal radiculopathy on needle electromyography (EMG) provides greater diagnostic certainty and modifies management decisions.

The value of EDX testing depends on multiple clinical factors. For whatever reason EDX testing is being done, the testing should have potential to alter patient management. These reasons are numerous and may include confirming a clinical situation (normal or abnormal) that is highly suspected, trying to identify which problem is causing the symptoms, looking for two problems that coexist, determining how much neuromuscular injury is present, assessing improvement or deterioration in the electrophysiology, and ruling out serious problems to reassure the patient and treatment team. EDX testing in the presence of motor weakness can establish time course, severity, and prognosis when establishing a prioritized diagnostic workup and treatment plan. If a patient with unilateral leg pain and foot drop (ankle dorsiflexion weakness) of unknown duration presents for EDX evaluation, testing in this scenario could provide evidence of a preexisting lumbar radiculopathy without signs of active axon loss. Combined with a recovery pattern, this may support a trial of nonsurgical management with close clinical monitoring. Another indication for EMG testing in a similar clinical presentation is one of an elderly diabetic patient with sciatica, having limited physical examination findings and equivocal or age-related magnetic resonance imaging (MRI) changes, presenting with an unclear clinical picture. In this case, EDX testing may be valuable in clarifying a complicated clinical presentation, putting in perspective the imaging findings and identifying diabetic polyneuropathy and/or lumbosacral radiculopathy if present. If a clinical diagnosis is obvious, additional testing is only indicated if it may alter management decisions. In a patient with acute-onset sciatica with evolving L5 muscle weakness, a positive straight leg raise, and MRI showing a large correlating extruded L4-L5 disc herniation, EDX testing may not be needed.

Nerve Injury Classification

Peripheral nerve injury is one of the most common types of pathology likely to be encountered during an EDX evaluation. It is necessary to be familiar with the various classification systems available to categorize an insult to neural tissue. The degree to which a nerve is damaged has obvious implications with regard to its present function and potential for recovery. There are essentially two general classification systems (Seddon and Sunderland) that provide a useful conceptual framework to interpret EDX testing results.[51] In the Seddon classification, there are three degrees or stages of injury to consider: neurapraxia, axonotmesis, and neurotmesis (Table 9.1).

Neurapraxia

Neuropraxia is used to designate a mild degree of action potential propagation slowing/failure in a portion of nerve fibers in an affected segment. Neurapraxia is almost always synonymous with the term *conduction block*. Conduction block is used to describe measurable nerve conduction features usually caused by mechanical myelin disruption, temporary ischemia, or local anesthetic. If myelin disruption is significant enough, neuropraxia is present. From a clinical standpoint, complete conduction block of all nerve fibers is infrequent. Most entrapment neuropathies involve varying degrees of myelin damage and corresponding degrees of conduction block. If all the axons are involved, there will be no measurable nerve conduction across an involved segment (i.e., complete conduction block). The most important feature of a

TABLE 9.1 Nerve Injury Classification

Type	Function	Pathologic Basis	Prognosis
Lundborg			
Physiologic Conduction Block			
Type a	Focal conduction block	Intraneural ischemia; metabolic (ionic) block; no nerve fiber changes	Excellent; immediately reversible
Type b	Focal conduction block	Intraneural edema; increased endoneurial fluid pressure; metabolic block; little or no fiber changes	Recovery in days or weeks
Seddon-Sunderland			
Neurapraxia			
Type 1	Focal conduction block; primarily motor function and proprioception affected; some sensation and sympathetic function may be present	Local myelin injury, primarily larger fibers; axonal continuity; no Wallerian degeneration	Recovery in weeks to months
Axonotmesis			
Type 2	Loss of nerve conduction at injury site and distally	Disruption of axonal continuity with Wallerian degeneration; endoneurial tubes, perineurium, and epineurium intact	Axonal regeneration required for recovery; good prognosis, because original end organs reached
Type 3	Loss of nerve conduction at injury site and distally	Loss of axonal continuity and endoneurial tubes; perineurium and epineurium preserved	Disruption of endoneurial tubes, hemorrhage, and edema produce scarring; axonal misdirection; poor prognosis; surgery may be required
Type 4	Loss of nerve conduction at injury site and distally	Loss of axonal continuity, endoneurial tubes, and perineurium; epineurium intact	Total disruption of guiding elements; intraneural scarring and axonal misdirection; poor prognosis; surgery necessary
Neurotmesis			
Type 5	Loss of nerve conduction at injury site and distally	Severance of entire nerve	Surgical modification of nerve ends required; prognosis guarded and dependent on nature of injury and local factors

From Lundborg G. *Nerve Injury and Repair.* Churchill Livingstone; 1988.

conduction block is its reversibility. Axon loss does not occur, by definition, in neurapraxia. Muscle wasting usually does not occur because muscle innervation is maintained, and recovery is typically rapid enough to avoid disuse atrophy. Fibrillation potentials should not be observed because axons are not disrupted.

Axonotmesis

Axonotmesis is a more severe type of nerve injury, where axons are physically disrupted while the enveloping endoneurial and other supporting connective tissue structures (perineurium and epineurium) are preserved. Severe neuronal compression and/or traction typically result in axon loss lesions. Once the axon has been disrupted, the characteristic changes of Wallerian degeneration occur distal to the lesion. Wallerian degeneration with axon disintegration and degeneration within 24 hours in small fibers and 48 hours in larger fibers in muscles just distal to the lesion progresses down longer nerves to distant muscles within 5 to 10 days depending on nerve length.[176] Muscle membrane changes occur in denervated muscles days after Wallerian degeneration has occurred, and fibrillation potentials are an indication of this process. The fact that the endoneurium can remain intact is a very important aspect of this type of injury. A preserved endoneurium implies that once the remnants of the degenerated nerve have been removed by phagocytosis (completed Wallerian degeneration), the regenerating axon can then follow its original course directly back to the appropriate end organ. Nerve lesions where the structural supportive connective and endoneurium remain intact have a much better prognosis for recovery. Incomplete axonotmesis lesions have two phases of recovery. Axons viable across the lesion that continue to innervate muscle can develop collateral sprouts to adjacent noninnervated muscle fibers in the first phase of recovery. The second recovery phase involves axon growth lengthwise down the structurally intact endoneurial tubes to the remaining noninnervated muscle fibers. Return of motor strength depends on the lesion distance, the metabolic/vascular environment in which the axon is attempting to regrow, and the degree of atrophy and fibrosis in the distal muscles.

Neurotmesis

Neurotmesis represents the greatest degree of nerve disruption in the Seddon classification. Neurotmesis describes complete nerve disruption (i.e., transection) involving all sensory, motor, and autonomic axons and all supporting connective tissue and vascular structures, whereby the endoneurium, perineurium, and epineurium are no longer in continuity. This lesion has a poor prognosis for complete functional recovery. Surgical reapproximation of the nerve ends, if possible, or nerve grafting will likely maximize the chance of recovery. Surgery does not guarantee proper endoneurial tube alignment, but it improves the chances that axonal growth will occur across the injury site. In the special case of acute nerve transection or severe nerve trauma, errors in EDX interpretation can occur if test timing is not considered. Complete conduction block will be observed across a lesion immediately across the transected/severe nerve injury, but normal nerve conduction will be preserved distal to the lesion for up to 7 days until Wallerian degeneration occurs. If any voluntary motor units are recorded by needle EMG in a muscle distal to the nerve lesion and innervated by that particular nerve, then the electrodiagnostician can definitively say that neurotmesis is not present. This assumes that anatomic variations in innervation are accounted for and are not involved in this recruitment.

Sunderland Classification

A second popular and more detailed classification is that proposed and modified by Sunderland. The Sunderland classification is divided into five types of injury based exclusively on pathologic appearance, of which connective tissue components are disrupted (see Table 9.1). Type 1 injury corresponds to the Seddon designation of neurapraxia. Axonotmesis is subdivided by Sunderland into three gradations of neural insult (types 2–4). Sunderland type 5 injury corresponds to Seddon neurotmesis (complete neural disruption). The majority of peripheral nerve injuries have multifactorial etiologies and involve variable degrees of myelin injury, conduction block, and axon loss at the point in time when EDX testing is performed. EDX interpretation and prognostication in this setting require knowledge of anatomy, nerve physiology, nerve conduction studies (NCS), EMG, nerve injury classification systems, and recovery patterns. Any motor responses (compound muscle action potentials [CMAPs]) that are recorded in muscles innervated by a damaged nerve (after Wallerian degeneration has occurred) indicate roughly the degree of motor axonal loss that has occurred. The presence of elicitable responses is a positive prognostic finding for recovery of damaged nerves.[150]

EDX Testing of Motor and Sensory Nerves

EDX testing consists of nerve conductions and EMG. Standard NCS used in patient evaluation include sensory nerve conductions, motor nerve conductions, F-waves, and H-reflexes. NCS are performed by administering a small electrical stimulus to peripheral nerves and recording the results along sensory nerves or on muscle. Reusable surface electrodes are used to record NCS responses and should be cleaned with a 1:10 dilution of household bleach and 70% isopropyl alcohol or dilute household soap solution between patients. Disposable surface electrodes are widely available and easy to use.[6] The popularity of disposable surface electrodes is limited by cost, duration of adhesive function, and other factors. The American Association of Neuromuscular and Electrodiagnostic Medicine (AANEM), concerned with the quality of reference values in our field, formed a task force that created a set of criteria that defined what constitutes a high-quality normative data study.[47] Using this set of criteria, the task force scoured the published scientific literature, identified the highest quality studies, and recommended these norms for use by EMCs.[35] These norms are shown in Table 9.2 and Fig. 9.2. These reference values consider age, sex, height, and body mass index. They provide the most useful and accurate reference values available today and reflect the highest contemporary standards of practice.

Patient safety during NCS is an important consideration. In a study involving patients with dual-chamber pacemakers and implanted cardiac defibrillators, NCS (including median motor testing with an Erb point stimulation on the left side of the body) resulted in no change in the function of these devices nor did any stimulation show up on any of the electrocardiogram tracings.[162] These investigators concluded that conventional NCS have no effects on these implanted devices, and their findings should diminish the anxiety of EMCs when working with such patients.[162] These findings further support the safety of performing such testing in persons with implanted cardiac devices and dispel some of the concern expressed in previous recommendations.[6]

Sensory nerve conductions should nearly always be performed when assessing a patient.[7] In the upper limb, multiple sensory

TABLE 9.2 Standardized Techniques for Major Motor and Sensory Nerve Conduction Studies in Adults

TECHNIQUES (RECOMMENDED) — Electrode Placement: Ground Electrode Always Placed Between the Stimulating and Recording Electrodes					Machine	Settings
Nerves	*E1*	*E2*	*Stimulating Site (SS)*	*Distance (E1–SS) (cm)*	*DisplaySensitivity (lV/div) Sensory, (mV/div) motor*	*Sweep (ms/div)*
Superficial radial sensory	Extensor pollicis longus tendon	Base of thumb	Along the radius	10	5–10	1
Median sensory	Index finger Slightly distal to the second MCP	4 cm distal to E1	Wrist: between the flexor carpi radialis and the palmaris longus tendons Palm: midway between the 14-cm stimulation point and G1	14 7	20	1
Ulnar sensory	Fifth digit Slightly distal to the fifth MCP	4 cm distal to E1	Slightly to the radial side of the flexor carpi ulnaris tendon	14	20	1
Medial cutaneous nerve of the forearm (medial antebrachial cutaneous) sensory	Medial forearm	Distal: 3–4 cm	Midway between the medial epicondyle and the distal biceps tendon	10	10	1
Lateral cutaneous nerve of the forearm (Lateral antebrachial cutaneous) sensory	On a line to the radial pulse	Distal: 3–4 cm	Just lateral to the distal biceps tendon	10	10	1
Median motor	Abductor pollicis brevis motor point Midpoint of wrist crease and the first MCP	Distal to first Elbow: medial to the brachial pulse MCP	Wrist: between the flexor carpi radialis and the palmaris longus tendons	8	5	2
Ulnar motor	Hypothenar eminence Halfway between the pisiform and the MCP	Slightly distal to the fifth MCP joint Elbow flexion to 90 degrees	Wrist: slightly radial to the flexor carpi ulnaris tendon Below elbow: 4 cm distal to the medial epicondyle Above elbow: 10 cm proximal to the below-elbow site, measured in a curve behind the medial epicondyle to a point slightly volar to the triceps muscle Axillary: 10 cm proximal to above elbow site	8	5	2
Sural sensory	Posteroinferior to the lateral malleolus	Distal: 3–4 cm	At or slightly lateral to the calf midline	14	2–5	1
Peroneal (fibular) motor	Midpoint of extensor digitorum brevis	Just distal to fifth MTP	Ankle: lateral to the tibialis anterior tendon Below fibular head: posteroinferior to the fibular head Above fibular head: 10 cm proximal to the below fibular head site and slightly medial to the tendon of the biceps femoris	8	5	5
Tibial motor	Medial foot (slightly anterior/inferior to the navicular tubercle)	Slightly distal to first MTP (medial aspect of joint)	Ankle: posterior to the medial malleolus Knee: midpopliteal fossa	8	5	5

For all studies, temperature was maintained above 32.8°C for the upper extremity and above 31.8°C for the lower extremity. The temperature was measured on the dorsum of the hand for all upper limb NCS, both motor and sensory. Temperature was recorded over the dorsum of the foot for the sural sensory, tibial motor, and peroneal (fibular) motor nerve conduction. Sensory nerve studies: Frequency filters at 20 HZ (low) and 2 kHZ (high). Motor nerve studies: Frequency filters at 2–3 Hz (low) and 10 kHz (high). *MCP*, Metacarpophalangeal joint; *MTP*, metatarsophalangeal joint; *NCS*, nerve conduction study.

From Chen S, Andary M, Buschbacher R, et al. Electrodiagnostic reference values for upper and lower limb nerve conduction studies in adult populations. *Muscle Nerve.* 2016;54:371-377.

AANEM Reference Values

Motor Nerve Reference Values

NERVE (muscle studied)	AGE	DISTAL LATENCY (ms)*/ GENDER)	AGE	AMPLITUDE (mv)**	SITE	AGE/HEIGHT	CONDUCTION VELOCITY (m/s)**
Median (APB) 8 cm	All	4.5	All	4.1		All	49
	19-49	4.6 M 4.4 F	19-39	5.9		19-39	49 M 53 F
			40-59	4.2			
	50-79	4.7 M 4.4 F	60-79	3.8		40-79	47 M 51 F
Ulnar (ADM) 8 cm	All	3.7	All	7.9	Below elbow	All	52
					Across elbow	All	43
					Above elbow	All	50
Fibular/ Peroneal (EDB) 8 cm	All	6.5	All	1.3	Ankle to fibular head	All	38
						19-39,<5'7"	43
			19-39	2.6		19-39,>5'7"	37
			40-79	1.1		40-79,<5'7"	39
						40-79,>5'7"	36
					Across fibular head	--	42
Tibial (AH) 8 cm	All	6.1	All	4.4		All	39
			19-29	5.8		19-49,<5'3"	44
						19-49,5'3"-5'7"	42
			30-59	5.3		19-49,≥5'7"	37
						50-79,<5'3"	40
			60-79	1.1		50-79,5'3"-5'7"	37
						50-79,≥5'7"	34

*Upper limits 97th% of the observed differences distribution **Low limits 3rd% of the observed differences distribution

Reference: Shan Chen, et al. AANEM Practice Topic: Electrodiagnostic reference values for upper and lower limb nerve conduction studies in adult populations. Muscle Nerve 2016;54: 371-377.

• **Fig. 9.2** AANEM motor neuron reference values. *ADM,* abductor digiti minimi; *AH,* abductor hallucis *AMP*, Amplitude; *APB,* abductor pollicis brevis; *CV*, conduction velocity; *EDB,* extensor digitorum brevis; *M,* male; *F,* female. (From Chen S, Andary M, Buschbacher R, et al. Electrodiagnostic reference values for upper and lower limb nerve conduction studies in adult populations. *Muscle Nerve*. 2016;54:371-377.)

Acceptable Amplitude/Conduction Velocity Drop

NERVE	SITE	AMP DROP	CV DROP
Ulnar Motor	Across Elbow		15 m/s *or* 23%
Fibular Motor	Ankle to below fibular head	32%	--
	Across fibular head	25%	6 m/s or 12%
Tibial Motor	Ankle to knee	10.3 mV or 71%	--

Sensory Nerve Reference Values

NERVE	AGE	ONSET LATENCY (ms)	PEAK LATENCY ( )	BMI	AMPLITUDE (μV)*	
					ONSET TO PEAK	PEAK TO PEAK
Superficial Radial (10 cm)	All	2.2	2.8	All	7	11
Median (second digit, wrist 14 cm, palm 7 cm)	All	3.3 (wrist)	4.0 (wrist)	All	11 (wrist)	13 (wrist)
		1.6 (palm)	2.3 (palm)		6 (palm)	8 (palm)
	19-49	--	--	< 24	17	19
		--	--	≥ 24	11	13
	50-79	--	--	< 24	9	15
		--	--	≥ 24	7	8
Ulnar (fifth digit, 14 cm)	All	3.1	4.0	All	10	9
	19-49	--	--	< 24	14	13
				≥ 24	11	8
	50-79	--	--	< 24	10	13
				≥ 24	5	4
MABC (10 cm)	--	--	2.6	--	4	3
LABC (10 cm)	--	--	2.5	--	5	6
Sural (14 cm)	--	3.6	4.5	--	4	4

*The lower limits of onset-to-peak and peak-to-peak amplitudes are shown as mean -2 SD, showing the statistically significant effects of age and BMI on the amplitudes of the median and ulnar sensory nerves at the wrist ($P < 0.01$).

Median and Ulnar Latency Differences

MEDIAN-ULNAR (wrist)	SENSORY (14 cm)		MOTOR (8 cm)*	
	ONSET	PEAK	AGE	
Changes in Latencies (ms)*	0.5	0.4	All	1.5
			19-49	1.4
			50-79	1.7

*Upper limits of normal is the 97th% of the observed differences distribution. These values only apply if the median is greater than the ulnar value, look up the normals if ulnar sensory is greater than median. The ulnar motor latency should not be longer than the median. If it is, then ulnar pathology at the wrist may be present.

Electrode placement: ground electrode always placed between the stimulating and recording electrodes. Temperature above 32°C for the upper extremities and above 31°C for the lower extremities, measured at the dorsum of the hands and feet.

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AANEM

Charts created 4/20/2024

• **Fig. 9.2, cont'd**

nerves are easily accessible and allow assessment for both entrapments and polyneuropathies. In the lower limbs, the most readily accessible sensory nerve is the sural nerve, which has been recommended as an excellent nerve for screening for distal symmetric polyneuropathy.[70] Sensory nerve responses—sural and fibular (peroneal)—should almost always be normal in persons with lumbosacral radiculopathy.[134] If there is a reduction of sensory nerve action potential (SNAP) amplitude, then causes other than radiculopathy should be considered.[134] In the rare circumstances that the sensory response is reduced or absent and a radiculopathy is clear from EMG findings, it may be caused by a lateral herniated disc, or the dorsal root ganglion is within the lumbar canal.[134]

Motor NCS should be performed in almost all situations.[7] Examiners should assess the morphology of the waveform, its latency, amplitude, and conduction velocity. Conduction should be performed across any suspected sites of entrapment or injury. A study pointed out the problems in identifying polyneuropathy that arise when different experts and technologists use their own laboratory standards when evaluating patients with and without diabetic sensorimotor polyneuropathy.[67] In this well-done prospective study, they found "interobserver variability" was attributed to differences in performance of nerve conduction testing.[67] In a companion study, Litchy et al.[67,123] instituted standard instructions and standard techniques and procedures, and significantly reduced the interobserver differences in nerve conduction. Together, these two investigations underscore the need for standard well-derived reference values for practice across the United States. These findings underscore the value of adopting the norms derived from the AANEM as described earlier and shown in Table 9.2 and Fig. 9.2.

H-Reflexes

The commonly used H-reflex is a late response that is the electrophysiologic equivalent of an Achilles muscle stretch reflex without the use of the muscle spindle. The H-reflex is called a late response because the tibial nerve is stimulated in the popliteal fossa, and the axonal depolarization wave travels proximally along the tibial and sciatic nerves through the S1 nerve root up to the spinal cord and back down the entire length of the nerve along the reflex arc to the gastrocnemius/soleus muscles. Care must be taken to perform this test correctly. It is a submaximal elicited reflex response. The stimulus duration is 1 ms, and the stimulus should be slowly increased in 3- to 5-mA increments. The patient should be relaxed, and the stimulus frequency should be less frequent than once per second. A pause between stimulations is optimal to decrease the patient's anticipation of the next stimulus. The H-reflex is consistent in latency and morphology. As the stimulus current is gradually increased, the H-reflex reaches its maximum amplitude and then extinguishes as the CMAP response becomes maximal. Many researchers have evaluated its sensitivity and specificity with regard to lumbosacral radiculopathies and generally found a range of sensitivities from 32% to 88%.[114] However, the specificity has been reported at 91% for H-reflexes in lumbosacral S1 radiculopathy.[128] The H-reflex can also help separate S1 radiculopathy from L5 radiculopathy, the latter being more likely to have a normal reflex.[128] Published AANEM guidelines suggest that H-reflexes are useful in evaluating for S1 radiculopathy.[39] H-reflexes are useful for assessing for demyelinating polyneuropathy and cauda equina syndrome and confirming a sciatic neuropathy. Remember that delay or decreased amplitude in the H-reflex can occur with lesions anywhere along its course, including the sciatic nerve, the lumbosacral plexus, or the S1 root. H-reflex studies of other nerves are more difficult to elicit, and they are not commonly used in clinical practice.

F-Waves

F-waves are late responses involving the motor axons and axonal pool at the spinal cord level. They can be elicited in most upper and lower limb muscles and are typically tested by stimulating the median, ulnar, fibular (peroneal), or tibial nerves when recording from muscles they innervate. In contrast to H-reflexes, they are elicited by maximal nerve stimulation. F-waves represent the rebound axon "backfiring" of the alpha motor neurons in a ready state to depolarize along a long arc from the stimulus location (i.e., wrist for median nerve) up to the anterior horn of the spinal cord and back down to the recording site (abductor pollicis brevis in thenar region). F-waves vary in morphology and latency because a different motor neuron pool is in a ready state to backfire at any given time. They should fall roughly within the same latency period when recording multiple F-waves. F-waves demonstrate low sensitivities for radiculopathy and thus have a limited role in such evaluations[161]; however, they are useful in the assessment of suspected polyneuropathy.[39]

A-Waves

A-waves or A-reflexes are also late responses involving motor units that are occasionally seen that may get confused with F-waves. They are characterized by identical or nearly identical waveforms at the same latency in contrast to F-waves. They typically occur with nearly every stimulation. A-waves are usually noted in the time between the M-wave and the F-wave, but they can be seen after the F-wave. When A-waves have the same latency as F-waves, they are very difficult to reliably separate from the F-waves. The two major theories for their presence are proximal nerve sprouting and ephaptic transmission from one axon to another. They are usually considered abnormal.[63,145]

Needle Electromyography

Needle EMG involves the placement of a needle electrode, monopolar or concentric, within muscle to evaluate electrical potentials that can be assessed auditorily and visually (waveforms). EMCs must be able to properly recognize, characterize, describe, and interpret various waveforms to establish EDX diagnoses. Disposable needles are recommended. Surface EMG has no clinical use over standard needle EMG.[132] EMG testing, a vital part of EDX evaluations, provides useful information and, although somewhat uncomfortable, poses minimal risks to the patient. There are several means for reducing the discomfort of needle EMG. Ibuprofen administered before the testing can decrease self-reported discomfort.[68] Mild cold spray over the skin can reduce discomfort. Care must be taken to use a coolant spray that will not burn the skin. Concentric needles are equal to monopolar needles in the degree of patient discomfort.[11]

The AANEM supports the Occupational Safety and Health Administration rule that mandates the use of gloves and universal precautions during testing. The majority of EMCs report at least one needlestick injury in their career.[130] The most common preventable reason for injury was a perceived lack of time.[130] Using a device to clamp the plastic sleeve that holds the needle and replacing the needle with a one-handed technique are recommended to help prevent needlesticks.

In most cases, it makes no difference whether the skin is antiseptically prepared before needle insertion.[6] Some examiners use alcohol as an antiseptic, and certainly if the skin is not clean then alcohol represents a useful skin preparation. With regard to lymphedema, use caution and sterilize the skin, and avoid traversing an infected space or an area of clearly taut skin that could weep serous fluid after the needle insertion. Care should also be used to avoid cellulitis.[8] The needle should not penetrate infected skin or open ulcerations or wounds. Needle EMG is not listed by the American Heart Association as a procedure requiring prophylactic antibiotic treatment to prevent endocarditis.[6]

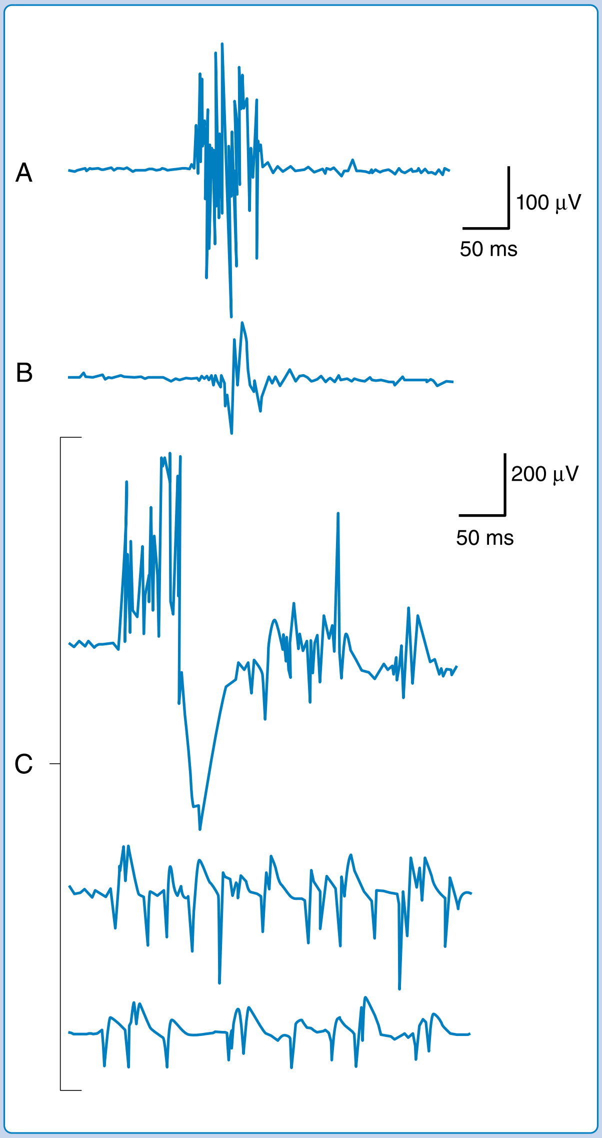
Hemorrhagic complications from needle EMG are rare and are asymptomatic when they occur.[81] In one case series, minor paraspinal muscle hematomas were noted on MRI performed shortly after needle EMG examination in 4 of 17 patients who were not receiving anticoagulants. These hematomas were small and of no clinical significance.[29] In this report, the authors state that there has never been a case report of paraspinal hematoma compressing the spinal roots or the epidural space.[29] Ultrasound evaluation of over 700 muscles showed only two small subclinical hematomas in patients who were anticoagulated.[19] However, since then, there has been at least one case of threatened compartment syndrome in the anterior tibialis of a patient who was supratherapeutically anticoagulated.[24] Although clinicians should always weigh carefully the risk-to-benefit ratio of testing, a person with appropriate levels of anticoagulation for venous thromboembolism can be safely studied with little risk of neurologic complications. Appropriate pressure should be applied to the area after testing to prevent intramuscular hematoma. The paraspinal muscles are such an important region to study, not only for persons with suspected radiculopathy but also in neuromuscular evaluations, that EMCs should examine them whenever reasonably possible.

As noted, the electrodes for standard EMG generally fall into two categories. Monopolar needles are those in which the needle is coated with Teflon except for the tip, and the differences in potential from the tip of the electrode to a nearby surface electrode are recorded. Concentric needles are those with a fine wire running through the center of an insulated shaft that is electrically referenced to an outer metal shaft. Concentric needles are useful for quantitative motor unit analysis and are commercially available in disposable form. Background interference is minimized with concentric needles, and the ground electrode can remain in one place for a given limb.

Needle EMG examination involves the assessment of four separate components of muscle electrical activity. These aspects are (1) insertion activity, (2) spontaneous activity, (3) the morphology and size of motor unit action potentials (MUAPs), and (4) motor unit recruitment.

Needle Insertional Activity

Placing a needle (monopolar or standard concentric) recording electrode into healthy muscle tissue and advancing it in quick but short intervals results in brief bursts of electrical potentials referred to as insertional activity (Fig. 9.3A and Video 9.1).[63] The observed electrical activity is believed to result from the needle electrode mechanically depolarizing the muscle fibers surrounding its leading edge as it pierces and deforms the tissue. Insertional activity and spontaneous activity are usually examined with three or four insertions in each of four muscle vectors or quadrants. Smaller muscles, such as the abductor pollicis brevis, can be adequately assessed with fewer needle movements. After a brief, small movement of the needle, insertional activity usually persists for no more than a few hundred milliseconds. Minimal and localized muscle tissue damage may occur from direct needle trauma and is the basis for the synonymous term of injury potentials.



• **Fig. 9.3** (A) Inserting a monopolar needle into healthy muscle tissue results in mechanical depolarization of muscle tissue, which generates a brief burst of electrical activity designated as insertional activity. (B) Inserting the same needle into fibrotic muscle or subcutaneous fatty tissue results in decreased insertional activity. (C) Inserting a monopolar needle into denervated muscle tissue produced not only the initial burst of electrical activity but also associated positive sharp waves and fibrillation potentials that abate over several hundred milliseconds. (Redrawn from Dumitru D, Amato AA, Zwarts MJ. *Electrodiagnostic Medicine.* 2nd ed. Hanley & Belfus; 2002.)

Decreased Insertional Activity

Atrophied muscle that has been replaced by fibrous or fatty tissue or that is otherwise electrically unexcitable can no longer generate electrical activity (Video 9.2). Consequently, the needle electrode is incapable of mechanically depolarizing this tissue. The result is that few, if any, electrical waveforms will be detected after needle movement and usually lasts less than 100 ms. This is usually considered abnormal. Care must be taken to ensure that the lack of electrical activity is not caused by placement of the needle into subcutaneous tissue instead of muscle (see Fig. 9.3B).

Increased Insertional Activity

Practitioners have noted that insertional activity can appear to persist after needle movement cessation. This finding has led to the term *increased insertional activity* (Video 9.3). In disease states in which the muscle is no longer connected to its nerve or the muscle membrane is inherently unstable from primary muscle pathology or abnormal ion channels, the increased insertional activity completes a temporal continuum from the previously normal insertional activity to the development of sustained membrane instability or waveforms that look like positive sharp wave (PSW)/fibrillation potentials (see Fig. 9.3C). There is no clear conventional cutoff time when persistent increased insertional activity becomes a run of fibrillation potentials. Many practitioners consider this cutoff time between 1 and 3 seconds after the needle has stopped moving. Very few authors specifically address this detail.[44] The use of the longer time of 2 or 3 seconds improves the specificity and limits false-positive results, but it slightly decreases the sensitivity of the needle EMG. Increased insertional activity with no fibrillations should be interpreted with caution and can be normal.

End-Plate Potentials

An active needle electrode located in the end-plate region can record two distinct normal waveforms: either miniature end-plate potentials (MEPPs) or end-plate spikes. It is imperative that the EMC properly characterize waveforms observed during EMG analysis to avoid errors in diagnosis.

Miniature End-Plate Potentials

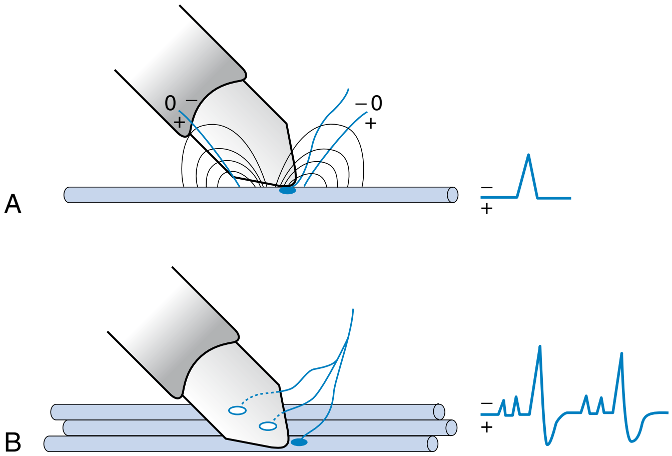
One of the waveforms is a short-duration (0.5–2 ms), small (10–50 μV), irregularly occurring, monophasic negative waveform (Video 9.4).[63] These potentials represent the random release of acetylcholine vesicles. Volume conduction theory would suggest that for a potential to be monophasic and negative, the current sink would have to start and finish within the very small recording region of the active electrode (Table 9.3).[66] Clinically, multiple MEPPs are the same as end-plate noise, and the sound is referred to as resembling a "seashell murmur," which sounds like noise generated by a seashell held close to the ear (Fig. 9.4).

End-Plate Spikes

A second waveform (end-plate spikes) can be detected with an active needle electrode placed in the end-plate region (Videos 9.5 and 9.6). These potentials are generated from the actual muscle fiber depolarization, similar to fibrillation potentials, and they are relatively short in duration (3–4 ms), of moderate amplitude (100–200 μV), irregularly firing, and biphasic with an initial negative deflection (Table 9.4).[63] End-plate spikes and MEPPs are frequently observed together because they arise from the same region (Fig. 9.5). They can be mistaken for fibrillation potentials (Table 9.5), but they fire irregularly, a distinction usually made visually on the display screen as well as by listening carefully. The needle should be moved to a new location after it enters an end-plate region to reduce pain and enter muscle more conducive to assessment for fibrillation potentials.

TABLE 9.3 Characteristics of End-Plate Noise (Video 9.4)

Characteristic	Details
Appearance	Monophasic negative waves
Rhythm	Irregular
Frequency	Usually so frequent and can be superimposed upon each other so that they cannot be individually seen
Amplitude	≤10 μV
Stability	Persist until the needle is moved
Observed in	Normal finding recorded from the end-plate zone



• **Fig. 9.4** (A) Monopolar needle electrode located over the end plate of a muscle records the spontaneous depolarization of a miniature end-plate potential (MEPP). As the electrode is located over the subthreshold central current sink of this potential, and hence does not propagate, a monophasic negative potential is recorded. (B) The large recording electrode is usually positioned over several end plates, thus recording multiple MEPPs and end-plate spikes. (Redrawn from Dumitru D, Amato AA, Zwarts MJ. *Electrodiagnostic Medicine.* 2nd ed. Hanley & Belfus; 2002.)

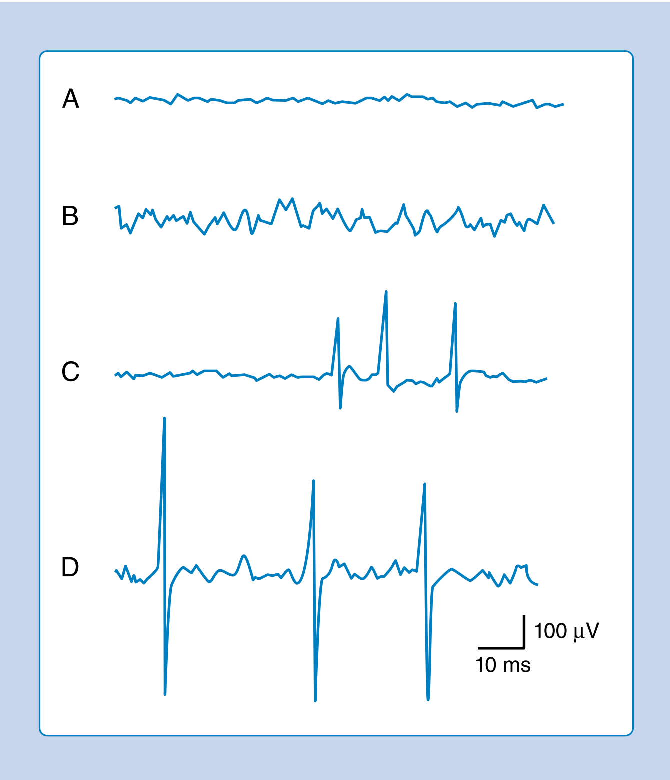
Fibrillation Potentials

Fibrillation potentials can have two basic forms, spike and positive sharp wave. These two forms likely represent the same electrophysiologic process and are interpreted similarly. It is important to recognize each waveform morphology to avoid confusion with other waveform types.

Fibrillation potentials have been characterized as the spike form or biphasic waveform (Videos 9.7, 9.8, and 9.9).[65] Characteristics are shown in Table 9.5. There is strong evidence that the PSW is a fibrillation that has a different shape primarily because the needle electrode is touching the fibrillating muscle cell, altering the muscle membrane and consequently the shape of the waveform. Fibrillation potentials are simply spontaneous depolarizations of a single muscle fiber and demonstrate waveform configurations similar to those of single muscle fibers that are voluntarily activated (Fig. 9.6). Fibrillation potentials are typically short in duration (<5 ms), less than 1 mV in amplitude, and fire at rates between 1 and 15 Hz. They have a typical sound, likened to a high-pitched tick similar to "rain on a tin roof," when amplified through a

TABLE 9.4 Characteristics of End-Plate Spikes (Videos 9.5, 9.6, and 9.10)

Characteristic	Details
Appearance	Biphasic or triphasic spikes or positive waves
Rhythm	Irregular
Frequency	Intervals frequently <50 ms
Amplitude	50–300 μV
Stability	Persists until the needle is moved
Observed in	Is a normal finding



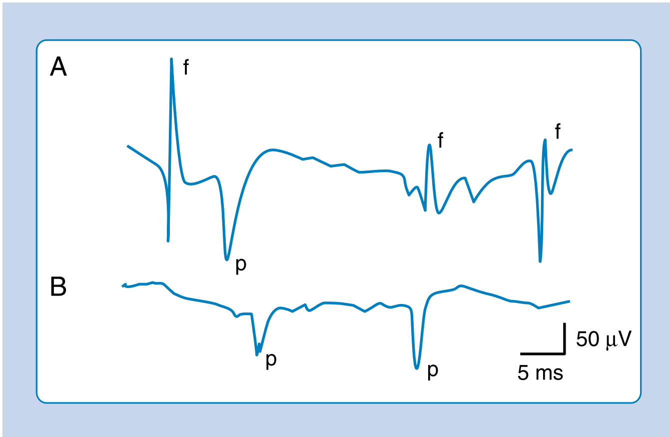
• **Fig. 9.5** (A) Monopolar needle located in a healthy muscle at rest. (B) Slight needle movement positions the electrode in an end-plate region with the recording of multiple miniature end-plate potentials of a negative spike configuration. (C) Repositioning the needle electrode to a slightly different region primarily records biphasic, initially negative end-plate spikes. (D) Advancing the needle electrode slightly permits the simultaneous recording of both potentials noted individually in (B) and (C). (Redrawn from Dumitru D, Amato AA, Zwarts MJ. *Electrodiagnostic Medicine.* 2nd ed. Hanley & Belfus; 2002.)

loudspeaker. When the recording electrode is located in the previous end-plate zone of a denervated muscle, fibrillation potentials can be biphasic with an initial negative deflection. A recording electrode outside the end-plate zone, but far from the tendinous region, will detect fibrillation potentials that are triphasic (positive-negative-positive).

It is possible to generate waveforms with similar configurations to fibrillation potentials while examining healthy muscle tissue.

TABLE 9.5 Characteristics of Fibrillation Potentials (Videos 9.7, 9.8, and 9.9)

Characteristic	Details
Appearance	Brief spike or positive sharp wave
Rhythm	Regular, rarely irregular
Frequency	0.5–15 Hz
Amplitude	20–300 μV
Stability	Gradually taper off
Observed in	Nearly all lower motor neuron disorders, severe myopathies, and occasionally severe neuromuscular junction disorders and in chronic spinal cord injuries and stroke

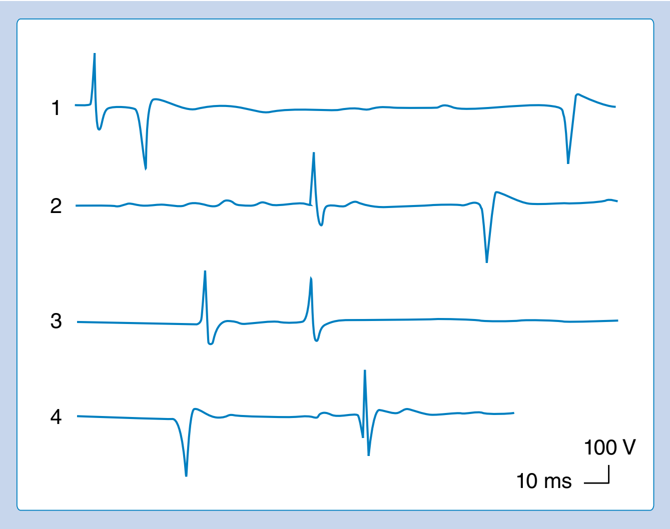


• **Fig. 9.6** (A) Monopolar needle recording of positive sharp waves (PSWs) *(p)* and fibrillation potentials *(f)*. (B) Only PSWs are depicted. (Redrawn from Dumitru D, Amato AA, Zwarts MJ. *Electrodiagnostic Medicine.* 2nd ed. Hanley & Belfus; 2002.)

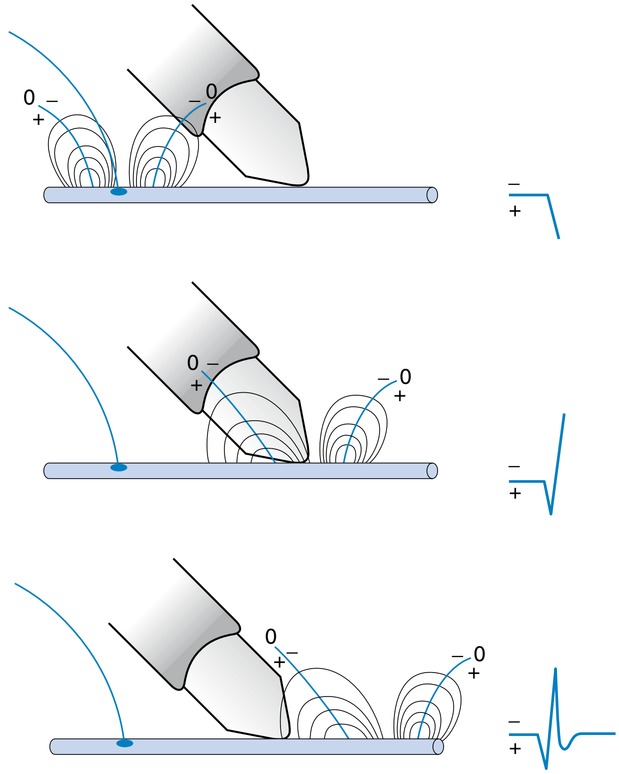
Specifically, inadvertently irritating the terminal axon or end-plate zone with the shaft of an electrode and evoking end-plate potentials while simultaneously recording at some distance from the end-plate zone will generate triphasic end-plate spikes that are identical in appearance to fibrillation potentials (Fig. 9.7).[63] This is understandable because both waveforms are single muscle fiber discharges. The key to identification is the rapid and irregular discharge of end-plate spikes, whereas fibrillation potentials usually fire slower and very regularly. Irregularly firing fibrillation potentials occur at times and are less well understood, but they are thought to arise from spontaneous depolarizations within the transverse tubule system.

Positive Sharp Waves

PSWs are fibrillation potential waveforms that can be recorded from a single muscle fiber having an unstable resting membrane potential secondary to denervation or intrinsic muscle disease. Typically, they have a large primary sharp positive deflection followed by a small negative potential (see Video 9.8). These potentials are called PSWs (see Fig. 9.6). Amplified through a



• **Fig. 9.7** A needle electrode is purposefully located in an end-plate region where prototypical and atypical end-plate spikes are evoked. Note that end-plate spikes may appear biphasic, initially negative, as anticipated, as well as triphasic and biphasic, initially positive, resembling fibrillation potentials and positive sharp waves, respectively. The important distinction between normal and abnormal waveforms in this case is not waveform configuration but rather firing rate (highly irregular for end-plate spikes). (Redrawn from Dumitru D, Amato AA, Zwarts MJ. *Electrodiagnostic Medicine.* 2nd ed. Hanley & Belfus; 2002.)



• **Fig. 9.8** A single muscle fiber action potential propagating past a needle recording electrode results in a triphasic waveform. This is because the voltage distribution creates an initial and terminal positive voltage source surrounding a negative current sink zone. (Redrawn from Dumitru D, Amato AA, Zwarts MJ. *Electrodiagnostic Medicine.* 2nd ed. Hanley & Belfus; 2002.)

loudspeaker, PSWs have a regular firing rate (1–15 Hz) and a dull thud sound. Their durations range from several milliseconds to 100 ms or longer. Although commonly observed to fire spontaneously, PSWs can be provoked by electrode movement, as can the spike form of fibrillation potentials.

A number of other waveforms can be observed that have configurations resembling a PSW. A MUAP recorded from the tendinous region can also have an initial positive deflection followed by a negative potential because the current sink cannot pass beyond the recording electrode.[63] It is also possible for the recording electrode to damage a number of muscle fibers in close proximity to the recording surface, again preventing an action potential from passing its recording surface, resulting in a primarily positive potential. These two potentials can be distinguished from a PSW in that they are MUAPs and subject to voluntary control, whereas a PSW is not. Asking the individual to contract and relax the muscle under investigation should demonstrate that the waveform has a variable firing rate. A PSW typically fires at a regular rate and is not under voluntary control. If any doubt remains, the electrode should be repositioned until successful recordings are obtained. Similar to end-plate noise and end-plate spikes, PSWs and fibrillations will usually disappear when the needle electrode is moved a small distance, such as 1 to 5 mm, whereas a motor unit will usually remain on the screen and change shape. This is because the fibrillation potential is generated by a very small electrical charge from only one muscle fiber, whereas a MUAP is generated by hundreds of muscle fibers covering an area of 1 to 3 cm^2.

Spontaneous Waveform Interpretation Clinical Caveats

The single muscle fiber generates three waveforms that are at times difficult to separate: end-plate spikes (see Videos 9.5 and 9.6) and the two types of fibrillation potentials (PSWs and spike fibrillation potentials) (see Videos 9.7, 9.8, and 9.9). Distant motor units can mimic the shape of the PSW but fire in a semirhythmic manner; by moving the needle closer, the positive waveform changes shape and looks more like a MUAP (Video 9.10). In contrast, moving the needle electrode 1 to 5 mm will almost always cause fibrillations to disappear. End-plate spikes can have the identical morphology of PSW and fibrillation potentials. The major key to separating them is that PSW and fibrillation potentials have a regular firing rate and can slowly trail off before they disappear. End-plate spikes usually fire irregularly and disappear when the needle is moved 1 to 5 mm. The extracellular waveform configuration of a single muscle fiber, similar to nerve tissue, depends on the characteristics of the intracellular action potential of a muscle. The action potential of a muscle is approximately 4 to 20 times longer than that of a nerve because of the prolonged repolarization process.[63] A triphasic waveform with a small terminal phase should be recorded from an extracellular active electrode placed adjacent to a propagating single muscle fiber action potential at some distance from the end-plate region (Fig. 9.8).

Transient runs of PSW-appearing potentials can be seen in healthy skeletal muscle, particularly in the paraspinal and hand or foot intrinsic muscles. The nonpathologic PSWs are thought to arise because the needle electrode is oriented in such a manner as to irritate the terminal axon of an end-plate but extend along the muscle fiber while compressing the tissue and preventing action potential conduction.[63] The induced end-plate spike appears similar to a PSW, but it displays a relatively rapid and irregularly firing rate characteristic of end-plate spikes (see Fig. 9.7). Sustained PSWs are more significant than unsustained PSWs.

Diffuse abnormal insertional activity with prolonged trains of PSWs in essentially every muscle, yet without any symptoms or disability, has been described as "EMG disease."[190] This is a rare condition that is infrequently encountered in clinical practice and is of unclear clinical significance; it may be a subclinical myotonic disorder or a mild type of channelopathy.

Fibrillations are seen in any condition causing denervation (axon loss), including nerve disease, inflammatory myopathies, and direct muscle trauma.[142] In persons with complete spinal cord injury, muscles innervated by roots below the level of the lesion also demonstrate fibrillation potentials.[180] Such prevalence of these findings in the legs of patients with spinal cord injuries can make EDX testing for such patients ineffective at identifying radiculopathies or entrapment neuropathies. Similar fibrillations and positive waves can also be seen in the hemiparetic limbs of patients with strokes, and such findings should be interpreted with caution.[97]

Fibrillation potentials (spike form fibrillation potentials and PSWs) are commonly graded with a 0 to 4 grading scheme. This grading scale is described as follows:

Grade 0: No reproducible abnormality

Grade 1: Transient but reproducible trains (>2–3 seconds) of discharges (fibrillations or PSWs) after moving the needle in more than one site or quadrant (vector)

Grade 2: Occasional spontaneous potentials in more than two different sites or quadrants (vectors)

Grade 3: Spontaneous potentials present in all sites or quadrants (vectors)

Grade 4: Abundant spontaneous potentials nearly filling the screen in all sites or quadrants (vectors)

The finding of fibrillations in only one area of the muscle that is not easily reproducible is probably of uncertain significance. There is little scientific evidence that there is a difference between the various grades of fibrillations. The density of fibrillation potentials does not necessarily correlate with the degree of nerve damage and loss of axons. CMAP gives a better estimate of the proportion of axons remaining. The innervation ratio is the average size of the motor unit expressed as a ratio between the total number of extrafusal muscle fibers and the number of innervating motor axons. For small extraocular muscles, this ratio is 1:3 (expressed as 3/1). For large muscles, such as the gastrocnemius, this ratio increases to approximately 1934 muscle fibers per motor axon (1934/1). This means that the loss of relatively few motor axons results in many fibrillating muscle fibers in these larger muscles.[105,106] The peak-to-peak amplitude of fibrillation potentials can be helpful in determining the time course of a direct nerve injury. Small fibrillation potentials suggest a more chronic problem that has lasted over 6 months.[111] In persons with traumatic injury, the fibrillation potentials averaged 612 μV for the first 2 months. By the sixth month postinjury, the fibrillations averaged 320 μV. After 1 year, there were no populations of fibrillations larger than 100 μV.[111] This reduction in amplitudes over the first year was identified in persons with a defined single-event severe trauma to specific nerves. It is less clear how well the decrease in fibrillation amplitude over time correlates with symptoms in persons with other conditions (besides nerve injuries), such as radiculopathy. Small fibrillation potentials are more suggestive of a chronic problem that has lasted 6 months to 1 year.[111]

Complex Repetitive Discharge

A complex repetitive discharge (CRD) is a spontaneously firing group of action potentials (formerly called bizarre high-frequency discharges or pseudo-myotonic discharges) and requires a needle recording electrode for detection (Videos 9.11 and 9.12).[63] CRD waveforms are continuous runs of simple or complex spike patterns (fibrillation potentials or PSWs) that regularly repeat at 0.3 to 150 Hz. The repetitive pattern of spike potentials has the same appearance at each firing and bears the same relationship with its neighboring spikes (Table 9.6).

A distinct sound, likened to that of heavy machinery or an idling motorcycle, is produced by CRD firing. In addition to the sound and repetitive pattern, a hallmark of these waveforms is that they start and stop abruptly. CRDs might begin spontaneously or be induced by needle movement, muscle percussion, or muscle contraction. Nerve block and curare do not abolish CRDs, suggesting that the origin of these potentials is within the muscle tissue. Single-fiber and standard electromyographic studies suggest that CRDs are generated by an ephaptic activation of adjacent muscle cells.[44] Detection of these waveforms usually suggests that a chronic abnormal process has resulted in a group of muscle fibers becoming separated from their neuromuscular junctions. They may be associated with a currently active disease or be the residual from a past disorder.

Myotonic Discharges

The phenomenon of delayed muscle relaxation after muscle contraction is referred to as myotonia or action myotonia (Videos 9.13 and 9.14).[63] The finding of delayed muscle relaxation after reflex activation or induced by striking the muscle belly with a reflex hammer is called percussion myotonia. Clinical myotonia is usually accentuated by energetic muscle activity after a rest period. Continued muscle contraction lessens the myotonia and is known as the "warmup." It is thought that cooling the muscle accentuates myotonia, but this finding has only been objectively documented in paramyotonia congenita.

Myotonic discharges present in one of two waveform types (Table 9.7). The myotonic potential induced by needle electrode insertion usually assumes a morphology like that of a PSW or a spike fibrillation potential. It is thought that the needle movement induces a repetitive firing of the unstable membranes of multiple single muscle fibers. This is because the recording needle is thought to have damaged that portion of the muscle fiber with which it is in contact. Regardless of the waveform type, the hallmark of myotonia is the waxing and waning in both frequency and amplitude. The myotonic discharge has a characteristic sound, likened to that of a dive bomber. Trains of PSWs that trail off and stop but do not restart (wane, but do not wax) can be confused with myotonia. Amplitudes range from 20 to 300 μV, and firing rates range from 20 to 100 Hz.[177] Myotonic discharges can occur with or without clinical myotonia. The observation of these potentials requires needle movement or muscle contraction. These waveforms persist after nerve block, neuromuscular block, or frank denervation. This suggests that their site of origin is the muscle membrane itself, and they appear to be attributable to a channelopathy. Although the exact mechanism of myotonic discharge production remains unclear, myotonia is found in channelopathies of both the sodium and potassium channels.[44] Myotonic discharges are not specific for any one disorder. In addition to the myotonic disorders of muscle, myotonic discharges also can be

TABLE 9.6 Complex Repetitive Discharge Characteristics (Videos 9.11 and 9.12)

Characteristic	Details
Appearance	May take any form, but this form is constant from one potential complex to the next
Rhythm	Regular
Frequency	10–100 Hz
Amplitude	50–1000 μV
Stability	Abrupt onset and cessation
Observed in	Myopathies: polymyositis, limb-girdle dystrophy, myxedema, Schwartz-Jampel syndrome Neuropathies: poliomyelitis, spinal muscular atrophy, amyotrophic lateral sclerosis, hereditary neuropathies, chronic neuropathies, carpal tunnel syndrome. Normal: iliopsoas, biceps brachii

Modified from Dumitru D. Needle electromyography. In: Dumitru D. ed. *Electrodiagnostic Medicine.* Hanley & Belfus; 1995.

TABLE 9.7 Characteristics of Myotonic Discharges (Videos 9.13 and 9.14)

Characteristic	Details
Appearance	Brief spikes, positive waveform
Rhythm	Wax and wane
Frequency	20–100 Hz
Amplitude	Variable (20–300 μV)
Stability	Firing rate alterations
Observed in	Myopathies: myotonic dystrophy, myotonia congenita, paramyotonia, polymyositis, acid maltase deficiency, hyperkalemic periodic paralysis Other: chronic radiculopathy, chronic peripheral neuropathy

Modified from Dumitru D. Needle electromyography. In: Dumitru D. ed. *Electrodiagnostic Medicine.* Hanley & Belfus; 1995.

detected occasionally in acid maltase deficiency, polymyositis, drug-induced myopathies, severe axonal neuropathies, and at times in any nerve or muscle disorder.

Fasciculation Potentials

Fasciculation potentials are a visible spontaneous contraction within a small portion of muscle. When these contractions are observed with an intramuscular needle recording electrode, they are called fasciculation potentials (Video 9.15).[63] A fasciculation potential is the electrically summated voltage of depolarizing muscle fibers belonging to all or part of one motor unit. Occasionally, fasciculation potentials can be documented only with needle EMG because they lie too deep in muscle to be seen from the skin. Recently, ultrasound has been used to visualize fasciculation potentials in skeletal muscle.

TABLE 9.8 Characteristics of Fasciculation Potentials (Video 9.15)

Characteristic	Details
Appearance	Single motor unit
Rhythm	Irregular and random
Frequency	Variable, absent to 1–2 Hz
Amplitude	Variable, similar to motor units, 20 μV–20 mV depending on needle location and motor unit characteristics
Stability	Each fasciculation is an individual motor unit and looks different. If the same unit is fasciculating frequently, it is usually stable unless that motor unit happens to be unstable and thus is likely to be pathologic
Observed in	Nearly all normal people. Is more frequent in nearly any lower motor neuron disorders. If they are accompanied by fibrillations, they are much more likely to be abnormal

Fasciculation waveforms can be characterized with regard to phasicity, amplitude, and duration (Table 9.8). Their discharge rates (absent or <1 per minute to 1–2 Hz) are irregular and occur randomly. They are not under voluntary control, nor are they influenced by mild contraction of the agonist or antagonist muscles. Fasciculation potential site of origin is unclear: The discharge might arise from the anterior horn cell, anywhere along the entire peripheral nerve (particularly the terminal portion), or at times within the muscle itself. Fasciculation potentials occur in almost all normal persons in many muscles, but they occur very commonly in the foot intrinsic, gastrocnemius, and soleus muscles in addition to patients with neuromuscular disorders. Typical diseases in which fasciculation potentials can be found include motor neuron disorders, radiculopathies, entrapment neuropathies, and cervical spondylotic myelopathy. Fasciculation potentials have also been described in metabolic disturbances, including tetany, thyrotoxicosis, and anticholinesterase overdoses. Benign (normal) and pathologic fasciculation potentials look the same. Perhaps the best way to evaluate fasciculation potentials is to analyze the "company they keep." A careful analysis of voluntary MUAP morphology, combined with a search for abnormal spontaneous potentials, is required before concluding that fasciculation potentials are either a normal or an abnormal finding. Their frequency can be measured by waiting for 1 minute and documenting the number of fasciculations per minute.

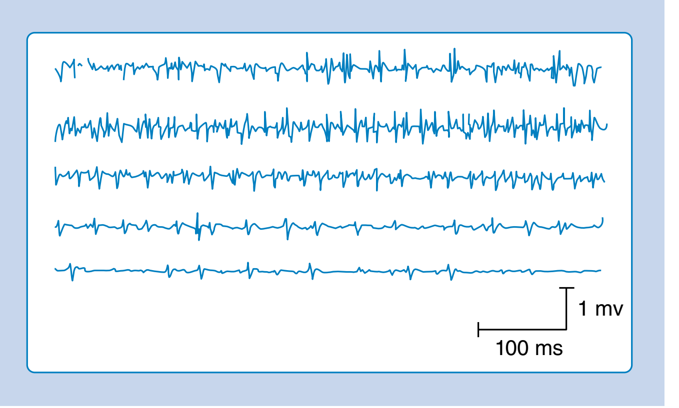
Myokymic Discharge

Myokymia is frequently readily observable as vermicular (bag of live worms) or rippling movement of the skin but can be invisible on the skin. It is usually associated with myokymic discharges (Video 9.16).[63] The myokymic discharge consists of bursts of normal-appearing motor units with interburst intervals of electrical silence, thus there are two frequencies. Typically, the slow firing rate is 0.1 to 10 Hz in a semirhythmic pattern. Each burst usually has 2 to 10 potentials within a single burst that can fire at 20 to 150 Hz. These potentials are not affected by voluntary contraction. The sounds associated with these potentials vary and

TABLE 9.9 Characteristics of Myokymic Discharges (Video 9.16)

Characteristic	Details
Appearance	Normal motor unit action potentials
Rhythm	Regular
Frequency	0.1–10 Hz
Burst frequency	20–250 Hz
Stability	Persistent firing or occasional abrupt cessation
Observed in	Facial: multiple sclerosis, brainstem neoplasm, polyradiculopathy, Bell palsy, normal. Extremity: radiation plexopathy, chronic nerve compression (carpal tunnel syndrome, radiculopathy), rattlesnake venom

Modified from Dumitru D. Needle electromyography. In: Dumitru D, ed. *Electrodiagnostic Medicine.* Hanley & Belfus; 1995.



• **Fig. 9.9** A characteristic muscle cramp as recorded with an intramuscular needle electrode. There is an initial burst of motor unit activity that eventually subsides as the cramp dissipates. (Redrawn from Daube JA. *AAEM Minimonograph 11: Needle Examination in Electromyography.* American Association of Neuromuscular & Electrodiagnostic Medicine; 1979.)

have been described as a type of sputtering often heard with a low-powered motorboat engine, marching soldiers, or bursts of machine gun. The actual discharge may be distinguished from CRDs in that myokymic discharges do not display a regular pattern of spikes from one burst to the next, nor do they typically start and stop abruptly. Myokymic discharges are groups of motor units, whereas CRDs represent groups of single muscle fibers. The groups of motor units within a burst may fire only once or possibly several times. The sputtering bursts of myokymic discharges usually sound very different from the continuous drone of a CRD. Myokymic potentials are rarely seen in healthy people. They are likely to originate from demyelinated motor axons either spontaneously or through ephaptic transmission from axon to axon.[145] Characteristics of myokymic discharges are seen in Table 9.9.

A sustained and possibly painful muscle contraction of multiple motor units, lasting seconds or minutes, can appear in normal individuals or specific disease states (Videos 9.17 and 9.18). In healthy individuals, a cramp usually occurs in the calf muscles or other lower limb muscles after exercise, abnormal positioning, or maintaining a fixed position for a prolonged period. Cramps might also be induced by hyponatremia, hypocalcemia, vitamin deficiency, ischemia, early motor neuron disease, and peripheral neuropathies. Familial syndromes have been reported that involve fasciculations and cramps; alopecia, diarrhea, and cramps; and simply an autosomal dominantly inherited cramp syndrome. A needle-recording electrode placed into a cramping muscle shows multiple motor units firing synchronously between 40 and 60 Hz and occasionally reaching 200 to 300 Hz (Fig. 9.9 and Table 9.10). Cramps are believed to arise from a peripheral portion of the motor unit. A cramp that results in a taut muscle with electrical silence is seen in McArdle disease.

Neuromyotonic Discharges

Neuromyotonia is a rare peripheral form of continuous muscle fiber activity originating in the peripheral motor axon, usually referred to as Isaac syndrome (Table 9.11 and Video 9.19). The continuous motor unit activity is eliminated by neuromuscular block but not peripheral nerve block, spinal or general anesthesia, or sleep. The motor unit activity usually begins in the lower limbs

TABLE 9.10 Characteristics of Cramp Potentials (Videos 9.17 and 9.18)

Characteristic	Details
Appearance	Single and multiple motor units
Rhythm	Can fire similar to motor units in groups
Frequency	Very fast, 40–200 Hz
Amplitude	Variable similar to motor units, 20 μV–20 mV depending on needle location and motor unit characteristics
Stability	Variable number of units that fire, but each unit is usually the same morphology
Observed in	Can be seen in normal people (nocturnal or after exercise). Wide range of neuropathic, endocrinologic, and metabolic conditions

TABLE 9.11 Characteristics of Neuromyotonia Potentials (Video 9.19)

Characteristic	Details
Appearance	Single and multiple motor units that decrease in size over the time of the contraction
Rhythm	Appear regular
Frequency	Very fast, 150–250 Hz
Amplitude	Variable similar to motor units but decrease over time, 20 μV–20 mV depending on needle location and motor unit characteristics
Stability	Stable and gradually decrease but each unit is usually the same morphology
Observed in	Isaacs syndrome, extremely chronic neuropathic diseases, familial neuromyotonia

in the late teens and progresses to all skeletal muscles. Needle recordings demonstrate motor unit discharges with frequencies of 150 to 300 Hz, including doublets, triplets, and multiplets. The neuromyotonic waveforms usually have a decrementing amplitude.[145] Intraoperative nerve mechanical stimulation can generate a form of neuromyotonia called neurotonic discharges. These bursts of motor units can alert the surgeon that the nerve is being mechanically disturbed and is at risk of being damaged.[63]

Waveform Analysis

Needle EMG waveform evaluation requires optimized waveform localization in the muscle for consistent analysis. The waveforms being analyzed should be close to the recording electrode to maximize recognition. Auditory feedback for crisp-sounding waveforms with sharp visual oscilloscope takeoffs from the baseline are indications of optimal needle tip placement within the muscle that facilitate waveform evaluation. Care must be taken to recognize signals that are extrinsic to muscle that may confound waveform analysis. The following videos are provided for reference:
Video 9.1 - Normal needle insertional activity
Video 9.2 - Decreased insertional activity
Video 9.3 - Increased insertional activity
Video 9.4 - Miniature end plate potentials
Video 9.5 - End-plate spikes
Video 9.6 - End-plate spikes
Video 9.7 - Typical single fibrillation
Video 9.8 - Positive sharp waves
Video 9.9 - Positive sharp waves and fibrillations
Video 9.10 - End-plate spike and motor unit
Video 9.11 - Complex repetitive discharge
Video 9.12 - Complex repetitive discharge
Video 9.13 - Myotonia
Video 9.14 - Myotonia
Video 9.15 - Fasciculations
Video 9.16 - Myokymia
Video 9.17 - Cramp
Video 9.18 - Familial cramp syndrome
Video 9.19 - Isaac syndrome myokymic
Video 9.20 - Normal recruitment
Video 9.21 - Stiff person syndrome
Video 9.22 - Doublet
Video 9.23 - Doublets and multiplets
Video 9.24 - Parkinsonism tremor
Video 9.25 - Nascent potentials reinnervation
Video 9.26 - Satellite potential
Video 9.27 - Normal motor unit action potential
Video 9.28 - Motor unit variability in Lambert-Eaton myasthenic syndrome
Video 9.29 - Amyotrophic lateral sclerosis decreased recruitment
Video 9.30 - Myopathy early recruitment
Video 9.31 - 60 Hz notch
Video 9.32 - Electrocardiographic artifact
Video 9.33 - Pacemaker artifact

Motor Units and Recruitment

Motor unit analysis and motor unit recruitment are important distinct components of the needle EMG examination (see eAppendix 9B for details on motor unit morphology). Motor unit morphology assessment is generally a multifactorial qualitative process and less binary than the presence or absence of fibrillations.

TABLE 9.12 Characteristics of Motor Units (Videos 9.10, 9.20, and 9.25–9.30)

Characteristic	Details
Appearance	Variable can be triphasic or polyphasic
Rhythm	Semirhythmic or variable regularity. Not randomly irregular
Frequency	5–50 Hz
Amplitude	Variable, 20 μV–20 mV depending on needle location and motor unit characteristics
Stability	Stable if normal, occasional alterations in shape in some diseases
Observed in	Normal and abnormal patients if they have motor units

However, such motor unit changes—when clear—can be very helpful, particularly in the proper clinical context (Table 9.12; also see eAppendix 9B). MUAPs that are large in peak-to-peak amplitude (>8 mV), particularly when they are the first- or second-order units recruited with a low force, indicate that reinnervation and reconsolidation of the motor unit have occurred, and each motor axon innervates more muscle fibers than normal; these fibers are more densely packed in the area of the recording needle. This indicates a neuropathic process in the past with substantial motor axonal loss, such as a severe nerve injury that has resolved. It may also indicate a slow chronic process (e.g., axonal polyneuropathy) in which motor axons are lost and the remaining axons reinnervate the denervated muscle fibers. Polyphasic potentials, particularly when long in duration, are indicative of a relatively recent motor axonal trauma and that a robust reinnervation process is underway. Small MUAPs on needle EMG can indicate a myopathy process, particularly when they are short in duration and show early recruitment.

An increase in muscle contraction results in an increase in force production through (1) an increase in motor unit firing rate (temporal recruitment) and (2) the orderly addition of more motor units (spatial recruitment or activation). The "rule of five" is a way of assessing this recruitment. To assess recruitment, the patient is asked to gradually increase the strength of the muscular contraction. The number of motor units and firing rate of these motor units both increase corresponding with the force of the contraction. The firing rate of the fastest firing motor unit is identified by freezing the screen and determining the number of milliseconds between successive depolarizations of the same motor unit. To determine the motor unit firing rate (Hz), 1000 is divided by the interval between the successive appearance of the same motor unit on the screen in milliseconds. For example, if the motor unit is 50 msec between events, then the calculation is 1000/50, which is 20 Hz. If the motor unit is 110 msec between events, then the firing rate is approximately 9 Hz. Another way to quickly calculate the firing rate is to set the display screen so it represents 100 ms (e.g., 10 ms per division with 10 divisions). That way, if there is one unit on the screen, you simply multiply by 10 (representing 1 second), and the firing rate is 10 Hz. If there is one distinct unit shown twice on the screen, it is firing at 20 Hz. Care must be taken using this technique because if there is only one unit on every other screen, the firing rate may actually be much slower (5 Hz). If there are two units on the screen, one at each edge of the screen, the firing rate may actually be closer to 12 to 13 Hz, which can confuse abnormal from normal results. The "rule of five" guideline suggests that, in normal situations, the first recruited motor unit will start at 5 Hz and will gradually increase to about 10 Hz. At 10 Hz, the second motor unit starts firing at 5 Hz. Thus you have two motor units, one firing at 10 Hz and the second firing at 5 Hz. As the contractile force increases, both of those motor units increase their firing rate proportionally so that one is firing at 15 Hz, the next is firing at 10 Hz, and a third newly recruited at 5 Hz. The recruitment ratio is calculated by identifying the firing rate of the fastest firing motor unit and dividing it by the number of different motor units on the screen. The number calculated should be near 5. For example, if the firing rate of the fastest firing motor unit is 20 Hz, approximately four distinct motor units should be seen.[106] In this example, the fastest firing motor unit is 20 Hz; if there are only two different motor units on the screen, then the recruitment ratio would be $20/2 = 10$. This would be abnormal because it indicates a dropout of motor units, which is reduced or neuropathic recruitment. In practice, this becomes difficult because counting the exact number of motor units can be problematic when there are distant and/or small motor units (see Video 9.29).

Reduced recruitment with a high firing ratio represents one end of a spectrum of recruitment findings, which is strongly suggestive of functional dropout resulting from conduction block (neurapraxia) or motor axonal loss (axonotmesis). At the other end of a spectrum of recruitment findings is a situation in which many units are rapid or recruited to generate a low level of force. This is termed *early recruitment* and is seen in myopathies and neuromuscular junction disorders. Such recruitment is often seen with small, short-duration MUAPs that are characteristic of myopathies (see Video 9.30). At these two extremes of recruitment, findings are relatively specific for the underlying types of neuromuscular pathology. Beyond this, many recruitment findings are less clear and nearly impossible to reliably differentiate from normal. In the case of pain, poor cooperation, strong muscles or two joint muscles such as the gastrocnemius, or upper motor neuron disorders, there can be slow firing rates with few units activated. This is frequently called "poor or reduced activation."[44]

Extent of Electrodiagnostic Testing

Guidelines from the AANEM, published in 1999,[5] provide valuable guidance regarding the extent of appropriate EDX testing in 90% of cases (see eAppendix 9C for instrumentation). The guidelines and recommendations are summarized in Table 9.13.[5] Guidelines in EDX medicine provide recommendations for testing and give clinicians and insurers crucial information regarding how many tests should be conducted in most cases to expeditiously arrive at a diagnosis (e.g., what is enough and what is excessive). These guidelines were developed to improve EDX patient care and to combat unscrupulous providers of EDX services in the United States who perform excessive studies beyond what is necessary to make a diagnosis simply for the purpose of increasing the EDX fee. It is clearly understood that these are guidelines and may well vary depending on the clinical question asked by the referring physician, the suspicions and insights of the EMC, and the findings as testing unfolds. An unsuspected set of findings may well prompt a far more extensive set of testing than would be expected for the initial referring question.

Electromyography and Pathology

EDX testing may assist in the decision to perform either a muscle or nerve biopsy. There are limited studies correlating EMG/NCS and muscle or nerve biopsies. Needle EMG findings and pathologic correlation have been poorly studied and not universally agreed upon. Separating a neuropathic disease from a myopathic neuromuscular disease can be helpful.

Muscle

A triad of elevated creatinine phosphokinase (CPK), proximal weakness, and positive EMG correlate to an abnormal muscle biopsy much better than elevated CPK alone.[167] Long-duration motor unit potentials correlate with fiber-type grouping but not fiber-type predominance.[135] Fibrillations and short-duration motor units predict inflammatory myopathies and muscular dystrophies, whereas absence of fibrillations suggests other myopathic etiologies, such as congenital myopathy.[166]

Nerve

Unfortunately, the Seddon and Sunderland classifications do not cover the most common pathologic issue with nerves: demyelination. The demyelination seen in many compression neuropathies and myelin abnormalities seen in acquired and hereditary neuropathies is not analogous to a focal entrapment neuropathy with either neurapraxia, axonotmesis, or neurotmesis. Neurapraxia and conduction block almost always exist together. These are usually caused by demyelination. However, nerve ischemia, local anesthetics, and local metabolic derangement can also cause neurapraxia and thus conduction block not associated with demyelination. Probably the only time that conduction block is not from neurapraxia is when there is an acute (0–10 days) axonal lesion between the two points of stimulation for NCS. This will turn into a classic axon loss lesion after Wallerian degeneration is complete. Slowed velocities, distal latencies, temporal dispersion, and late responses are correlated with demyelination or dysfunctional myelin. Small or absent responses—fibrillations—are mostly correlated with axonal loss. Nerve biopsy can be helpful to identify inflammatory, infectious, or toxic causes for the neuropathy, many of which may be treatable. The indications for nerve and muscle biopsies in the age of molecular genetics and antibody identification are constantly evolving and are well beyond the scope of this chapter.

Limitations of Electrodiagnosis

Unfortunately, very few EDX findings are clearly specific for any single diagnostic entity. Repetitive nerve stimulation at 2 or 3 Hz can reveal decrements not only in neuromuscular junction disorders but also in motor neuron disease, myopathies, peripheral neuropathies, and myotonic disorders.[101] Fibrillations are seen in any disease that causes motor axon loss, including polyneuropathies, motor neuron disease, inflammatory myopathies, radiculopathies, and entrapment neuropathies. Marked facilitation of the CMAP to more than 400% of the baseline amplitude after a brief contraction in persons with Lambert-Eaton myasthenic syndrome (LEMS) is one finding that is unique to this uncommon disease.[157]

The time course over which a disease process progresses and the time at which EDX testing is conducted both play major roles in determining whether the EDX testing can provide a reasonably certain diagnosis. It is frequently necessary to repeat the study if the diagnosis remains in question or if the clinical situation

TABLE 9.13 **Recommendations Representing a Generally Accepted Study for Addressing a Suspected Condition**

Suspected Condition	Sensory Testing	Motor Testing	Electromyography	Proximal Conductions	Other Special Tests
Radiculopathy	One sensory NCS to exclude polyneuropathy	One motor NCS to exclude polyneuropathy	Sufficient number of muscles representing all myotomes with paraspinal muscles	None	None
Carpal tunnel syndrome	Median sensory NCS and another for comparison	Median motor NCS and another for comparison	Optional, although useful to determine severity of CTS and exclude concomitant C8 radiculopathy	None	None
Ulnar neuropathy	Ulnar sensory NCS and another for comparison	Ulnar motor NCS and another for comparison; need to examine conduction velocity across elbow	Several ulnar-innervated muscles, always the first dorsal interosseus, as it is most frequently abnormal, plus median muscles for comparison to exclude C8 radiculopathy	None	Inching motor NCS across the elbow
Fibular (peroneal) neuropathy	Superficial fibular (peroneal) sensory NCS and another for comparison	Fibular (peroneal) motor with stimulation below and above the fibular head, assessing for conduction velocity and amplitude changes; test another for comparison	Fibular (peroneal) nerve–innervated muscles and tibial nerve muscles for comparison; suggest paraspinal and L4–L5 muscles to assess for radiculopathy	None	None
Entrapment neuropathies	Two sensory NCS, one in the distribution and a comparison	Two motor NCS, one in the distribution and a comparison	Muscles in the involved distribution and comparison muscles	None	None
Myopathy	One or two sensory NCS in a clinically involved limb	One or two motor NCS in a clinically involved limb	Two muscles (proximal and distal) in two limbs, one of which is symptomatic	—	Consider repetitive nerve stimulation
Neuromuscular junction disorders	One sensory NCS in a clinically involved limb	One motor NCS in a clinically involved limb	One proximal and one distal muscle in clinically involved limbs	—	Repetitive nerve stimulation at 2–3 Hz in clinically weak muscle and, if normal, other weak muscles; single-fiber electromyography if high suspicion and repetitive nerve stimulation is negative—can be first test in ocular myasthenia gravis
Polyneuropathy	Sensory NCS in at least two extremities: if abnormalities in one limb, study the contralateral limb; four or more may be necessary to classify the polyneuropathy	Motor NCS in at least two extremities: if abnormalities in one limb, study the contralateral limb; four or more may be necessary to classify the polyneuropathy	One distal muscle in both legs and a distal muscle in one arm	Consider proximal NCS (H-reflexes, F-waves, and blink reflexes) to assess demyelination	—
Motor neuron diseases	One sensory NCS in at least two clinically involved limbs	One motor NCS in at least two clinically involved limbs; proximal stimulation sites to exclude conduction block in the case of multifocal motor neuropathy	Several muscles in three extremities, *or* two extremities and cranial nerve–innervated muscles as well as lumbar or cervical paraspinal muscles; thoracic paraspinal muscles may be considered an extremity; sample distal and proximal muscles. Assess for fasciculations	Consider F-waves	Consider repetitive nerve stimulation

CTS, Carpal tunnel syndrome; *NCS,* nerve conduction study.
Modified from the American Association of Electrodiagnostic Medicine. Guidelines in electrodiagnostic medicine. *Muscle Nerve.* 1999;8:S1-S300.

changes.[90] One important issue related to EDX medicine is the possibility of false-positive results. The probabilities of finding false-positive results on the basis of chance alone increase as the number of independent measures also increases.[59] It is important to realize that if five measurements are performed, there is a 12% chance of having one false-positive result. If nine measurements are made, there is a 20% chance of a spuriously false-positive result. However, if there are two abnormal results when six tests are conducted, the likelihood that they are false-positive findings is very low, less than approximately 1%. This underscores the need for EDCs to critically examine their findings and not overdiagnose a disorder based on one subtle abnormality. If one EDX parameter is markedly abnormal, far beyond the upper limit of normal, this can be compelling, particularly when that abnormality is consistent with the clinical impression. The study should be repeated to make certain of its validity. However, two abnormalities indicating the same diagnosis are far more likely to represent true findings and a clear underlying disorder.

Other common conditions, such as diabetes and its corresponding polyneuropathy, can make EDX testing more difficult to interpret. With a diffuse polyneuropathy, the consultant is often unable to identify focal lesions against the backdrop of this disorder. In this scenario, the EMC must assess the magnitude of the background polyneuropathy and make a judgment as to whether findings for the nerve in question are out of proportion to the background polyneuropathy. Caution is urged in this scenario to avoid overcalling entrapment neuropathies when a generalized polyneuropathy is present. These issues are another reason why it is critically important that the EMC have sufficient training to interpret the findings appropriately in the clinical context of the patient.

Anatomic variations in muscle and nerve pathways can also confound EDX interpretation. The Martin-Gruber anastomosis (MGA) is when motor nerve fibers running with the median nerve at the elbow cross over to the ulnar nerve in the forearm (often from the anterior interosseous nerve [AIN]) and enter the hand through Guyon canal with the ulnar nerve.[152,170]

The MGA fibers (C8–T1) that run with the median nerve at the elbow commonly result in dual innervation to the first dorsal interosseus, abductor digiti minimi, and other hand muscles.[170] Thus a median nerve injury at the elbow will cause fibrillation potentials in ulnar innervated muscles in the hand. In this situation, the median CMAP with wrist stimulation is smaller than that with proximal stimulation. The ulnar nerve demonstrates a larger amplitude with wrist stimulation than with below-elbow or above-elbow stimulation. If a suspected conduction block is found in the forearm when testing the ulnar nerve, then median nerve conduction should be performed to exclude MGA as the cause of CMAP reduction with proximal stimulation.

The most overlooked and clinically relevant, in our opinion, is the ulnar to median anastomoses in the hand that may result in the appearance of the so-called all ulnar hand. Ulnar anastomoses include the most commonly seen Riche-Cannieu anastomosis (RCA) 55.5% (motor fibers) and the Berrettini anastomosis 60.9% (sensory fibers).[152,164,165] The "all ulnar hand" is a misnomer because the RCA has not been clearly reported to innervate 100% of the thenar muscles, and the RCA only shares (does not replace) innervation with the median nerve. Imprecise EDX evaluation in patients with RCA can result in (1) incorrectly identifying a proximal lesion (C8–T1, lower trunk, or medial cord) when the lesion is only at the ulnar nerve, (2) missing a median nerve lesion attributing fibrillation potential or MUAPs in the abductor pollicis brevis to RCA, (3) missing a complete median neuropathy at the wrist (see earlier), or (4) erroneously concluding that both ulnar and median nerves have lesions or that it is a polyneuropathy (in the setting of RCA and only ulnar lesion). Distal radial nerve needle EMG (extensor indicis) may help avoid this potential diagnostic error.

An accessory deep fibular (peroneal) nerve is an anomalous part of the deep fibular (peroneal) nerve that remains with the superficial fibular (peroneal) nerve and innervates the extensor digitorum brevis after passing behind the lateral malleolus. It should be suspected if stimulation at the fibular head yields a larger CMAP than ankle stimulation when recorded over the extensor digitorum brevis. Stimulation of the accessory deep fibular (peroneal) nerve behind the lateral malleolus will confirm the presence of such an anatomic variant.

Standards of Practice for Electrodiagnostic Medicine

The AANEM has addressed the need to set standards for the evaluation and interpretation of studies in response to the burgeoning and disquieting practice of decoupling nerve conductions and needle EMG. It is the contention of the organization that EDX studies should be performed by physicians properly trained in EDX medicine, that interpretation of NCS data alone (absent face-to-face patient interaction and control over the process) provides substandard care, and that performance of NCS without needle EMG has the potential of compromising patient care.[9] Considerable concern exists that there has been a dramatic increase in NCS without EMG testing.[9] EMG testing is crucial for the assessment of radiculopathies and plexopathies as well as to complement NCS and give additional information about entrapments and nerve injuries. Interpreting NCS findings without the benefit of a focused history and examination is also inappropriate.[9] Communication of the consultation results to the referring physician and other health care providers is essential. The AANEM recommends that the EDX report contain (1) a description of the problem and the reason for referral, (2) a focused history and examination, (3) tabular NCS and EMG information with normative reference values, and (4) temperature and limbs studied.[8] It is recommended that the study results are defined as normal or abnormal, and a probable physiologic diagnosis is presented. Any limitations to the study should be included.[8] An optional section after this summary can be a synthesis of the clinical and electrophysiologic information into clinical recommendations or a stronger set of conclusions. Such a section should be separate from the electrophysiologic testing. The EMG and NCS data (and the conclusions they support) should stand alone. Visual depictions of the waveforms, although not required, are an ideal means of documenting the nerve conduction findings and are usually available on modern EDX instruments.

Pediatric Electrodiagnosis

The pediatric examination poses challenges for even the most experienced EDX physician. Diagnosis of pediatric neuromuscular disorders is a specialized and rapidly evolving area of medicine in which genetic testing plays an ever-greater role in diagnosis. Interested readers are encouraged to examine specialized textbooks and recent articles because much of this material is beyond the scope of this chapter. In terms of equipment required for pediatric electrodiagnosis, a pediatric bipolar stimulating probe is available and is useful for small children.[186] Monopolar needles are generally used because their diameter is smaller and insertional resistance is less, which is attributable to their Teflon coating. It is critically important to maintain a skin surface temperature of 36°C to 37°C to avoid spurious results.[186] There are strong advocates for and against the use of sedation and analgesia during EDX examination.[186] Performance of all conscious sedation and anesthesia should be done by specialists in pediatric anesthesia. When general anesthetic is required, children with possible neuromuscular diseases should not be given halogenated inhalation agents, such as halothane, because of the risks for malignant hyperthermia.[177] Nerve conduction testing should include both sensory and motor studies. The needle examination, although difficult, should always be part of the diagnostic evaluation. Despite the challenging nature of the needle examination, a complete needle EMG is necessary to reach appropriate conclusions. A single muscle exploration is insufficient for diagnosis in a child.[186]

The conduction velocities in newborns are approximately half of those found in adults. There is a rapid increase in values during the first year of life. Median values for conduction velocities in children and adults are equalized by 5 years of age.[186] Tables 9.14 and 9.15 show values for infants and newborns.[138] Motor conduction velocities in full-term infants should be no less than 20 m/s.[186] Other investigators in China found that the motor conduction velocities, sensory conduction velocities, F-wave velocities, and H-reflex velocities in newborns were approximately 50%, 40%, 40% to 45%, and 40% to 45% of adult values, respectively. At age 3 years, normal values were in the adult range for all motor conduction velocities and for the sensory conduction velocities in the upper limbs. The sensory conduction velocities, H-reflex velocities, and F-wave velocities in the lower limbs did not reach adult values until age 6 years, whereas the upper limb F-wave velocities reached adult levels between 6 and 14 years in different nerves. These results indicate that adult values for nerve conduction velocities, including the late responses, are reached earlier in the lower extremities; however, conduction velocities at any age are always faster in the upper limbs and in the proximal compared with the distal segments. The maturation process occurs most rapidly during the first 3 to 6 years of life, especially in the first year.[26] Late responses evaluate an infant's proximal peripheral nervous system. They offer several advantages, including reduced problems with temperature control, longer conduction distance, reduced measurement error, and a single stimulation site. In contrast to adults, the H-reflex can be elicited from any muscle during infancy, but most responses are gradually and variably suppressed by 1 year of age.[186] However, the tibial nerve retains its electrophysiologic H-reflex throughout life.

The needle EMG examination in infants is challenging and should begin with the muscles of highest diagnostic yield, such as those in a weak limb. Needle manipulation and repositioning are often necessary because of the greater relative amount of adipose tissue and reduced muscle activation. Muscle relaxation to examine for spontaneous or normal insertional activities is best obtained by placing the muscle at its shortest length from origin to insertion.[186] The infant's foot and hand intrinsic muscles can best be used to study insertional and spontaneous activity because they are usually not very active.[186] However, these distal muscles exhibit high levels of end-plate noise because of the relatively larger end-plate regions, which can be confused with fibrillation potentials.[186] Extensor muscles are not activated as often by the neonate and therefore are good choices as well for evaluation of spontaneous and insertional activity.[186]

Assessment of motor unit morphology and recruitment is difficult. Motor unit potentials in the pediatric population differ from those in the adult population. Motor unit potentials in the newborn are usually biphasic or triphasic; their amplitudes range from 100 to 700 μV, and they are shorter in duration at 5 to 9 ms.[103,186] Recruitment of motor units is often difficult to interpret because there is typically no voluntary control. When looking for motor unit morphology and recruitment, it is advisable to examine flexor muscles because there is usually strong flexor tone and muscle activation. It is important to remember that this is a difficult examination, and overinterpretation is common for the inexperienced practitioner.

Repetitive nerve stimulation is used to evaluate presynaptic and postsynaptic neuromuscular disorders. Immobilization of the arm or leg and maintenance of warm body and limb temperatures are crucial for optimizing sensitivity for this testing. Use of 20-Hz stimulation to assess for postactivation decrement repair or to look for postexercise facilitation is occasionally necessary. This is painful, and an anesthetized, immobile infant is optimal to derive a meaningful study. Stimulated single-fiber EMG can be performed and is an ideal, sensitive means of testing for neuromuscular junction disorders.[34,42]

The hypotonic infant poses diagnostic challenges, with hypotonia being the most common reason infants are referred for EDX examination. Within this category, the most common cause for generalized hypotonia is a central nervous system disorder.[186] In fact, only 10% to 20% of infants and children with hypotonia have a neuromuscular etiology.[186] The most common neuromuscular diagnosis in hypotonic infants is spinal muscular atrophy.

Several peripheral nerve entities occur in the newborn. Partial facial paralysis or asymmetric faces might occur, and motor NCS and needle EMG of the facial nerve and its corresponding innervated muscles are useful to clarify the mononeuropathy. Side-to-side comparisons of the CMAPs can give an estimate of the degree of axonal loss and some prognostic information. Brachial plexopathies at birth are uncommon, occurring in 1.5 per 1000 live births.[57] These brachial plexus palsies are difficult to predict and are usually unrelated to identifiable obstetric characteristics, such as gestational age, oxytocin augmentation, delivery mode (forceps, spontaneous, breech, or cesarean), or birth weight.[57] However, the duration of labor and presence of shoulder dystocia were significantly related to the occurrence of brachial plexus palsy.[57] Erbo palsy involves the upper plexus and C5 or C6 roots. A Klumpke or lower plexus palsy primarily demonstrates C8 or T1 and lower trunk involvement. The needle EMG can help confirm the site of pathology and location of primary involvement.[186] In one series, supracostoclavicular space narrowing caused by anatomic variations, such as cervical ribs or fibrous bands, was identified as a predisposing factor that rendered the brachial plexus more prone to injury.[14] For situations in which a neurologic deficit is found at the time of delivery, an EDX study shortly afterward (within 24 hours) can help elucidate whether such a deficit developed intrauterine or during delivery. Intrauterine onset of a focal neuropathy or plexopathy may demonstrate fibrillation potentials in the weak muscles.[99]

A particularly interesting case report discussed the intrauterine onset of a fibular (peroneal) neuropathy with fibrillations found within 20 hours of birth in the affected muscles, underscoring the onset before delivery and arguing against obstetric trauma as the cause.[99] In this case report, the authors discuss other cases in which onset predated the delivery, as evidenced by fibrillations in clinically affected muscles shortly after birth. These findings have

TABLE 9.14 **Motor Conduction Studies in 155 Children[a]**

	MEDIAN NERVE[b]				FIBULAR (PERONEAL) NERVE[c]			
Age (*n*)	Distal Motor Latency (ms)	Conduction Velocity (m/s)	F-Latency (ms)	Amplitude (mV)	Distal Motor Latency (ms)	Conduction Velocity (m/s)	F-Latency (ms)	Amplitude (mV)
7 days–1 month (20)	2.23 (0.29)	25.43 (3.84)	16.12 (1.50)	3.00 (0.31)	2.43 (0.48)	22.43 (1.22)	22.07 (1.46)	3.06 (1.26)
1–6 months (23)	2.21 (0.34)	34.35 (6.61)	16.89 (1.65)	7.37 (3.24)	2.25 (0.48)	35.18 (3.96)	23.11 (1.89)	5.23 (2.37)
6–12 months (25)	2.13 (0.19)	43.57 (4.78)	17.31 (1.77)	7.67 (4.45)	2.31 (0.62)	43.55 (3.77)	25.86 (1.35)	5.41 (2.01)
1–2 years (24)	2.04 (0.18)	48.23 (4.58)	17.44 (1.29)	8.90 (3.61)	2.29 (0.43)	51.42 (3.02)	25.98 (1.95)	5.80 (2.48)
2–4 years (22)	2.18 (0.43)	53.59 (5.29)	17.91 (1.11)	9.55 (4.34)	2.62 (0.75)	55.73 (4.45)	29.52 (2.15)	6.10 (2.99)
4–6 years (20)	2.27 (0.45)	56.26 (4.61)	19.44 (1.51)	10.37 (3.66)	3.01 (0.43)	56.14 (4.96)	29.98 (2.68)	7.10 (4.76)
6–14 years (21)	2.73 (0.44)	57.32 (3.35)	23.23 (2.57)	12.37 (4.79)	3.25 (0.51)	57.05 (4.54)	34.27 (4.29)	8.15 (4.19)

[a]Mean (standard deviation).
[b]Median motor recorded from abductor pollicis brevis and stimulated at wrist.
[c]Fibular (peroneal) motor recorded over extensor digitorum brevis and stimulated at ankle.
Modified from Parano E, Uncini A, DeVivo DC, et al. Electrophysiologic correlates of peripheral nervous system maturation in infancy and childhood. *J Child Neurol.* 1993;8:336-338.

TABLE 9.15 **Sensory Conduction Studies in 155 Children**[a]

	MEDIAN NERVE[b]		SURAL NERVE	
Age (*n*)	Conduction Velocity (m/s)	Amplitude (μV)	Conduction Velocity (m/s)	Amplitude (μV)
7 days–1 month (20)	22.31 (2.16)	6.22 (1.30)	20.26 (1.55)	9.12 (3.02)
1–6 months (23)	35.52 (6.59)	15.86 (5.18)	34.63 (5.43)	11.66 (3.57)
6–12 months (25)	40.31 (5.23)	16.00 (5.18)	38.18 (5.00)	15.10 (8.22)
1–2 years (24)	46.93 (5.03)	24.00 (7.36)	49.73 (5.53)	15.41 (9.98)
2–4 years (22)	49.51 (3.34)	24.28 (5.49)	52.63 (2.96)	23.27 (6.84)
4–6 years (20)	51.71 (5.16)	25.12 (5.22)	53.83 (4.34)	22.66 (5.42)
6–14 years (21)	53.84 (3.26)	26.72 (9.43)	53.85 (4.19)	26.75 (6.59)

[a]Mean (standard deviation).
[b]Median sensory recorded at wrist with stimulation of index fingers.
Modified from Parano E, Uncini A, DeVivo DC, et al. Electrophysiologic correlates of peripheral nervous system maturation in infancy and childhood. *J Child Neurol.* 1993;8:336-338.

implications regarding medicolegal issues surrounding brachial plexus palsies and other mononeuropathies in newborns.

Mononeuropathies and Entrapment Neuropathies

Suspected mononeuropathies are the most common reasons for EDX laboratory referral. Top among these conditions is median neuropathy at the wrist or carpal tunnel syndrome. Other entrapments include ulnar neuropathy at the elbow, common fibular (peroneal) neuropathy at the fibular head, and tibial nerve entrapment at the ankle (tarsal tunnel syndrome). In addition to these entrapments, focal nerve injuries can occur with penetrating trauma and fractures. The common entrapments and nerve injuries, their anatomic causes, and their EDX correlates are given in Table 9.16. The approach to assessing for mononeuropathies is to fully test the nerve in question and another nerve in the affected limb for comparison. Stimulation both below and above the suspected site of injury can help assess for conduction block or conduction slowing. EMG of selected muscles innervated by the particular nerve is required. If these muscles are abnormal, the study should be expanded to ensure that these EMG findings are confined to a single peripheral nerve distribution and that they are not attributable to a radiculopathy, plexopathy, or polyneuropathy. The EMC should have a low threshold for testing the same nerves in the opposite limb. The examiner must sort out a focal process versus a generalized process such as polyneuropathy.

Plexopathies

Brachial plexopathies are important causes of shoulder and neck pain in adults.[40] The EMG examination is crucial and should be extensive, with emphasis on proximal muscles with clinical weakness.[140] SNAPs are a key part of the evaluation. SNAPs are readily available for C6 (radial to the thumb), C7 (median to digit 3), and C8 (ulnar to digit 5) root levels in the hand, and L5 (superficial fibular [peroneal] sensory) and S1 (sural sensory) in the leg. Other SNAPs, such as the lateral cutaneous nerve of the forearm (lateral antebrachial cutaneous) (C5, C6) or the medial cutaneous nerve of the forearm (medial antebrachial cutaneous) (C8, T1), can provide additional information to the study. SNAPs are generally low in amplitude or absent with postganglionic plexopathies and help differentiate between radiculopathies and plexopathies. The exception to this rule is in the case of traumatic root avulsion. This is a severe injury with little chance of recovery that should be recognized. In this case, the patient's weakness is profound with associated numbness, the CMAP amplitudes are low or absent, but the SNAPs are preserved.

CMAP amplitudes in weak muscles should be documented when possible and compared with the uninvolved side. After Wallerian degeneration has occurred, this gives a rough quantitative estimate of axonal loss. It is thought that CMAP amplitudes less than 10% to 20% of the uninvolved side portend poor prognosis; however, the literature supporting this statement is weak and in large measure derived from Bell palsy experience.[49] Idiopathic brachial neuritis, neuralgic amyotrophy, and Parsonage-Turner syndrome are special types of nerve injuries that are frequently termed a plexopathy or radiculoplexopathy and may involve cervical roots and cranial nerves. All of these can clinically mimic cervical radiculopathy and should be included in differential diagnosis of persons with limb symptoms, particularly with symptoms in the shoulder girdle region. However, the distribution may not be a radiculoplexopathy and is more accurately localized to multiple mononeuropathies and occasionally only one nerve. The suprascapular, long thoracic, anterior interosseous, and axillary nerves are the most commonly involved, in that order of frequency.[73] Because of the common C5 to C6 innervated muscle patterns, neuralgic amyotrophy can mimic upper trunk plexopathy or radiculopathy. In this retrospective investigation, all patients demonstrated the two clinical features of neuralgic amyotrophy: (1) shoulder region pain, described as sudden in onset, severe in degree, and transient (typically lasting 1–2 weeks), and (2) the presence of muscle weakness and wasting in the arm.[73] The finding of hourglass deformities in some symptomatic nerves has raised the possibility of multiple etiologies for this clinical presentation. Inflammation has been the most cited cause, but there may be subgroups of nerve constriction and/or torsion that

TABLE 9.16 **Entrapment Neuropathies**

Nerve	Causes	Clinical Features	Suggested Electrodiagnostic Evaluation	Potential EMG Findings
Median Neuropathies				
Carpal tunnel syndrome: Median entrapment at wrist	The most common entrapment neuropathy Incidence: 55–125 cases/100,000 Causes: Repetitive trauma, obesity, pregnancy, lupus, etc.[51]	Aching pain in forearm and wrist with insidious onset, tingling, paresthesias of thumb, index, and long finger, thenar weakness, nocturnal pain	Sensory NCS across the wrist and another sensory nerve in the affected limb to compare Motor NCS of median nerve and another motor nerve (usually ulnar) Needle EMG is optional[5]	Mild CTS: Prolonged SNAP and/or slightly reduced SNAP amplitude Moderate CTS: Abnormal median SNAP as above, plus prolonged median motor latency[175] Severe CTS: Prolonged motor and sensory distal latencies; evidence of axon loss (small or absent SNAPs, CMAP, fibrillations, polyphasic or large motor unit potential)
There are a group of median neuropathies at or around the elbow. Median nerve entrapment can occur at the ligament of Struthers, bicipital aponeurosis, heads of pronator teres, or FDS arch in the forearm.	Trauma from repetitive overuse, tight casting, penetrating injuries such as intravenous catheter, carrying a bag with the arm flexed ("grocery bag neuropathy")[51]	Aching pain in the volar forearm, hand numbness, symptoms worsened by repeated pronation, clumsiness, loss of dexterity, weakness, flexor muscle and thenar muscle wasting if severe, tender pronator muscle	Motor nerve conduction across this area Median sensory study distally EMG of median muscles Evaluation of ulnar nerve for comparison Radiculopathy screen to exclude this condition EMG findings should be confined to median distribution	SNAP: Normal unless axonal loss CMAP: Normal or reduced if motor axonal loss present; NCV slowed across the pronator area or conduction block across this site Pronator teres spared with distal (FDS and pronator heads) compression
Anterior interosseus: Entrapment at forearm (Kiloh-Nevin syndrome)	Direct trauma, inflammation, strenuous exercise, fractures, variant of brachial neuritis, compression by anomalous fibrous bands in this region[51]	Acute pure motor syndrome with pain in forearm and elbow, weak flexor pollicis longus and flexor digitorum profundus of index and long finger, weak pronator quadratus; "OK" sign No sensory complaints	Motor nerve conduction across this area Median sensory study distally EMG of median muscles Evaluation of ulnar nerve for comparison Radiculopathy screen to exclude this condition EMG findings should be confined to anterior interosseous nerve distribution	Median SNAPs are normal Normal median nerve conduction studies to the APB With surface or needle recordings from the pronator quadratus, there can be slowed median NCV On EMG, findings will be limited to FPL, PQ, and FDP (index and long)
Median nerve entrapment beneath the ligament of Struthers	Caused by trauma or inflammation Supracondylar process seen in 0.7%–2.7% of population[37] Associated with fibrous band from supracondylar process to medial epicondyle (Supracondylar process 5 cm above medial epicondyle on radiograph is noted.)	Weak grip, weakness in flexion of wrist, deep aching pain in forearm, numbness of 1–3 digits, weak hand pronation, second- and third-digit flexion, wasting of APB	Motor nerve conduction across this area Median sensory study distally EMG of median muscles Evaluation of ulnar nerve for comparison Radiculopathy screen to exclude this condition EMG findings should be confined to median distribution Pronator teres often involved	Decreased or absent SNAP amplitude Reduced motor NCV or conduction block across this segment Decreased CMAP from APB EMG findings in median nerve distribution, including pronator teres[51]
Radial Neuropathies				
Radial nerve entrapment at axilla	Improper use of crutches, falling asleep with arm over a sharp chair back or edge	Weakness in all radial nerve–innervated muscles, including triceps, decreased sensation in posterior arm and forearm	Superficial radial sensory Radial motor testing with recording from EIP with either surface or needle electrodes. Should stimulate at Erb point Testing at least one additional motor and sensory nerve in the involved limb EMG of radial nerve–innervated muscles EMG screen for radiculopathy (e.g., other muscles not innervated by radial nerve)	Reduced or absent SNAP if axonal loss Normal or reduced motor CMAP if axonal loss occurred Conduction block across the axilla EMG findings in radial nerve–innervated muscles, including triceps when motor axonal loss is present

Radial nerve entrapment at spiral groove (Saturday night palsy or honeymooner palsy)	Acute prolonged compression of the nerve in the humeral region caused by a person sleeping on the arm or falling asleep in an improper position Frequently associated with alcohol use Other causes include an injection or compression with a tourniquet	Involves all radial nerve–innervated muscles except triceps brachii and anconeus Wrist extension weakness and finger extension weakness Sensory loss over dorsal hand and posterior forearm	Same as above	Reduced or absent SNAP if axonal loss; normal or reduced motor CMAP if axonal loss occurred Conduction block across the spiral groove region EMG findings in radial nerve–innervated muscles excluding triceps and anconeus when motor axonal loss is present
Radial nerve with posterior interosseous nerve compression	Radial tunnel syndrome encompasses entrapment by fibrous bands at the radial head or the sharp tendinous margin of the ECRB Posterior interosseus syndrome or supinator syndrome where the PIN is entrapped at the fibrous band at the origin of supinator muscle, called the arcade of Frohse[51] Other causes include dislocation of elbow, Monteggia fracture, fall on outstretched hand, surgical resection of radial head, mass lesions	Deep pain at elbow Weakness with wrist extension Weakness of MCP extension yet preserved ability to extend IP joints resulting from interossei No loss of sensation	Superficial radial sensory Radial motor testing with recording from EIP with either surface or needle electrodes. Consider stimulation at Erb's point Testing at least one additional motor and sensory nerve in the involved limb EMG of radial nerve–innervated muscles EMG screen for radiculopathy (e.g., other muscles not innervated by radial nerve)	Normal superficial radial sensory response Slowed motor conduction across the elbow; low CMAP over EIP EMG findings in PIN-innervated muscles; brachioradialis and ECRL usually spared
Radial nerve entrapment at the wrist (Wartenberg syndrome or handcuff neuropathy)	Bracelets and handcuff neuropathy Ganglions, overuse syndrome	Numbness, paresthesias of dorsoradial aspect of hand and dorsum of first three digits, tender to palpation in this area No motor weakness	Same as above In certain compelling clinical circumstances, the examination can be abbreviated and focused on the superficial radial nerve, with another nerve for comparison	Superficial radial sensory response delayed in latency or low in amplitude
Ulnar Neuropathies				
Ulnar nerve compression at Guyon canal at wrist	Chronic pressure from using tools, bicycle riding, ganglion, rheumatoid arthritis	Type 1: Hypothenar and deep ulnar branch Type 2: Deep ulnar branch Type 3: Superficial ulnar sensory branch Numbness, paresthesias, weakness, intrinsic muscle atrophy, nocturnal pain Painless wasting of hand intrinsics	Ulnar sensory testing Ulnar motor testing with recording electrodes over ADM and FDI and conduction across the elbow Dorsal ulnar cutaneous Another motor and sensory nerve in the affected limb for comparison EMG ulnar muscles ADM and FDI	Prolonged motor latency Dorsal ulnar cutaneous nerve spared Ulnar sensory response to fifth digit prolonged in latency or normal if mostly motor fibers are involved CMAP decreased if axonal loss present EMG demonstrates fibrillations or motor unit changes in hand intrinsics but not in thenar (median innervated) muscles unless Riche-Cannieu anastomosis (ulnar to median innervated muscles in the hand) is present

Continued

TABLE 9.16 Entrapment Neuropathies—cont'd

Nerve	Causes	Clinical Features	Suggested Electrodiagnostic Evaluation	Potential EMG Findings
Ulnar nerve compression at the elbow	Trauma, cubitus valgus, bony spurs, tumors, overuse, previous fracture Proximal to elbow the nerve may be entrapped by the arcade of Struthers, a fibrous structure associated with the medial triceps muscle Cubital tunnel made of the fibrous entrance to the FCU is the most common site	Vague dull aching forearm, intermittent paresthesia, hypoesthesia on ulnar side of hand, weakness of abduction of little finger, ulnar clawing in severe cases, wasting of first dorsal interosseus and hypothenar eminence, wasting of ulnar border of forearm	Ulnar motor conduction across the elbow to ADM and/or FDI and in forearm; if abnormal, additional motor and sensory to exclude diffuse process Consider an inching study across elbows[5] Ulnar sensory Dorsal ulnar cutaneous EMG of ulnar muscles Consider ulnar mixed nerve conduction compared with median from wrist to midhumerus[91]	The ulnar sensory nerve may be reduced in amplitude or absent The dorsal ulnar cutaneous nerve may be reduced or absent Ulnar motor amplitude might be reduced Conduction block across the elbow can occur Slow NCV across elbow (>15 m/s CV drop or NCS <43 m/s) Short segmental inching at 1-cm intervals can reveal >0.4-ms segmental change or a clear discontinuity[20]
Lower Limb Mononeuropathies				
Femoral neuropathy	Trauma, retroperitoneal hematoma caused by anticoagulation, cardiac catheterizations[5]	Weak quadriceps muscle, weakness of knee extension, absent knee jerk, groin pain, and decreased sensation over medial and anterior thigh and lower leg in the saphenous nerve distribution	Motor nerve conduction to quadriceps Saphenous sensory study EMG of quadriceps as well as other L3 and L4 muscles (e.g., the adductor muscles); consider an EMG screen for radiculopathy	Reduced amplitude or absent saphenous nerve response Reduced CMAP over rectus femoris On EMG, fibrillations in the femoral nerve–innervated muscles
Lateral femoral cutaneous nerve entrapment at thigh (meralgia paresthetica)	Repeated low-grade trauma, obesity, pregnancy, tight clothing most commonly under the lateral end of the ilioinguinal ligament	Pure sensory syndrome at the lateral thigh, including unpleasant paresthesias, burning, or a dull ache; no motor symptoms Can be aggravated by prolonged standing or walking	Lateral femoral cutaneous study Femoral evaluation as above should be considered	Reduced amplitude in lateral femoral cutaneous nerve This nerve is technically challenging to study, and side-to-side comparisons are useful
Fibular (peroneal) nerve entrapment at the proximal fibula (head/neck)	Fractures, plaster casts, tight stockings, improper positioning, excessive weight loss, farm work, tumors, crossing legs for a long time	Foot drop, weakness of eversion, numbness on the dorsum of the foot, and pain	SPS testing Motor nerve conduction below and across the fibular head recorded from EDB or tibialis anterior if EDB is atrophied Exclude L5 radiculopathy by EMG Test an additional motor and sensory in the same leg	Reduced or absent SPS response Conduction block across the fibular head Conduction velocity reduced in fibular head segment by >10 m/s compared with leg segment Fibrillations in muscles innervated by fibular (peroneal) nerve
Tibial nerve entrapment at tarsal tunnel under flexor retinaculum of medial malleolus	Compression from shoes, casting, posttraumatic fibrosis, overuse, ganglion cysts	Pain in foot and ankle, wasting and weakness in feet, sensory impairment at toes and sole of the foot	Plantar sensory (mixed) nerve conductions Tibial motor to AH muscle Another motor and sensory to exclude polyneuropathy EMG of intrinsic foot muscles Consider EMG screen for radiculopathy or sciatic neuropathy	Prolonged or absent latencies or low amplitudes in plantar nerves Prolonged, absent, or small distal responses across tarsal tunnel Fibrillations in the AH or other intrinsic foot muscle innervated by the tibial nerve Normal EDB EMG and sural sensory

ADM, Abductor digiti minimi; *AH,* abductor hallucis; *APB,* abductor pollicis brevis; *CMAP,* compound muscle action potential; *CTS,* carpal tunnel syndrome; *CV,* conduction velocity; *ECRB,* extensor carpi radialis brevis; *ECRL,* extensor carpi radialis longus; *EDB,* extensor digitorum brevis; *EIP,* extensor indicis proprius; *EMG,* electromyography; *FCU,* flexor carpi ulnaris; *FDI,* first dorsal interosseus; *FDP,* flexor digitorum profundus; *FDS,* flexor digitorum superficialis; *FPL,* flexor pollicis longus; *IP,* interphalangeal; *MCP,* metacarpophalangeal; *NCS,* nerve conduction study; *NCV,* nerve conduction velocity; *PIN,* posterior interosseous nerve; *PQ,* pronator quadratus; *SNAP,* sensory nerve action potential; *SPS,* superficial fibular (peroneal) sensory nerve.

complicate and contribute to this. EDX abnormalities are seen in the vast majority of patients with neuralgic amyotrophy.[80]

Electrodiagnosticians should understand the peripheral neuroanatomy of the lumbosacral and brachial plexuses. A thorough appreciation for muscular anatomy, root innervations, and needle localization is a prerequisite. The consultant should specify, with as much precision as the study will allow, the location of the lesion, the magnitude (severity), and (when possible) rough estimates of probable outcomes. There are frequent deviations from accepted anatomy. Perneczky et al.[143] described an anatomic study of 40 cadavers. In all cases, there were deviations from the accepted cervical root and brachial plexus anatomy. In a prospective study of 100 patients with lumbosacral radiculopathy who underwent lumbar laminectomy, EMG precisely identified the involved root level 84% of the time.[195] EMG failed to accurately identify the compressed root in 16%; however, a least half of the failures were attributable to anomalies of innervation.[195] Another component to this study involved stimulating the nerve roots intraoperatively with simultaneous recording of muscle activity in the lower limb using surface electrodes. These investigators demonstrated that the L5 nerve root innervates the soleus and medial gastrocnemius in 16% of cases.[195] It is important to also appreciate that different parts of the brachial plexus can be involved with differing severities.

Radiculopathies

Cervical, thoracic, and lumbosacral radiculopathies are conditions involving a pathologic process affecting the spinal nerve root. There are multiple causes for radiculopathies. One common cause is a herniated disc that anatomically compresses a nerve root within the spinal canal or foramen. Another common etiology for radiculopathy is spinal stenosis, resulting from a combination of degenerative spondylosis, ligament hypertrophy, and/or spondylolisthesis. Inflammatory radiculitis is another pathophysiologic process that can cause radiculopathy. However, it is important to remember that other more ominous processes, such as malignancy and infection, can manifest the same symptoms and signs of radiculopathy as those of the more common causes.

The dorsal root ganglion lies in the intervertebral foramen. This anatomic arrangement has implications for the clinical electrodiagnosis of radiculopathy—namely, that SNAPs are preserved in most radiculopathies because the nerve root is affected proximal to the dorsal root ganglion. A recent study examined the sensory findings in persons with documented herniated discs causing lumbosacral radiculopathy.[134] In this study of 108 patients with lumbosacral radiculopathy, sensory responses in the leg (sural, fibular [peroneal], or saphenous) were reduced in amplitude in only 7% of patients. None of these patients had an absent sensory response. This well-designed prospective trial supported the long-held notion that sensory responses are almost always preserved in lumbosacral radiculopathy. An abnormality in the NCS of a sensory nerve should prompt consideration of other causes of a patient's symptoms besides radiculopathy.[134]

In the lumbar spine, the dorsal and ventral lumbar roots exit the spinal cord at about the T11 to L1 bony level and travel in the lumbar canal as a group of nerve roots in the dural sac. This is termed the cauda equina, or "horse's tail." This poses challenges and limitations to the EMG examination. A destructive intramedullary (spinal cord) lesion at T11 can produce EMG findings in muscles innervated by any of the lumbosacral nerve roots and manifest the same findings on needle EMG as those seen with a herniated disc at any of the lumbar segmental levels. For this reason, in the presence of a spinal cord lesion injuring anterior horn cells at that level or below, the electromyographer cannot determine with certainty the anatomic location of the lumbar intraspinal lesion producing distal muscle EMG findings in the lower limbs.[46] The needle EMG examination can identify only the root or roots that are physiologically involved but not the precise anatomic site of pathology in the lumbar spinal canal. This is an important limitation requiring correlation with imaging findings to determine the most likely anatomic lesion location. A normal study can also help guide treatment, for example, in the case of musculoskeletal mimics of cervical[38] or lumbar[12] radiculopathy. It helps the clinician and patient that there is no significant nerve injury, and they can be assured that exercise and nonoperative care is a reasonable approach. Even a person with documented radiculopathy by EMG has an excellent prognosis for symptom resolution with active conservative therapy.

EDX testing is frequently used to identify and manage patients with radiculopathy. For the purposes of this discussion, radiculopathy is defined broadly as any pathology or dysfunction of the nerve root. The symptoms caused by radiculopathy are not necessarily associated with a positive or negative test of any kind, including EMG or MRI. The etiology of the radiculopathy may be compressive, inflammatory, metabolic, infectious, unknown, or other. Radiculitis is considered a subgroup of radiculopathy associated with inflammation of the nerve root. In electrodiagnostic testing, the most useful test for confirming the presence of a radiculopathy is needle EMG. Needle EMG testing with a sufficient number of muscles and at least one motor and one sensory NCS should be performed in the involved limb.[7] NCS tests are necessary to exclude polyneuropathy. An EMG study is considered diagnostic for radiculopathy if EMG abnormalities are found in two or more muscles innervated by the same nerve root and different peripheral nerves, yet muscles innervated by adjacent nerve roots are normal.[192] This assumes that other generalized conditions, such as polyneuropathy, are not present. The need for EMG, particularly in relation to imaging of the spine, has been highlighted.[151] Needle EMG is particularly helpful given that the false-positive rates for MRI of the lumbar spine are high, with 27% of normal individuals having a disc protrusion.[96] For the cervical spine, the false-positive rate for MRI is lower, with 19% of individuals demonstrating an abnormality, but only 10% showing a herniated or bulging disc.[17] Radiculopathies can occur without structural findings on MRI and likewise without EMG findings. The EMG evaluates only motor axonal loss or motor axon conduction block (reduced recruitment is seen), and for these reasons a radiculopathy affecting the sensory root will not yield abnormalities by EMG. If the rate of denervation is balanced by reinnervation in the muscle, then spontaneous activity is less likely to be seen with EMG. The reported sensitivity for EMG ranges from 49% to 92%.[185] The sensitivity of EMG for cervical and lumbosacral radiculopathies has been examined in multiple studies. Reported sensitivity for lumbosacral radiculopathy is modest, ranging from 49% to 86%.[88,104,110,114,120,136,163,184,188] For cervical radiculopathies, the sensitivities are similar, from 50% to 71%.[16,88,93,141,171,178,195] These sensitivities vary because the diagnostic standard for radiculopathy varied in each of these studies. The science of sensitivity and specificity has major limitations because the concept of gold standard is missing in most diagnostic tests. Therefore the different reference standards used for each study are variable and imperfect. If the reference standard for radiculopathy is a pathologic process of the root that is associated with motor axon loss, then the needle EMG should have a higher sensitivity and specificity. However, a reference standard of radiculopathy based on symptoms and many patients not having any motor axon loss would, by definition, lead to a lower sensitivity.[79] Some clinicians use the presence or absence of axon loss to assist in guidance in management.

Determining the prevalence rate of fibrillations in presumably normal people is an imperfect science and dependent on how the electromyographer interprets waveforms that look like fibrillations or end-plate spikes. The specificity of needle EMG was addressed in a group of asymptomatic persons in comparison with a radiculopathy group. When considering the typical distribution of findings to identify a radiculopathy (two abnormal muscles not innervated by the same peripheral nerve) and using six muscle screens and considering only spontaneous activity as abnormal, the needle EMG had 100% specificity. There were no false-positive tests. This is valuable information that supports the usefulness of needle EMG as a confirmatory test that complements spinal imaging.[182]

Cervical and lumbar paraspinal muscles are important to examine with EMG when assessing for radiculopathy. If positive, they localize the site of pathology to the spine or root level. When assessing paraspinal muscles, only fibrillations, CRDs, or myotonic discharges have clinical relevance. Recruitment findings and motor unit morphology for paraspinal muscles have not been well established. There are no normative data regarding motor unit morphology in either cervical or lumbar paraspinal muscles to which precise recruitment comparisons can be made. Examiners should not overcall radiculopathies based on reduced "recruitment" or "increased polyphasicity" in paraspinal muscles. Paraspinal muscles show either spontaneous activity or other abnormal discharges (CRDs), and therefore they localize the lesion to a root but not to a specific level. There is considerable overlap in paraspinal muscles, with single roots innervating muscle fibers above and below their anatomic levels. The root level is best defined by the distribution of EMG findings in limb muscles, minimizing the number of root levels that tie together and explain the EMG findings in abnormal muscles. Given the multiple root innervations of virtually every muscle, sometimes two root levels are equally likely to explain all the muscle denervation seen. The multifidi (deepest levels) may have more specific innervation patterns, thus positioning the needle to assess these muscles is preferred. Because the paraspinal muscles have so much root overlap, the level of radiculopathy cannot be delineated by paraspinal EMG alone; rather, it is based on the root level that best explains the distribution of limb muscles with EMG abnormalities.

The lumbar paraspinal muscle examination can be easily performed and should be studied in persons with suspected radiculopathies. Over the past 3 decades, Haig et al.[82-86,181-183] pioneered a means of assessing the lumbar paraspinal muscles with needle EMG. Haig[83] derived a means of quantifying paraspinal findings with an index derived by adding the fibrillations found in a standard set of muscle insertions. Such quantitative assessment distinguishes patients with radiculopathies and spinal stenoses with greater precision. This mini-paraspinal mapping (Mini-PM) screen uses four insertions with the pin directed in three quadrants for each site. This yields 12 rating areas. The spontaneous activity is rated at each of the 12 sites on a scale of 0 to 4 fibrillation levels. These individual scores are then combined into a score for one side of the lumbar spine. Considerable work was done to validate this screening lumbar exam relative to different lumbar spine disorders—radiculopathy, low back pain (LBP), and spinal stenosis. Fibrillations with a score greater than 3 on the Mini-PM scale are abnormal. Furthermore, this technique was validated in a group of asymptomatic older adults.[36] The stricter criterion of abnormal spontaneous activity on needle examination, and in paraspinal mapping scores greater than 6 being considered abnormal, serve to lower the risk of false-positive EMG testing. These findings, from a well-conducted and blinded study, suggest that EMG is less likely to be abnormal (false positive) in asymptomatic adults than MRI when quantified by paraspinal muscles mapping.[36] Although this technique is rigorous, it is relatively easy to perform and seems to improve the accuracy and reproducibility of paraspinal EMG.

Persons with lumbar stenosis compared with normal patients all had cutoff scores greater than 4. This cutoff score of greater than 4 on the Mini-PM yielded 100% specificity for lumbar stenosis. This study showed stronger correlations to lumbar stenosis symptoms than MRI findings.[85] Now it is known that some normal subjects have a smattering of fibrillations, and patients with clinical disease have extensive findings medially.[83] Using the Mini-PM screen of the lumbar spine, it is clear that there is a continuum of what would generally be considered normal across different ages and disease categories. In general, persons with scores of less than 2 are normal, those with scores from 2 to 4 are borderline, and those persons with scores over 4 are usually considered abnormal; these latter findings are representative of a lumbosacral radiculopathy. This still requires clinician interpretation and diagnostic decision making for individual patients.

Dumitru et al.[64] have also done groundbreaking research on the presence of fibrillations in normal people. They examined lumbosacral paraspinal muscles and intrinsic foot muscles with monopolar EMG. These investigators recorded potentials and found that there were irregularly firing potentials with similar waveform characteristics as fibrillations and PSWs. By excluding irregularly firing potentials (atypical end-plate spikes) and considering only regularly firing potentials with appropriate morphology consistent with fibrillations, they found that only 4% of these individuals had false-positive paraspinal EMG findings. This well-designed quantitative lumbosacral study underscores the need to assess both firing rate and rhythm, as well as discharge morphology, when evaluating for fibrillations in the lumbar paraspinal muscles. Fibrillations fire regularly, and this should be confirmed by listening carefully and seeing consistent intervals between fibrillations on the display screen. Electrodiagnosticians should take care not to overcall paraspinal muscle EMG findings by mistaking irregularly firing end-plate spikes for regularly firing fibrillations.

How Many and Which Muscles to Study

A screening EMG study involves determining whether the radiculopathy can be confirmed by EMG. The concept of a screening EMG encompasses identifying the possibility of an electrodiagnostically confirmable radiculopathy. If one of the muscles in the screen is abnormal, the screen must be expanded to exclude other diagnoses and to fully delineate the radiculopathy level. Because of the screening nature of the EMG examination, electrodiagnosticians with experience should look for more subtle signs of denervation and, if present in the screening muscles, then expand the study to determine whether these findings are limited to a single myotome or peripheral nerve distribution. If they are limited to a

TABLE 9.17 Six-Muscle Screen Identifications of Patients With Cervical Radiculopathies

Muscle Screen	Neuropathic[a] (%)	Spontaneous Activity[b] (%)
Six Muscles Without Paraspinals		
Deltoid, APB, FCU, triceps, PT, FCR	93	66
Biceps, triceps, FCU, EDC, FCR, FDI	87	55
Deltoid, triceps, EDC, FDI, FCR, PT	89	64
Biceps, triceps, EDC, PT, APB, FCU	94	64
Six Muscles With Paraspinals		
Deltoid, triceps, PT, APB, EDC, PSM	99	83
Biceps, triceps, EDC, FDI, FCU, PSM	96	75
Deltoid, EDC, FDI, PSM, FCU, triceps	94	77
Biceps, FCR, APB, PT, PSM, triceps	98	79

APB, Abductor pollicis brevis; *EDC,* extensor digitorum communis; *FCR,* flexor carpi radialis; *FCU,* flexor carpi ulnaris; *FDI,* first dorsal interosseus; *PSM,* cervical paraspinal muscles; *PT,* pronator teres.
[a]The "neuropathic" column indicates the identification rates when looking for all types of subtle neuropathic findings.
[b]The "spontaneous activity" column indicates identification rates when only fibrillations or positive sharp waves are considered.
Modified from Dillingham TR, Lauder TD, Andary M, et al. Identification of cervical radiculopathies: optimizing the electromyographic screen. *Am J Phys Med Rehabil.* 2001;80:84-91.

TABLE 9.18 Six-Muscle Screen Identifications of Patients With Lumbosacral Radiculopathies

Muscle Screen	Neuropathic[a] (%)	Spontaneous Activity[b] (%)
Six Muscles Without Paraspinals		
TA, TP, MGAS, RFEM, BFSH, LGAS	89	78
VMED, TFL, LGAS, TP, ADD, MGAS	83	70
VLAT, BFSH, LGAS, ADD, TFL, PT	79	62
ADD, TFL, MGAS, TP, TA, LGAS	88	79
Six Muscles With Paraspinals		
TA, TP, MGAS, PSM, VMED, TFL	99	93
VMED, LGAS, TP, PSM, BFSH, MGAS	99	87
VLAT, TFL, LGAS, PSM, TA, BFSH	98	87
ADD, MGAS, TP, PSM, VLAT, BFSH	99	89
VMED, TA, TP, PSM, BFSH, MGAS	100	92
VMED, TFL, LGAS, PSM, TA, TP	99	91
ADD, MGAS, TPB, PSM, TA, BFSH	100	93

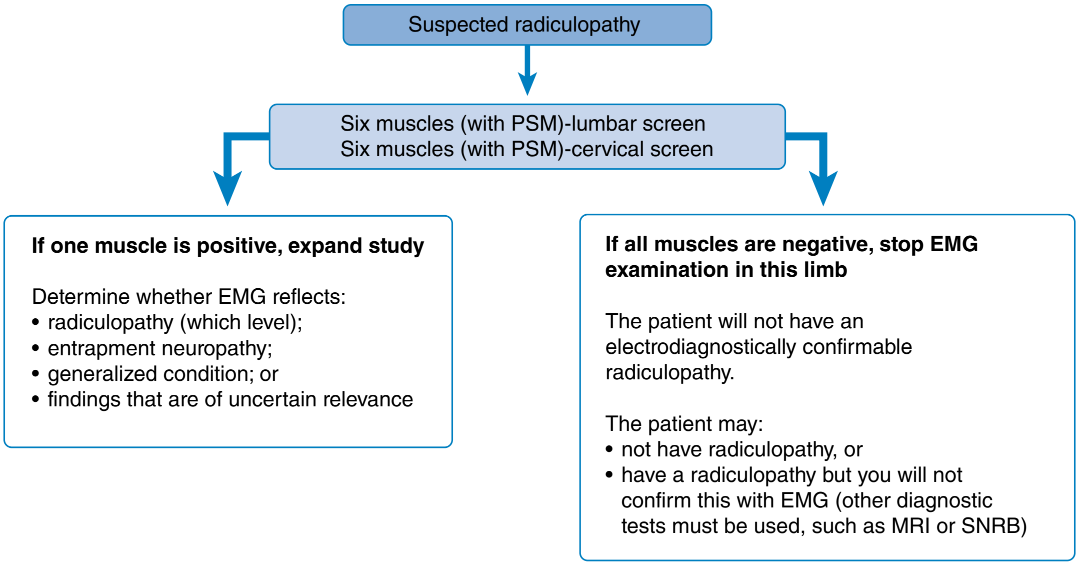
ADD, Adductor longus; *BFSH,* biceps femoris short head; *LGAS,* lateral gastrocnemius; *MGAS,* medial gastrocnemius; *PSM,* lumbar paraspinal muscles; *RFEM,* rectus femoris; *TA,* tibialis anterior; *TP,* tibialis posterior; *TFL,* tensor fascia lata; *VLAT,* vastus lateralis; *VMED,* vastus medialis.
[a]The "neuropathic" column indicates the identification rates when looking for all types of subtle neuropathic findings.
[b]The "spontaneous activity" column indicates identification rates when only fibrillations or positive sharp waves are considered.
Modified from Dillingham TR, Lauder TD, Andary M, et al. Identifying lumbosacral radiculopathies: an optimal electromyographic screen. *Am J Phys Med Rehabil.* 2000;79:496-503.

single muscle, their clinical significance is uncertain, so some clinicians will increase the number of muscles studied to look for another abnormal muscle to include three or more muscles that cover each root level.

A prospective multicenter study evaluating patients referred to participating EDX laboratories with suspected cervical radiculopathy was conducted to address the issue regarding how many muscles are sufficient to confidently identify an electrodiagnostically confirmable cervical radiculopathy.[50] There were 101 patients with electrodiagnostically confirmed cervical radiculopathies representing all cervical root levels. When paraspinal muscles were one of the screening muscles, five-muscle screens identified 90% to 98% of radiculopathies, six-muscle screens identified 94% to 99%, and seven-muscle screens identified 96% to 100% (Table 9.17). When paraspinal muscles were not part of the screen, eight distal limb muscles recognized 92% to 95% of radiculopathies. Six-muscle screens, including paraspinal muscles, yielded consistently high identification rates, and studying additional muscles led to marginal increases in identification. Individual screens useful to the electromyographer are listed in Table 9.17. These findings were consistent with those derived from a large retrospective study.[119]

A similar prospective multicenter study was conducted to assess the optimal screening EDX assessment for persons with suspected lumbosacral radiculopathy.[51,52] In this study, there were 102 patients with electrodiagnostically confirmed lumbosacral radiculopathies representing all lumbosacral root levels. When paraspinal muscles were one of the screening muscles, four-muscle screens identified 88% to 97%, five-muscle screens identified 94% to 98%, and six-muscle screens identified 98% to 100% (Table 9.18). When paraspinal muscles were not part of the screen, identification rates were lower for all screens, and eight distal muscles were necessary to identify 90%. Other retrospective studies are consistent with these findings.[118]

Dillingham and Dasher[48] reanalyzed data from a study published by Knutsson almost 40 years earlier.[110] In this detailed study, 206 patients with sciatica underwent lumbar surgical exploration. All subjects underwent a standardized 14-muscle EMG evaluation using concentric needles. Screens of four muscles, one being the lumbosacral paraspinal muscles, yielded (1) an identification rate of 100%, (2) a 92% sensitivity with respect to the intraoperative anatomic nerve root compressions, and (3) an 89% sensitivity with respect to the clinical inclusion criteria.[48] This retrospective study further underscores the utility of streamlined screening EMG evaluations for persons with suspected lumbosacral radiculopathy. In summary, for both cervical and lumbosacral radiculopathy screens, the optimal minimum number of muscles appears to be six, which includes the paraspinal muscles and muscles that represent all root level innervations. When paraspinal muscles are not reliable, then eight nonparaspinal muscles must be examined. Another way to think of this is as follows: "To minimize harm, six in the leg and six in the arm." If one of the six muscles studied in the screen is positive, there is the possibility of confirming electrodiagnostically that a radiculopathy is present. In this case, the examiner must study additional muscles to determine the radiculopathy level and to exclude a mononeuropathy.

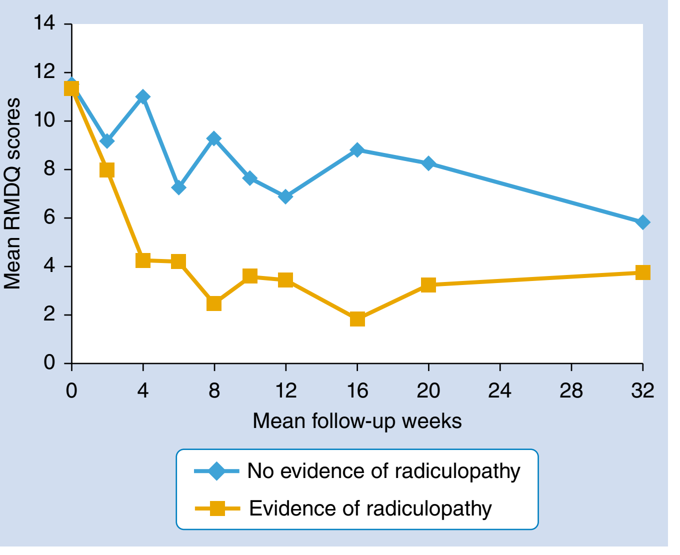


• **Fig. 9.10** Algorithm for suspected radiculopathy. *EMG,* Electromyography; *MRI,* magnetic resonance imaging; *PSM,* paraspinal muscle; *SNRB,* sensory nerve root block.

If the findings are found in only a single muscle, they remain inconclusive and of uncertain clinical relevance. If none of the six muscles are abnormal, the examiner can be confident there is a very low probability of missing the opportunity to confirm electrodiagnostically that a radiculopathy is present, and the EMG examination can be stopped. The patient might still have a radiculopathy (purely sensory and not associated with motor axon loss), but other tests such as MRI will be necessary to confirm this clinical suspicion. This logic is illustrated in Fig. 9.10.

Previously, a well-defined temporal course of events was thought to occur with radiculopathies despite the absence of studies supporting such a relationship. It was a commonly held notion that in acute lumbosacral radiculopathies, the paraspinal muscles denervated first, followed by distal muscles, and that reinnervation started with paraspinal muscles and then with distal muscles. This paradigm was addressed with a series of investigations.[52-54,144] For both lumbosacral and cervical radiculopathies, symptom duration had no significant relationship to the probability of finding spontaneous activity in paraspinal or limb muscles. This simplistic explanation, although widely quoted in the older literature, does not explain the complex pathophysiology of radiculopathies. When reporting EDX conclusions regarding the presence or absence of a radiculopathy, the EMC uses the pattern of muscle involvement to identify the minimum number of root levels to explain these EMG findings.

A pattern of unequivocal fibrillations in a myotomal distribution is compelling for radiculopathy. Other softer findings (e.g., reduced recruitment) should inspire less confidence in the diagnostic conclusions. More subjective findings (reduced recruitment) are open to interpretation and are best appreciated by skilled electrodiagnosticians. If polyphasic MUAPs are clearly abundant (>30%) in a myotomal distribution, this finding may indicate radiculopathy, but care must be used when making a diagnosis of radiculopathy without other correlating findings. In such cases, the contralateral limb should be examined to ensure there is not a generalized process going on. The results of a novel cohort study suggested that in patients with symptoms related to sciatica, the presence of radiculopathy confirmed with EMG



• **Fig. 9.11** Comparison of average disability scores based on patient's electrodiagnostic status from baseline data collection through posttreatment follow-up period. Roland-Morris Disability Questionnaire *(RMDQ).* (Redrawn from Savage NJ, Fritz JM, Kircher JC, et al. The prognostic value of electrodiagnostic testing in patients with sciatica receiving physical therapy. *Eur Spine J.* 2015;24:434-443.)

testing was associated with a significantly more favorable prognosis.[160] A positive EMG in this study was associated with statistically significant and clinically meaningful improvements in LBP-related disability at 6 months follow-up regardless of the type of physical therapy treatment received (Fig. 9.11). For those persons with nonspecific LBP or sciatica without a positive EMG, they likely had a heterogeneous set of causes of LBP that could be considered chronic or multifactorial with overuse and deconditioning playing a role. Those patients with a positive EMG showing physiologic lumbosacral nerve root impingement have a clear cause for their symptoms, which, in this study by Savage et al.,[160] responded well to physical therapy measures. Identifying a precise lumbosacral radiculopathy by needle EMG correlated with a more favorable outcome with conservative care. Rigler and Podnar[148] conducted a follow-up survey study of 300 consecutive persons in Slovenia who underwent EDX testing. In the final sample of 186 persons, they showed that those who had an abnormal EDX study had significantly better outcomes at 3 years than those who had a normal EDX study. Having an abnormal EDX study (carpal tunnel syndrome, radiculopathy, or other entrapment neuropathy) was a strong independent predictor of a good outcome in their multivariate statistical models. Interestingly, the subgroups having the worst outcomes were those persons identified with polyneuropathy and those with normal electrodiagnostic studies. The positive association with better outcomes in this group was largely driven by the positive outcomes for persons who had median neuropathies and radiculopathy identified by EDX.[148]

Overall, outcomes for persons with lumbosacral radiculopathies because of herniated discs is very favorable, with over 90% resolving with symptom management and physical therapy and with an active exercise program.[25,32,77,87,98,102,124,154,155,188] Similar to lumbosacral radiculopathy, persons with cervical radiculopathy resolve with conservative care in over 90% of cases.[92,121,127,129,147,156,173] One landmark study showed that persons with cervical radiculopathy with positive EMG and imaging findings, and even with extruded discs, did not require surgery in the majority (92%) of cases and successfully resolved their symptoms and returned to work.[156] EDX findings have been increasingly recognized as diagnosing the root-level problem, and the findings have implications for treatment outcomes. Positive EMG findings indicating a lumbosacral radiculopathy, if present, are associated with significantly more favorable responses to lumbar epidural steroid injections.[10,13,74,131]

Traumatic Nerve Injuries

Peripheral nervous system injuries are often associated with musculoskeletal trauma, surgical intervention, and anesthesia and are considered secondary to multiple factors such as stretch, pressure (tourniquet), crush, ischemia, and chemical stresses. Electrodiagnosis is valuable in identifying lesion location, characterizing injury severity, and recovery prognosis.[62] EDX testing can identify electrophysiologic evidence of spontaneous regeneration that precedes clinical findings of recovery.[125] The demonstration of actively firing motor units across the zone of injury can determine with certainty that there is not complete neurotmesis (nerve transection). The pertinent issues in electrodiagnosis of peripheral nerve injuries are localization of the injury, pathophysiology of the lesion, severity of the dysfunction, and the presence of reinnervation.[50] Frykman et al.[76] presented an algorithmic approach to the assessment and management of traumatic peripheral nerve injuries based on the clinical examination and EDX findings (Fig. 9.12). For acute injuries (between 0 and 7 days after injury), the electromyographer can only comment on nerve continuity if there are voluntary motor units or conduction is found across the area of suspected injury. Any conduction across the area of injury documents at least one axon intact and rules out neurotmesis. At this time, the electromyographer cannot differentiate neurotmesis from 100% neurapraxia; thus in the first 7 days, the electromyographer can only give good news.

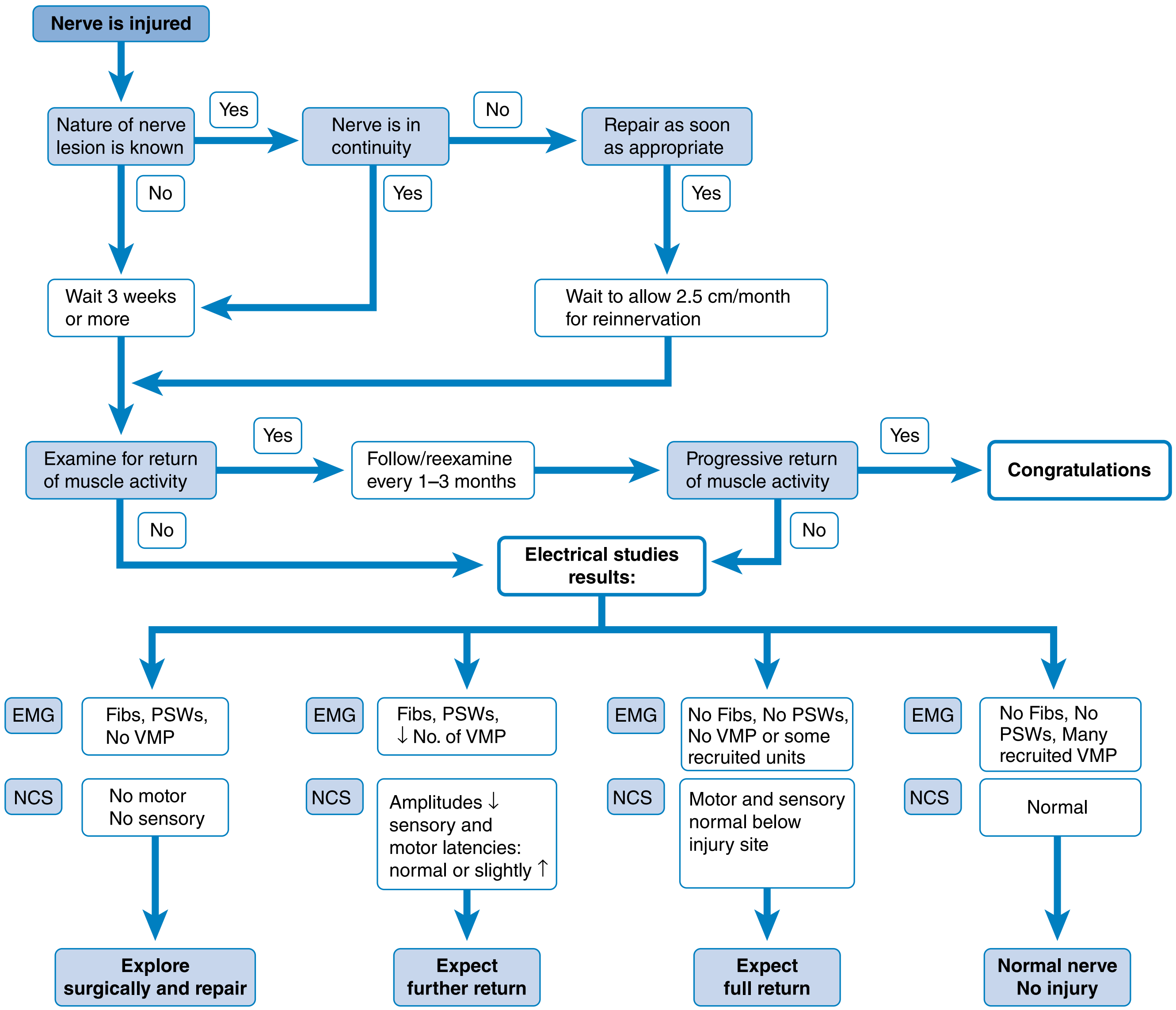
In those patients for whom little or no recovery is evident at 3 weeks by clinical examination, EDX studies are recommended. If no voluntary motor units and no motor or sensory responses are elicited, surgical exploration can be considered. If fibrillations are present, yet some voluntary units are noted along with reduced but present sensory or motor responses, then a partial nerve injury can be diagnosed, and further recovery can be reasonably expected. In a mild or neurapraxic injury with no fibrillations and normal sensory and motor amplitudes, a full recovery can be expected.[76] Although somewhat simplistic, this approach provides a conceptual framework for consolidating EDX and clinical information into a treatment plan. The algorithm by Frykman et al.[76] has been updated and slightly modified regarding the presence of voluntary MUAPs. Nerve injury classification based solely upon EDX testing has limitations; other imaging modalities, such as MRI and ultrasound, where indicated, provide clinically useful supplemental information.[21,30,69,191,193] Some newer surgical approaches to peripheral nerve injury are based on intraoperative NCS. Intraoperative nerve action potential assessment across individual fascicles can further define whether a fascicle has axons in continuity.[100,107-109] However, the Frykman approach is a reasonable conceptual framework for managing nerve injuries.

Neurapraxic injuries are expected to recover in days to weeks. More significant traumatic nerve injury exhibits mixed nerve component (fascicle, large or small axon) involvement with varying degrees of demyelination/axon shearing. Depending on the severity of axonotmesis, months may be required for recovery in cases where the nerve remains in continuity. NCS distal to the site of the lesion after Wallerian degeneration has taken place can help delineate the degree of axonal loss. Right to left motor and sensory NCS amplitudes can be compared, yielding a semiquantitative means of determining the degree of axonal loss. Care must be used to set up NCS studies comparably to minimize side-to-side amplitude variability.[150]

In partial nerve injuries, regeneration occurs through axonal sprouting from undamaged axons in proximity to denervated muscle fibers. Axonal sprouting can reinnervate muscle fibers within days and is demonstrated by polyphasic motor units. On the other hand, axonal regeneration occurring at the site of injury requires months to reinnervate the distal denervated muscle fibers, which is demonstrated by nascent potentials (Fig. 9.13; also see Video 9.25).[14]

Chaudhry and Cornblath[33] studied Wallerian degeneration in humans. After complete nerve transection, failure at the neuromuscular junction occurs in 2 to 5 days in animals.[78] They found that motor-evoked responses were absent in 9 days, sensory responses were absent by 11 days, and denervation potentials were seen 10 to 14 days after injury. These results indicate that the motor and sensory responses reliably assess the status of the nerve after 11 days. It takes 3 to 4 weeks for fibrillations to fully develop and be consistently recorded by needle EMG.[150]

Kraft[111] studied fibrillation potential amplitudes and their relationship to time after injury. The mean fibrillation amplitude in the first 2 months after injury was 612 μV. This important study showed that 1 year after traumatic nerve injury, no population of fibrillation potentials was greater than 100 μV in amplitude. These findings can help determine whether fibrillations are the result of recent or remote denervation. Fibrillation potentials in areas of direct muscle trauma should be interpreted with caution. An informative study regarding fibrillation potential generation resulting from direct muscle injury was reported by Partanen and Danner.[142] They studied 43 patients after muscle biopsy. At 6 to 7 days after biopsy, approximately 50% of the patients showed fibrillations on needle EMG testing. At 16 days, all patients revealed fibrillation potentials in the biopsied muscles, and these fibrillations persisted for up to 8 months. These findings suggest



• **Fig. 9.12** Algorithm for management of peripheral nerve injuries. *EMG,* Electromyography; *Fibs,* fibrillation potentials; *NCS,* nerve conduction study; *PSWs,* positive sharp waves; *VMP,* voluntary motor unit potentials (interference pattern). (Modified from Frykman GK, Wolf A, Coyle T. An algorithm for management of peripheral nerve injuries. *Orthop Clin North Am.* 1981;12:239-244.)

that direct muscle trauma can result in denervation potentials and can occur in the absence of any nerve injury.[142] Such information is important when evaluating traumatically injured limbs. Areas of muscle damage or surgical scars should be avoided because fibrillation potentials in these areas are uninterpretable.

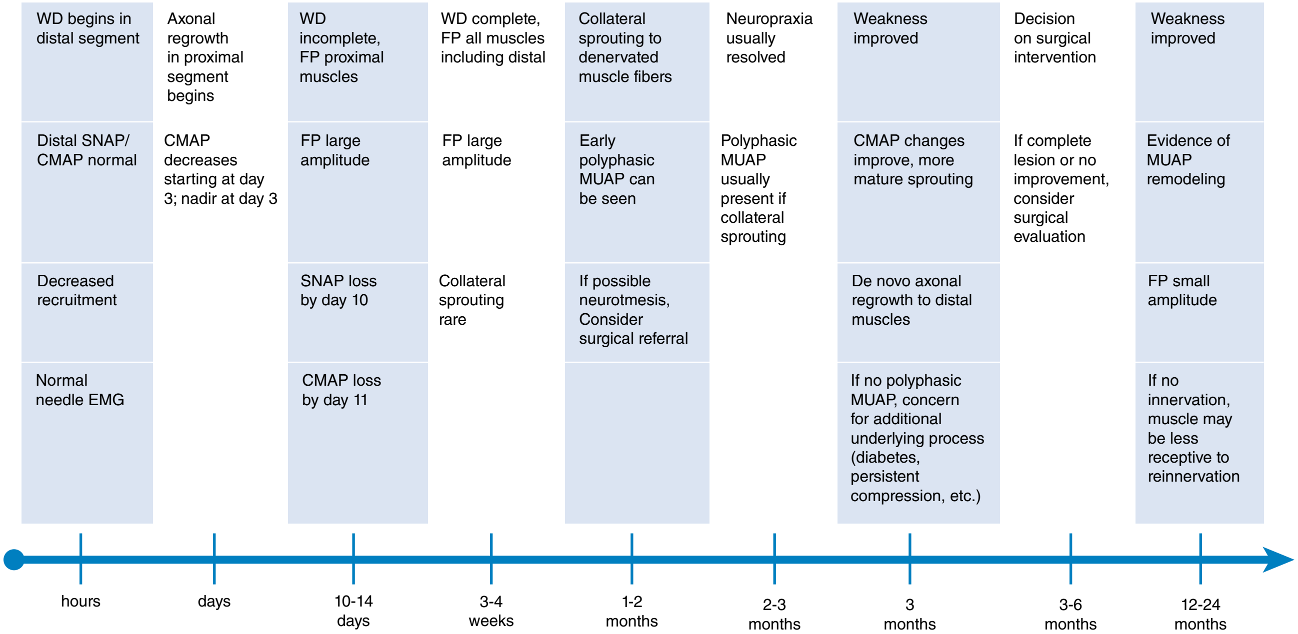
Generalized Disorders

EDX testing is an extension of the physical examination and, as such, should be placed within a conceptual framework of physical findings and patient symptoms. Fig. 9.1 provides a conceptual framework for using the clinical scenario and examination findings to address the highest probability disorders. For the purposes of this discussion, generalized disorders are those affecting more than one limb. The guidelines in Table 9.13 provide a framework for structuring the examination to evaluate for these disorders but is by no means a perfect taxonomy.

The science of diagnosing and managing persons with neuromuscular disorders is proceeding at a rapid pace. Genetic and laboratory testing continues to advance, becoming more precise and sensitive. However, EDX testing has remained an important evaluation for assessing most patients with neuromuscular disorders. A major aspect of EDX testing is to exclude other disorders and refine the differential diagnostic possibilities.

Polyneuropathy

Evaluation for polyneuropathy involves identifying peripheral nervous system involvement. The EMC should also strive to categorize the type of neuropathy based on its EDX features. Donofrio and Albers[58] compiled a comprehensive list of polyneuropathies classified by their characteristic EDX findings. Determining whether sensory, motor, or both types of nerves are involved, and whether the primary type of pathology is demyelination or axonal



• **Fig. 9.13** Timeline for electrodiagnostic studies in suspected PPNI: SNAP propagation failure can be seen as early as 7 to 9 days and is considered complete by day 10. Alpha motor neuron axon damage results in neuromuscular junction transmission failure in 6 days and WD is detectable in motor nerve axons by day 11 postinjury. WD can be observed in muscle fibers demonstrating FP immediately distal to the lesion within 7 days, middistance muscle fibers after 14 days, and in all distally innervated muscle fibers by 28 days. *CMAP*, Compound motor action potential; *EMG*, electromyography; *FP*, fibrillations/sharp waves; *MUAP*, motor unit action potential; *PPNI*, perioperative peripheral nerve injuries; *SNAP*, sensory nerve action potential; *WD,* Wallerian degeneration. Modified from Seidel GK, Vocelle AR, Ackers IS, et al. Electrodiagnostic Assessment of Peri-Procedural Iatrogenic Peripheral Nerve Injuries and Rehabilitation. *Muscle Nerve*. Published online February 12, 2025. doi:10.1002/mus.28364

loss, allows the EMC to narrow the list of potential etiologies for the polyneuropathy (Box 9.1). The EDX changes seen with demyelinating and axonal neuropathies are shown in Table 9.19. Persons with polyneuropathies involving acute or recent motor axonal loss show fibrillations and PSWs in predominantly distal muscles. The distribution of EMG findings in polyneuropathies is such that distal lower limb muscles often demonstrate the greatest involvement. In persons with a more chronic process with either old axonal loss or slow axonal loss that is balanced with reinnervation and reorganization of the motor unit, large-amplitude, long-duration MUAPs can be seen.[112]

Absent SNAPs and CMAPs can reflect either axonal loss or conduction block. EMG can provide supporting information about motor axonal loss. Nerve conduction velocity is an important parameter to measure from multiple nerves. Temporal dispersion suggests an acquired demyelinating polyneuropathy. Care must be taken to study a sufficient number of nerves to confidently determine whether a diffuse process is present rather than a single entrapment neuropathy. If one limb is affected, the EMC should study the contralateral limb. If this is abnormal, another limb should be examined. For example, a patient with abnormalities in both legs and possible polyneuropathy can be quickly and easily diagnosed by finding fibrillations in the intrinsic hand muscles.

F-waves measure longer neurologic pathways and are important in diagnosing polyneuropathies. They are helpful in identifying diabetic polyneuropathy, acute inflammatory demyelinating polyneuropathy (AIDP), and chronic inflammatory demyelinating polyradiculoneuropathy (CIDP).[75] F-waves and H-reflexes are important tests for assessing proximal segments of the peripheral nervous system and can be helpful for neuropathy evaluations.

AIDP, also known as Guillain-Barré syndrome, is the most common, acute, rapidly progressive polyneuropathy in clinical practice and must be recognized when present.[3] Patients typically present with sensory symptoms and weakness, with disease progression over 2 to 4 weeks. Sensory nerve conduction abnormalities, reduced amplitude, or prolonged distal latency, particularly in the median nerve, are observed but can take 4 to 6 weeks to peak.[63] Motor NCS reveal prolonged distal latencies, temporal dispersion and conduction block, or slowed velocities in 80% to 90% of these patients. Reduced CMAPs in ulnar nerve (and median nerve) innervated muscles that are 10% to 20% of normal values signify a poorer prognosis and can provide early information to guide clinical management.[133] In severe cases, EMG findings of fibrillations and PSWs can be seen if motor axonal loss has occurred. Facial muscle weakness and bulbar muscle involvement are commonly found in patients with AIDP.[189] Nerve conduction findings of slowed conduction velocity, temporal dispersion, prolonged distal latencies, or late responses should be found in two or more nerves to support a diagnosis of AIDP.[2] In some cases, late responses (F-waves and H-reflexes) are the only findings noted on EDX testing early on in persons with AIDP.

CIDP represents a demyelinating polyneuropathy of chronic nature with fluctuating weakness and stepwise progression. Weakness and sensory symptoms similar to those of AIDP are found, but the chronicity is longer. In CIDP, sensory responses are frequently absent in both upper and lower limbs, motor distal latencies are

• BOX 9.1 Classification of Polyneuropathies by Their Electrodiagnostic Characteristics

Uniform, Demyelinating, Mixed Sensorimotor Polyneuropathy

- Hereditary motor sensory neuropathy types 1, 3, and 4
- Metachromatic leukodystrophy
- Krabbe globoid leukodystrophy
- Adrenomyeloneuropathy
- Congenital hypomyelinating neuropathy
- Tangier disease
- Cockayne syndrome
- Cerebrotendinous xanthomatosis

Segmental, Demyelinating, Motor Greater Than Sensory Polyneuropathy

- Acute inflammatory demyelinating polyneuropathy
- Chronic inflammatory demyelinating polyneuropathy
- Multifocal demyelinating neuropathy with persistent conduction block
- Osteosclerotic myeloma
- Waldenström macroglobulinemia
- Monoclonal gammopathy of undetermined significance
- Gamma heavy chain disease
- Angiofollicular lymph node hyperplasia
- Hypothyroidism
- Leprosy
- Diphtheria
- Acute arsenic polyneuropathy
- Pharmaceuticals: amiodarone, perhexiline, high-dose cytarabine (Ara-C)
- Lymphoma
- Carcinoma
- Acquired immunodeficiency syndrome (AIDS)
- Lyme disease
- Acromegaly
- Hereditary neuropathy with susceptibility to pressure palsies
- Systemic lupus erythematosus
- Glue-sniffing neuropathy
- Postportacaval anastomosis
- Neuropathy associated with progressive external ophthalmoplegia
- Ulcerative colitis
- Marinesco-Sjögren syndrome
- Cryoglobulinemia
- Pharmaceuticals
  - Amiodarone
  - Perhexiline
  - High-dose Ara-C
- Immunosuppressant biologics (rare)

Axon Loss, Motor Greater Than Sensory Polyneuropathy

- Porphyria
- Axonal Guillain-Barré syndrome: acute motor axonal neuropathy
- Hereditary motor sensory neuropathy types 2 and 5
- Lead neuropathy
- Dapsone neuropathy
- Vincristine and other vinca alkaloids
- Remote effect motor neuronopathy associated with lymphoma
- Remote effect motor neuronopathy associated with carcinoma
- Hypoglycemia or hyperinsulinemia
- West Nile

Axon Loss Sensory Neuronopathy or Neuropathy

- Hereditary sensory neuropathy types 1–4
- Friedreich ataxia
- Spinocerebellar degeneration
- Abetalipoproteinemia (Bassen-Kornzweig disease)
- Primary biliary cirrhosis
- Acute sensory neuronopathy
- Cisplatin and other platinum chemotherapeutic toxicity
- Carcinomatous sensory neuronopathy
- Lymphomatous sensory neuronopathy
- Chronic idiopathic ataxic neuropathy
- Sjögren syndrome
- Fisher variant Guillain-Barré syndrome
- Paraproteinemias
- Pyridoxine (vitamin B_6) toxicity
- Idiopathic sensory neuronopathy
- Styrene-induced peripheral neuropathy
- Crohn disease
- Thalidomide toxicity
- Nonsystemic vasculitic neuropathy
- Chronic gluten enteropathy
- Vitamin E deficiency
- Medications
  - Linezolid (also optic neuropathy)
  - Bortezomib (also small-fiber neuropathy)
  - Epothilones
  - Isoniazid (interferes with B6 synthesis)
  - Nucleoside reverse transcriptase inhibitors
  - Levodopa (mild)
  - Triazole drugs (voriconazole, itraconazole)

Axon Loss, Mixed Sensorimotor Polyneuropathy

- Amyloidosis
- Chronic liver disease
- Nutritional disease
- Vitamin B_{12} deficiency
- Folate deficiency
- Whipple disease
- Postgastrectomy syndrome
- Gastric restriction surgery for obesity
- Thiamine deficiency
- Alcoholism
- Sarcoidosis
- Connective tissue diseases
- Rheumatoid arthritis
- Periarteritis nodosa
- Systemic lupus erythematosus
- Churg-Strauss vasculitis
- Temporal arteritis
- Scleroderma
- Behçet disease
- Hypereosinophilia syndrome
- Cryoglobulinemia
- Toxic neuropathy
- Acrylamide
- Carbon disulfide
- Dichlorophenoxyacetic acid
- Ethylene oxide
- Hexacarbons
- Carbon monoxide
- Organophosphorus esters
- Glue-sniffing neuropathy
- Metal neuropathy
  - Chronic arsenic intoxication
  - Mercury
  - Thallium
  - Gold
- Pharmaceuticals
  - Colchicine
  - Phenytoin
  - Ethambutol (also optic neuropathy)
  - Amitriptyline
  - Metronidazole (also optic and autonomic neuropathy)
  - Misonidazole

• BOX 9.1 Classification of Polyneuropathies by Their Electrodiagnostic Characteristics—cont'd

- Nitrofurantoin
- Chloroquine
- Disulfiram
- Glutethimide
- Nitrous oxide (B_{12} deficiency)
- Lithium
- Vincristine and other vinca alkaloids
- Thalidomide
- Taxanes (paclitaxel, docetaxel)
- Leflunomide
- Fluoroquinolones
- Carcinomatous axonal sensorimotor polyneuropathy
- Chronic obstructive pulmonary disease
- Giant axonal dystrophy
- Olivopontocerebellar atrophy
- Neuropathy of chronic illness
- Acromegaly
- Hypophosphatemia
- Lymphomatous axonal sensorimotor polyneuropathy
- Hypothyroidism
- Myotonic dystrophy
- Necrotizing angiopathy
- Lyme disease
- AIDS
- Jamaican neuropathy
- Tangier disease
- Gouty neuropathy
- Polycythemia vera
- Typical multiple myeloma
- Pharmaceuticals
  - Colchicine
  - Phenytoin
  - Ethambutol (also optic neuropathy)
  - Amitriptyline
  - Metronidazole (also optic and autonomic neuropathy)
  - Misonidazole
  - Nitrofurantoin
  - Chloroquine
  - Disulfiram
  - Glutethimide
  - Nitrous oxide (B_{12} deficiency)
  - Lithium
  - Vincristine and other vinca alkaloids
  - Thalidomide
  - Taxanes (paclitaxel, docetaxel)
  - Leflunomide
  - Fluoroquinolones

Mixed Axon Loss, Demyelinating Sensorimotor Polyneuropathy

- Diabetes mellitus
- Uremia
- AIDS; AIDS-related complex
- Arsenic trioxides
- Immunosuppressant biologics (rare)
- Interferons (rare)

TABLE 9.19 Electrodiagnostic Findings in Peripheral Neuropathies

Parameter	Early Demyelinating	Chronic Demyelinating	Acute Axonal	Chronic Axonal
Distal latency	Increased	Increased	Normal or slightly increased	Normal or slightly increased
Nerve conduction velocity	Decreased	Decreased	Normal or slightly decreased	Normal or slightly decreased
F-latency	Increased or absent	Increased or absent	Normal or absent	Normal or absent
H-reflex	Increased latency, low amplitude or absent	Increased latency, low amplitude or absent	Low amplitude or absent	Low amplitude or absent
Sensory nerve action potential amplitude	Normal or decreased	Normal, decreased, or absent	Decreased or absent	Decreased or absent
Compound muscle action potential amplitude	Normal or decreased	Normal or decreased	Decreased	Decreased
Motor unit action potential duration	Normal	Normal (hereditary) or increased (acquired)	Normal	Increased
Motor unit action potential amplitude	Normal	Normal or increased	Normal	Increased
Polyphasics	Normal or decreased	Normal or increased	Normal	Increased
Recruitment	Normal or decreased	Normal or decreased	Decreased	Decreased
Abnormal spontaneous activity	None	None, or fibrillations, complex repetitive discharges	Fibrillations complex repetitive discharges	None, or fibrillations, complex repetitive discharges

Modified from Krivickas L. Electrodiagnosis in neuromuscular diseases. *Phys Med Rehabil Clin N Am.* 1998;9:83-114.

prolonged, nerve conduction velocities are reduced, and motor conduction block and temporal dispersion are evident with associated prolonged or absent F-waves.[4,133] EMG findings depend on the rate of disease progression and can reveal reduced recruitment, fibrillation potentials, polyphasic potentials, and reorganized MUAPs with large amplitudes and increased durations. Sensory nerve conduction slowing was recently shown to be a highly specific, yet less sensitive marker for differentiating CIDP from axonal polyneuropathy.[20]

Nerve ultrasound is a rapidly emerging complementary technique to electrophysiology and is beyond the scope of this chapter. However, it is worth mentioning the new literature showing that in CIDP the large upper limb nerves at sites not associated with entrapment neuropathies are larger in cross-sectional area on ultrasound or MRI than normal nerves or nerves found in diabetic polyneuropathy.[72,179] Imaging is proving to be a useful technique that, when coupled with electrodiagnosis, can provide more accurate diagnosis for CIDP.

Multifocal motor neuropathy with conduction block is a disorder that is sometimes confused with motor neuron disease. The two have markedly different prognoses. Patients present with asymmetric weakness in a single body region, frequently the hand, but without sensory symptoms. Progression is slow and spans many years.[115,180] Sensory nerves are usually normal on EDX testing. Motor nerve testing reveals conduction block in multiple nerves at sites not prone to focal entrapments: mid-forearm, mid-leg, arm, and brachial plexus. Distal motor latencies and amplitudes are usually normal. Proximal motor nerve stimulations at the Erb point are important to exclude conduction block over proximal nerve segments.

Diabetes mellitus has an increasing incidence and prevalence in the United States.[89] It often confounds the accurate diagnosis of radiculopathy and spinal stenosis.[1,41] Inaccurate recognition of sensory polyneuropathy, diabetic amyotrophy, or mononeuropathy can lead to unnecessary surgical interventions. The pattern of involvement can take the form of symmetric polyneuropathies or focal and multifocal patterns of mononeuropathies. The proximal lower limb can be involved, with pain and weakness in patients with diabetic amyotrophy or other radiculoplexus lesions. The sensory nerve involvement, particularly in the legs, often results in complaints of numbness, tingling, burning, aching, and pain. Up to 80% of patients with diabetes having clinical polyneuropathies will demonstrate an abnormality of the SNAP.[142] Sensory findings usually precede motor findings. Motor nerve conduction is 15% to 30% below normal. EMG can show fibrillations and motor unit changes along with reduced recruitment.[60] Accurate assessment for polyneuropathy requires heightened suspicion coupled with sufficient testing. Nonphysicians perform 17% of studies in the United States.[55] In a sample of 6381 patients with diabetes undergoing EDX testing in 1998, identification rates for polyneuropathy were highest for physiatrists, osteopathic physicians, and neurologists (12.5%, 12.2%, and 11.9%, respectively).[56] Podiatrists and physical therapists identified 2.4% and 2.1%, respectively, as having polyneuropathy; these are rates approximately one-sixth that of physiatrists and neurologists, despite controlling for case-mix differences. Nonphysician providers who did not recognize polyneuropathy in this group of patients with diabetes performed almost exclusively EMG testing (>90%) at the expense of NCS. These findings underscore the need for high-quality consultations of sufficient scope by well-trained EDX physicians using the latest in normative reference data and the techniques associated with these clinical benchmarks[35] to accurately diagnose patients with complex disorders.[56]

Alcoholic polyneuropathy is a common peripheral nerve disease accounting for almost 30% of all cases of generalized polyneuropathy.[169] This clinical entity usually occurs in the setting of long-standing alcoholism and nutritional deficiency. Substantial weight loss often occurs before or is concurrent with the development of polyneuropathy. Presenting symptoms typically include pain, dysesthesia, and weakness in the feet and legs. Alcoholic polyneuropathy is an axonal loss disorder affecting both sensory and motor nerves (see Box 9.1).[58] Low SNAP amplitudes are seen in the legs, and EMG often reveals fibrillations and positive waves in distal muscles.

Hereditary sensory and motor neuropathies (HSMNs) are a diverse group of polyneuropathies that are rapidly becoming better characterized through genetic studies and are reviewed elsewhere in this book.[60] From an electrophysiologic standpoint, they are distinctly different from acquired polyneuropathies because of the relative lack of temporal dispersion and conduction block. HSMN type 1 is a hypertrophic variety with onion bulb formation and axonal atrophy.[105] Symptoms begin in the first 2 decades, and markedly reduced nerve conduction velocities are seen but without substantial temporal dispersion. HSMN type 2 presents in adult life or later. Patients have severe distal muscle atrophy and weakness. NCS show mild slowing, but EMG reveals large, reorganized MUAPs with fibrillations. Dejerine-Sottas (HSMN type 3) is a severe type of neuropathy appearing in infancy with delayed motor development and reveals the slowest nerve conduction velocities (<10 m/s and often as low as 2–3 m/s).[109,112]

The AANEM, in conjunction with the American Academy of Neurology and the American Academy of Physical Medicine and Rehabilitation, determined that a formal case definition for polyneuropathy was needed.[70] They described that no single reference standard was most appropriate for distal symmetric polyneuropathy. The most accurate diagnosis comprised a combination of clinical signs, symptoms, and EDX findings. EDX findings should be included as part of the case definition because of their higher level of specificity. EDX studies are sensitive and specific validated measures for the presence of polyneuropathy. This panel recommended a simplified NCS protocol for screening for the presence of distal symmetric polyneuropathy.[70] Sural sensory and fibular (peroneal) motor nerve conductions performed in one lower limb, taken together, were thought by this group to be the most sensitive tests for detecting a distal symmetric polyneuropathy. If both studies are normal, there is no evidence of typical distal symmetric polyneuropathy, and in such a situation no further NCS are necessary. However, if one of these tests is abnormal, then additional NCS are recommended. There are retrospective reports of abnormalities in NCS of the superficial fibular sensory, dorsal sural sensory, medial plantar mixed nerve, and needle EMG of foot intrinsic muscles that increase the sensitivity of electrodiagnosis diagnosing polyneuropathy. There have not been evidence-based studies looking at accuracy of EDX in patients with neuropathy symptoms and more sensitive techniques listed earlier. Prevalence of denervation in the intrinsic foot muscles in patients with distal predominantly small fiber neuropathy and normal NCS was increased and also correlated with smaller sural SNAP amplitude. This strongly suggests needle EMG of foot intrinsics will increase sensitivity of the diagnosis of polyneuropathy. Specificity is not addressed.[139]

Myopathies

Myopathic disorders include acquired inflammatory types such as polymyositis and dermatomyositis, as well as congenital myopathies, metabolic myopathies, muscular dystrophies, and mitochondrial myopathies. A detailed discussion is beyond the scope of this chapter and is found elsewhere in this textbook. The interested reader is directed to other specialized references for a complete description of the rarer forms of myopathy. Unfortunately, the EDX examination is less sensitive or specific for detecting myopathies than for any other group of neuromuscular diseases.[112] Myopathies are one of the most challenging disorders to identify and classify by electrodiagnosis because no findings are completely specific for myopathy. There is often patchy involvement of proximal muscles, and the EMG findings can vary depending on the severity and duration of the myopathy. Characteristic findings on EMG can indicate myopathy, yet the precise etiology of the myopathy requires other testing, such as muscle biopsy and genetic testing. The electromyographic findings for myopathies are shown in Table 9.20. NCS are usually normal except when a muscle is atrophic, and then the CMAP can be reduced. Sensory NCS are normal unless a concomitant polyneuropathy exists.

Needle EMG is the most helpful part of the examination. In the acute inflammatory myopathies, polymyositis, and dermatomyositis, the characteristic findings are fibrillations and PSWs, CRDs, and early or increased recruitment of short-duration, polyphasic, low-amplitude MUAPs.[149] Early recruitment or increased recruitment refers to the case in which more motor units are recruited than would be expected to generate a low muscular force. This is because in myopathies, more diseased muscle fibers are necessary to generate force than in the normal situation. This can also result from the motor units having fewer remaining muscle fibers. At low force levels, many units are present on the display screen. Proximal limb girdle and paraspinal muscles are frequently involved. In progressive muscular dystrophies such as Duchenne and Becker, fibrillations are widespread, with occasional CRDs and myotonic discharges.[112] Inclusion body myositis accounts for 30% of all myopathies and requires muscle biopsy to diagnose the condition. The findings are similar to those seen in polymyositis, with fibrillations and PSWs more widespread and prominent.[112]

Neuromuscular Junction Disorders

Disorders of the neuromuscular junction can be classified as either presynaptic or postsynaptic. The presynaptic disorders are LEMS and botulism. Myasthenia gravis is a postsynaptic disorder.[157] These unusual causes of generalized weakness demonstrate characteristic EDX features (Table 9.21), making them readily identifiable to the EMC with a heightened suspicion for these entities.

Myasthenia gravis is an autoimmune disorder caused by antibodies directed at the acetylcholine receptors of skeletal muscle. EDX techniques most useful for identifying this disorder are repetitive nerve stimulation and single-fiber EMG.[37,94,158] Elevated acetylcholine receptor antibodies have been demonstrated in patients with generalized and ocular myasthenia gravis in 81% and 51% of cases, respectively. Elevated acetylcholine receptor antibody levels can be seen in those with penicillamine-induced myasthenia, in some elderly patients with autoimmune diseases, and in first-degree relatives of patients with myasthenia gravis. There is no clearly agreed upon sensitivity or specificity of testing in myasthenia gravis. Repetitive nerve stimulation and single-fiber EMG sensitivity and specificity are approximately 79% and 97%, respectively, in detecting generalized myasthenia gravis, and 29% and 94%, respectively, in ocular myasthenia.[122] Decrementing responses in a distal hand muscle on repetitive nerve stimulation at 2 or 3 Hz are seen in 68% of persons with definite generalized myasthenia gravis and in 31% with mild myasthenia gravis.[101] Repetitive nerve stimulation of a proximal muscle increases the sensitivity to 89% for definite and 68% for mild myasthenia gravis. Postexercise exhaustion repetitive nerve stimulation adds limited value to these numbers.[153] Single-fiber EMG is the most sensitive test for myasthenia gravis primarily because of restrictions in using resterilized material. The traditional single-fiber electrode has been replaced with the smallest disposable concentric needle electrode ("facial" needle). This needle is usually 25 mm in length and 30 gauge. New normal values for jitter have been reported. Unfortunately, fiber density cannot be accurately measured with concentric needles.[158] Reported sensitivity and specificity for single-fiber EMG and myasthenia gravis are reported as high as 78% to 98% sensitivity and 98% specificity in generalized myasthenia gravis from the extensor digitorum communis. Single-fiber EMG performed at the frontalis or orbicularis oculi has 86% to 97% sensitivity and 73% to 92% specificity in ocular myasthenia gravis.[117] Single-fiber EMG is a nonspecific test, and increased jitter can also be seen in many other neuromuscular conditions that can mimic myasthenia gravis, including amyotrophic lateral sclerosis (ALS), peripheral neuropathies, remote botulinum toxin injections, and some myopathies.[35,158]

LEMS is a unique condition characterized by weakness and fatigability of proximal limb muscles with sparing of ocular muscles. Dry mouth is often reported, and the patients demonstrate hyporeflexia and normal sensation.[101] There is a strong association with malignancy, most commonly oat cell carcinoma of the lung. There is reduced release of acetylcholine quanta from the presynaptic nerve terminal. A unique feature of this disorder is that with rapid repetitive nerve stimulation (20 Hz) or a brief voluntary maximal contraction (10 seconds), a postactivation CMAP increase of more than 60% to 200% is seen. Although such potentiation can be seen in myasthenia gravis and occasionally botulism, it is usually to a much lesser degree than that seen in LEMS. This profound potentiation is the hallmark EDX finding in LEMS. Other features of LEMS that are similar to those found in myasthenia gravis include normal sensory conductions, decrementing response to 2 or 3 Hz stimulation, and increased jitter on single-fiber EMG.[112,117]

Botulism is a rare disorder caused by the potent toxin of the bacterium *Clostridium botulinum*, through both oral ingestion and wound infection.[101] A rapid-onset paralysis of the eye muscles, followed by rapid spread to other parts of the body, is seen. The toxin irreversibly blocks acetylcholine release from presynaptic nerve terminals. EDX findings reveal low or absent CMAPs, with decrementing responses on 2 or 3 Hz repetitive nerve stimulation. The incremental increase in CMAP amplitude with exercise or rapid frequency stimulation (20–50 Hz) is less dramatic than in LEMS but should be greater than 40%.[95] With severe involvement, the neuromuscular junction is completely blocked, and no facilitation with rapid stimulation is seen. In these cases, the end plates break down, and fibrillation potentials are seen. In this group of disorders, SNAPs are normal. CMAPs are normal or of low amplitude, particularly in LEMS and botulism. Needle EMG reveals normal or polyphasic MUAPs with low amplitudes and short durations, similar to those found in myopathies. With EMG, MUAP variation in size can also be seen. Fibrillation potentials are seen only in severe disease with complete disintegration of the neuromuscular junction (see Table 9.21).

TABLE 9.20 **Electrodiagnostic Findings in Myopathies (Video 9.30)**

Parameter	Muscular Dystrophy	Congenital	Mitochondrial	Metabolic	Inflammatory	Channelopathy
Distal latency	Normal	Normal	Normal	Normal	Normal	Normal
Nerve conduction velocity	Normal	Normal	Normal	Normal	Normal	Normal
H-reflex	Normal or absent	Normal or absent	Normal	Normal	Normal or absent	Normal
Sensory nerve action potential amplitude	Normal	Normal	Normal	Normal	Normal	Normal
Compound muscle action potential amplitude	Normal or decreased	Normal or decreased	Normal or decreased	Normal or decreased	Normal or decreased	Normal
Motor unit action potential duration	Decreased and/or increased	Decreased or normal	Decreased or normal	Decreased or normal	Decreased and/or increased (inclusion body myositis)	Decreased or normal
Motor unit action potential amplitude	Decreased and/or increased	Decreased or normal	Decreased or normal	Decreased or normal	Decreased and/or increased (inclusion body myositis)	Decreased or normal
Polyphasics	Increased	Increased or normal	Increased or normal	Increased or normal	Increased	Increased or normal
Recruitment	Increased	Increased or normal	Increased or normal	Increased or normal	Increased	Increased or normal
Fibrillations and positive sharp waves	Yes	Centronuclear myopathy	No	Yes	Yes	Occasionally
Complex repetitive discharges	Yes	Centronuclear myopathy	No	Yes	Yes	Occasionally
Myotonic potentials	Myotonic dystrophy	Centronuclear myopathy	No	Acid maltase deficiency	No	Yes
Electrical silence	No	No	No	Contractures in McArdle disease	No	During attacks of paralysis

Modified from Krivickas L. Electrodiagnosis in neuromuscular diseases. *Phys Med Rehabil Clin N Am.* 1998;9:83-114.

TABLE 9.21 **Electrodiagnostic Findings in Neuromuscular Junction Transmission Disorders (Video 9.28)**

Parameter	Myasthenia Gravis	Lambert-Eaton Myasthenic Syndrome	Botulism
Distal latency	Normal	Normal	Normal
Nerve conduction velocity	Normal	Normal	Normal
Sensory nerve action potential amplitude	Normal	Normal	Normal
Compound muscle action potential amplitude	Usually normal	Decreased	Normal or decreased
Slow repetitive stimulation	Decrement	Decrement	± Decrement
Postexercise facilitation or 20–30 Hz repetitive stimulation	± Mild increment	Large increment (lasting 20–30 s)	Intermediate increment (lasting up to 4 min)
Postactivation exhaustion	Yes	Yes	No
Motor unit action potential configuration	Usually normal amplitude but can be variable in weak muscles. Occasionally decreased amplitude and duration	Usually normal amplitude	Amplitude variability in weak muscles. Decreased amplitude and duration, increased polyphasia In severe cases, no motor units
Recruitment	Normal or increased	Increased	Increased
Spontaneous activity	Fibrillation in severe disease	None	Fibrillation in severe disease
Single-fiber electromyography	Increased jitter and blocking (increased with increased firing rate)	Increased jitter and blocking (decreased with increased firing rate)	Increased jitter and blocking (decreased with increased firing rate)

Modified from Krivickas L. Electrodiagnosis in neuromuscular diseases. *Phys Med Rehabil Clin N Am.* 1998;9:83-114.

Stimulated single-fiber EMG offers advantages over conventional single-fiber EMG. Stimulated single-fiber EMG can be performed in patients who cannot maintain a voluntary contraction or in children and infants.[34] Stimulated single-fiber EMG is less time consuming, and by controlling the rate of stimulation, better quantification of the neuromuscular junction disorder can be obtained.[34] Ertas et al.[71] reported the use of concentric needle electrodes for single-fiber EMG. They showed that when the low-frequency filter is set at 2 kHz, concentric needle electrodes are comparable with single-fiber EMG electrodes and yield the same jitter values determined with a single-fiber needle for both normal individuals and for persons with myasthenia gravis. Additional advantages include that they are disposable and thus do not need resterilization. Other investigators have confirmed the usefulness and comparability of single-fiber EMG using a concentric electrode in the evaluation of persons with suspected neuromuscular disorders, and the concentric needle may be even less painful than the conventional single-fiber electrode.[159] Because of a larger recording area, somewhat more single muscle fiber discharges were identified with concentric needle electrodes than with standard single-fiber EMG electrodes.[71]

Motor Neuron Disease

- ALS is the most common adult motor neuron disease. ALS presents with a mix of upper motor neuron findings from spinal white matter involvement and lower motor neuron findings on needle EMG from anterior horn cell loss.[61,113] There are different classifications based on the predominance of these characteristics. ALS classification systems represent broad consensus efforts to identify all patients with rapidly progressive motor neuron disease. From an EDX standpoint, needle EMG is the most helpful part of the study. Confirming ALS, using the Gold Coast criteria, in the appropriate clinical circumstances requires demonstration of the following EDX findings:
- Signs of ongoing lower motor neuron denervation: fibrillation potentials, PSWs, and/or fasciculation potentials in a minimum of one limb/body region could include cranial with associated upper motor neuron findings (Gold Coast criteria; see later[126,146])
- Signs of chronic lower motor neuron reinnervation as defined by large motor unit potentials of increased amplitude and/or increased duration (Gold Coast criteria[146])
- Ruling out other problems is essential to diagnose ALS. Other criteria include:
  1. Normal sensory NCS
  2. Normal motor conduction velocities, except if the CMAP is less than 30% of the mean, in which case the conduction velocities may not be less than 70% of normal
  3. Reduced recruitment of MUAPs is a key/common EMG finding in ALS. In early/rapidly progressive ALS, it is possible that not enough time has transpired for motor unit remodeling to occur (increases in MUAP amplitude and/or duration).[112,126]

Makki and Benatar[126] examined the accuracy of the El Escorial criteria for the diagnosis of ALS. The El Escorial criteria divide the body into four segments: cranial, cervical, thoracic, and lumbosacral. Evidence of lower motor dysfunction in the cervical and lumbosacral segments is defined by the presence of both active denervation (fibrillations) and chronic reinnervation (large MUAPs, [polyphasic units supportive] reduced interference pattern, or unstable motor units) in at least two muscles.[22] Active denervation and chronic denervation in a single cranial or thoracic muscle is sufficient to declare these segments abnormal. If three of the four segments are abnormal by EMG, this is considered supportive evidence of definite ALS. If two segments are abnormal, this suggests possible ALS.[126] Their study was supportive of optimizing the sensitivity and specificity by requiring EMG changes in two muscles in the arm (cervical) or leg (lumbar) segments. These two muscles should demonstrate lower motor neuron findings.[126] There have been modifications to the El Escorial criteria that enhance the sensitivity of identifying correctly persons with ALS.[18] For example, revised El Escorial criteria rated EMG data equivalent to clinical findings in a category called "laboratory-supported ALS" and deleted the suspected ALS category.[23,168] The modified Awaji criteria require only one region of abnormality, EMG neurophysiologic evidence is considered equivalent to clinical evidence, and the presence of fasciculation potentials was considered as important as fibrillations. Several studies have shown the Awaji criteria showed increased sensitivity to the early diagnosis of ALS especially in bulbar onset disease.[137]

Pugdahl et al.[146] validated the diagnostic accuracy of the ALS Gold Coast criteria proposed by a joint initiative of the International Federation of Clinical Neurophysiology, the World Federation of Neurology, the ALS Association, and the MIND Association at Gold Coast, Australia, in September 2019.[168] This multicenter study compared the Gold Coast criteria, the revised El Escorial criteria,[23] and the Awaji criteria.[31] The key difference between Gold Coast and both the revised El Escorial and Awaji criteria is that the Gold Coast criteria include "possible" ALS disease patients as equal to ALS, increasing sensitivity.[146] The Gold Coast criteria, similar to the Awaji criteria, require only one region with concomitant lower and upper motor neuron involvement for diagnosis of ALS. The Gold Coast criteria defines only two categories of patients (having ALS or not), includes patients with isolated lower motor neuron signs (i.e., no upper motor neuron signs) in greater than or equal to two regions (progressive muscular atrophy [PMA] patients), and excludes patients with isolated upper motor neuron signs in two regions (primary lateral sclerosis).[168] The logical rationale for including PMA patients as a form of ALS is that, in some PMA patients, upper motor neuron signs are not always evident[168] and that PMA is a progressive disease with the same risk factors for survival as ALS.[146] Patients that meet Gold Coast criteria on clinical examination (without EMG) can be diagnosed with ALS because the Gold Coast criteria now define clinical evidence of lower motor neuron involvement as paresis and atrophy. Clinically, it is not clear that any criteria are the best, and many clinicians use overlapping features to make the diagnosis. For research purposes, ALS diagnostic specificity was high for all diagnostic criteria discussed earlier, and ALS sensitivity with Gold Coast criteria in this multicenter study was 88.2%, while the next highest sensitivity was for Awaji criteria (77.6%). The primary reason for the increase in sensitivity was the inclusion of progressive muscular atrophy patients.[146]

In examining cranial nerves, the trapezius is a high-yield muscle[172] for establishing the presence of cranial segment involvement but also can be involved with spinal cord tumors in the upper cervical region. Facial muscles innervated by the seventh cranial nerve or the masseter muscle, innervated by the trigeminal nerve, are also easily accessible cranial nerve–innervated muscles to examine. The thoracic segment can be examined by means of EMG of the thoracic paraspinal muscles or by testing the rectus abdominis.[194]

Accurate identification of persons with ALS is important not only for instituting appropriate counseling and multidisciplinary care but also in minimizing unnecessary surgical interventions. In one retrospective study, 21% of persons with ALS had surgeries within the previous 5 years of diagnosis.[174] It was estimated that 61% of these surgeries were inappropriate and likely to be related to early manifestations of ALS.[174] These included surgeries of the knee and spine despite the absence of pain or paresthesias.[174] Recognition of ALS mimics (191) and the use of the more sensitive Gold Coast criteria have potential to decrease this risk of unnecessary surgeries.[146]

Final Electrodiagnostic Conclusions and Report

The EDX report is a crucial document that conveys the findings and conclusions derived from testing. The EDX report should include a brief history regarding the presenting complaint, a focused physical examination, the tabulated EDX findings, and the final assessment and conclusions. There should be internal consistency within the report, such that the conclusions are supported by electrophysiologic data. The EMC can then comment on how these EDX findings correlate with the clinical impression, the limitations of the test, and any other observations. This then gives a more complete clinical picture. In EDX reports, the referring question(s) should be addressed. Conditions that were excluded also can be listed. It is best to state that there is "no electrodiagnostic evidence of" a particular disorder if the testing for that disorder was normal. This reflects the fact that EDX testing can be normal in persons with the particular condition that the referring physicians suspect. EDX testing is not as sensitive, yet is more specific for some conditions (e.g., radiculopathy), and for this reason the findings are very helpful when positive.

It is often useful to the referring physician if the EMC comments on other clinical conditions that were identified during the workup, such as shoulder impingement syndrome or lateral epicondylitis. Such observations can reveal other treatment alternatives for the patient that the referring physician can consider.

The limitations of a particular study, as well as confounding issues, should be clearly stated. There are multiple complicated issues. One such issue is pedal edema, which can make it difficult to obtain normal sural and fibular (peroneal) sensory responses. Morbid obesity can impede adequate stimulus of a peripheral nerve or accurate measurement of conduction distance. Previous surgeries in the spine, carpal tunnel decompression, or ulnar nerve transpositions can complicate interpretation of studies in these areas. Diabetes and aging can confuse interpretation of mild abnormalities. Limited patient tolerance for testing can compromise the completeness of testing. These statements can alert the referring physician or other provider to the limitations of the study and better place into context the strength of the results.

Summary

EDX medicine is a complex consultation that relies on clinical insights, technical skills, solid normative data, and sound clinical judgment. Such consultations can have a profound influence on subsequent diagnostic and therapeutic interventions. Only experienced and well-trained physicians should perform such diagnostic procedures. A healthy appreciation for the spectrum of normal values seen in people of different ages and heights should be maintained, and the AANEM normative values and nerve conduction techniques should be used for more precise judgments as to normality of a given nerve conduction parameter. Normative values should be well derived and interpreted correctly by electrodiagnosticians to avoid overcalling common disorders. Sufficient testing should be performed to confidently identify the suspected disorder and to eliminate the differential of confounding conditions.

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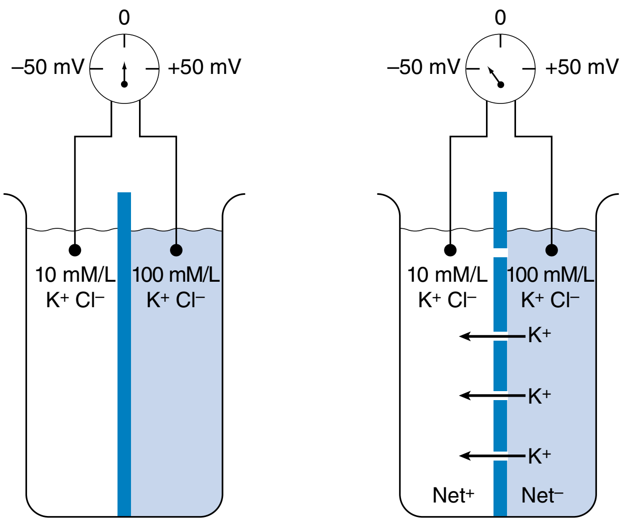
eAppendix 9A

Physiologic Basis for Electrodiagnostic Testing

Nerve and muscle tissues are capable of generating an electrical difference across their cellular membranes and can generate and sustain a propagating action potential. Interestingly, action potential generation is relatively similar between these two dramatically different tissues.[11,14]

All living cells generate an electrical potential across their membranes, with the intracellular region relatively negative compared with the extracellular region.[14] This potential difference across the cell membrane is referred to as the resting membrane potential. The development and maintenance of the resting membrane potential can be explained by a relatively simple model shown in eFig. 9A.1.

This model can be applied to all cells in the body, and in particular to nerve and muscle cells. The axon of a nerve will be used as a prototypical example, although the same principles apply to muscle cells. The nerve cell is known to have a specialized cell membrane permeable primarily to potassium and chloride ions in the resting state because of proteinaceous transmembranous ion channels rendering the membrane selectively permeable (i.e., semipermeable).[15] The potassium channels are referred to as passive because they are always open and permit the free flow of this ion, while the chloride channels are passive in the resting state. Contained within the axon but incapable of crossing the cell membrane are large, negatively charged protein molecules. This negative charge attracts potassium ions from outside the cell to enter through the passive potassium channels, resulting in a buildup of potassium ions within the axon. This process continues until there is so much potassium within the axon that the continued entry of more potassium ions is prevented by the high intracellular potassium concentration, even though all the negative charges have not yet been electrically balanced. The reason is that the force of the concentration gradient of the accumulated potassium ions now attempting to drive potassium out of the cell is just large enough to balance the electrical force of the intracellular anions attracting potassium ions into the cell.



• **eFig. 9A.1** A beaker containing two different concentrations (10 mM/L and 100 mM/L) of a potassium chloride (KCl) solution existing as potassium (K^1) and chloride (Cl−) ions. An impermeable partition containing closed potassium channels separates the two different concentration solutions. A voltmeter placed across the partition fails to register any voltage difference. If the partition is now made selectively permeable to just the potassium ions (potassium channels are opened), the concentration gradient difference will drive potassium ions into the lower concentration side of the beaker until the electrical attraction from the accumulating negatively charged chloride ions prevents any further net K^1 ion movement, thus establishing a dynamic equilibrium. At this point, a potential difference exists across the partition and represents the equilibrium potential.(Redrawn from Dumitru D, Amato AA, Zwarts MJ. *Electrodiagnostic Medicine.* 2nd ed. Hanley & Belfus; 2002.)

Action Potential Generation

The all-or-none phenomenon is dependent on voltage-gated channels that open and close depending on the transmembrane voltage.[11] These voltage-gated sodium and potassium ion channels are closed at the resting membrane potential. If the transmembrane voltage changes toward a more depolarized direction (less negative) and reaches approximately 15 to 20 mV less negative than the resting membrane potential, the voltage-gated sodium channels open. This results in an increased permeability of the sodium ion, a process known as sodium activation. This massive shift in transmembrane potential is referred to as depolarization. After staying open for a short period, the sodium gates automatically close (sodium inactivation), with a return of the resting membrane potential.

The nodes of Ranvier in myelinated nerves lack voltage-gated potassium channels and contain only voltage-gated sodium channels.[21,22] The action potential actually "jumps" from one node to the next, creating an efficient means of action potential propagation that is referred to as saltatory conduction. Repolarization in myelinated nerve does not require a delayed potassium current. As noted, the resting membrane potential is restored once the permeability of sodium is reduced. Passive "back-leak" sodium and potassium currents are thought to mediate the discharge of the capacitance of the membrane, which accumulates over time with multiple action potential discharges.

Physiologic Factors Affecting Action Potential Propagation

A number of physiologic factors have a direct effect on action potential propagation. These factors can be divided into those that can be altered by the practitioner and those intrinsic to the subject and beyond control. The most important factor that is readily amenable to change is the surface temperature of the limb. Physiologic variables beyond the control of the clinician include sex, age, height, and digit circumference.

Sex

Only a few studies have attempted to investigate the difference in nerve conduction studies (NCS) between males and females.[1] Noted is a slight increase in the antidromic sensory nerve amplitudes for both the median and ulnar nerves recorded from the digits in females. Also, females demonstrate a greater nerve conduction velocity (NCV) for upper and lower limb nerves compared with males. However, both of these differences are eliminated when one considers limb length and digit circumference (see later).[19]

Aging

Several generalizations can be made regarding peripherally evoked sensory nerve action potentials (SNAPs) and aging. The conduction velocity demonstrates a consistent decline of approximately 1 to 2 m/s per decade.[17] The duration of SNAPs is approximately 10% to 15% longer in persons aged 40 to 60 years, and 20% longer in those aged 70 to 88 years, compared with persons aged 18 to 25 years.[2] Compared with the 18- to 25-year-old group, the SNAP amplitude is one-half and one-third, respectively, for the 40- to 60-year-old and 70- to 88-year-old groups. The distal sensory latencies reveal a similar prolongation with age. There is a suggestion that the median and radial nerves do not demonstrate considerable alteration with age.[6] At present, there is disagreement regarding the magnitude of change in SNAP parameters induced by the aging process. However, the healthy older adult population has been relatively well studied.[6,7] This is addressed in more detail later in the clinical testing section.

After the age of 50 years, the conduction velocity of the fastest motor fibers progressively declines by approximately 1 to 2 m/s per decade. There is a concurrent increase in the distal motor latency and decrease in the amplitude of the motor response with advancing age. H-reflex latency demonstrates little alteration with aging in the healthy older adult population.[7] The decrease in amplitude is difficult to ascertain clinically because of the wide range of normal H-reflex amplitudes and the central effects on amplitude that are difficult to control and quantify.

Digit Circumference

Females consistently demonstrate significantly higher antidromic SNAP amplitudes for the median and ulnar nerves recorded from the second and fifth digits.[1] A negative linear correlation exists between finger circumference and amplitude for these two nerves. It is known that as the distance between the recording electrode and neural generator increases, the amplitude precipitously declines. Increasing the circumference of the finger displaces the electrode further from the nerve. Because males have significantly larger finger circumferences than females, this appears to explain the difference in SNAP amplitudes. There is no evidence that this difference is attributable to an intrinsic neural difference between male and female nerves.

Height

Several investigations have documented slower lower limb NCVs in taller compared with shorter individuals.[3,16] This difference is independent of the temperature of the limb or the age of the individual. The etiology of this finding is unknown. Distal nerve tapering or an abrupt change in axon diameter has been speculated to account for this finding, but the exact etiology has not been definitively identified.[5,20]

Temperature

Temperature is one of the most important factors influencing NCS. As the temperature of the nerve is lowered, the amount of current required to generate an action potential increases. This decreased excitability is a direct temperature effect on the action potential–generating mechanism of the nerve at the nodes of Ranvier, and not a result of membrane resistance changes. The transmembrane resistance is not increased by a drop in temperature.[12,13] In addition to excitability, the configuration of an action potential is profoundly affected by a drop in temperature.

The amplitude, rise time, and fall time of the action potential all increase as the temperature of the nerve declines. The time required for the action potential of a cold nerve to reach its peak depolarization from the resting membrane level increases approximately 33%.[18] Similarly, the time necessary for the action potential to return to its resting level is also increased, but much more so than the rise time (69%). Because both the SNAP duration and the spike height increase, the area of the SNAP increases dramatically at lower temperatures.

The compound muscle action potential (CMAP) arising from cooled muscle tissue demonstrates similar changes as those noted for SNAPs. The amplitude, duration, rise time, and area of the CMAP all increase as the temperature of the muscle reduces. Intramuscular recordings also reveal that those motor units in close proximity to the recording electrode are also increased in the same parameters noted earlier.

Temperature also has an impact on NCVs. Decreased temperature results in an increase in the amount of time necessary to reach the peak of the action potential at each node of Ranvier, thereby reducing conduction velocity.

In the upper limb, the relationship between temperature and NCV has been investigated for the surface temperature range of 26°C to 33°C, measured at the midline of the distal wrist crease. Calculations reveal that for median motor and sensory nerves, NCV is altered 1.5 and 1.4 ms/°C, respectively, whereas the distal latency for both nerves changes approximately 0.2 ms/°C.[8-10] The ulnar nerve demonstrated motor and sensory temperature relationships of 2.1 and 1.6 m/s per °C, respectively, and a distal motor and sensory latency correlation of 0.2 ms/°C.

Because of the profound effects of temperature on NCV, it is clear that reliable NCS requires temperature control. A cool limb, irrespective of the ambient room temperature, can result in latencies, NCVs, and amplitudes that are not in the "normal" range. Although applying a correction factor is less time consuming than heating the limb, it is doubtful how accurately correction factors can be applied to abnormal nerves. Until more data are available regarding the best correction factor for diseased nerves, warming of the limb should be considered the best choice for clinical nerve conduction testing. The surface temperature approximating the recording electrodes should be at least 32°C in the upper limbs and 30°C in the lower limbs. Caution must be exercised when attempting to warm the limbs of patients with ischemic limbs, or those with altered sensation, to avoid injury.

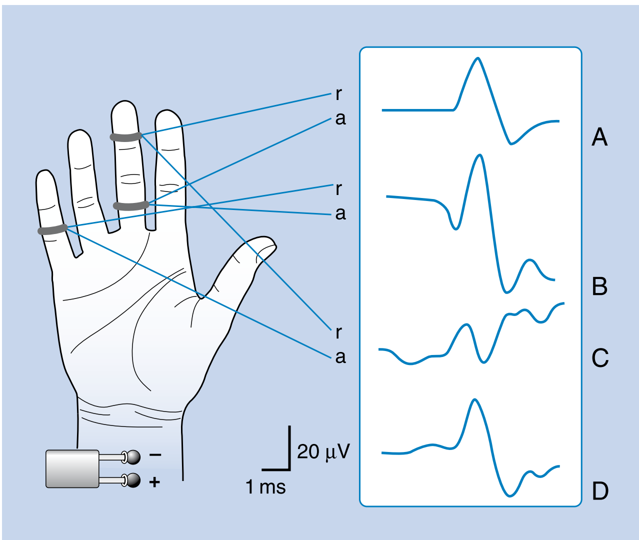
Physiology of Sensory and Motor Responses

SNAPs can be obtained with either antidromic or orthodromic techniques.[2] The term *antidromic* implies that the induced neural impulse propagates along the nerve in a direction opposite to its physiologic direction; for example, stimulating the median nerve at the wrist and recording the ensuing response from the second or third digit. A nerve will conduct an impulse proximally and distally from the point of stimulation by a depolarizing current. On the other hand, stimulating the median sensory fibers on the second digit and recording from the wrist is an example of an orthodromic technique. Antidromically obtained SNAPs are usually larger and hence easier to assess because the recording electrodes on the finger are in close proximity to the proper digital nerves than when a recording electrode is placed over the median nerve at the wrist for orthodromic techniques. Mixed nerve responses can have a component of both orthodromic and antidromic waveforms. For example, midpalm stimulations recorded at the wrist are a combination of orthodromic sensory and antidromic motor responses.

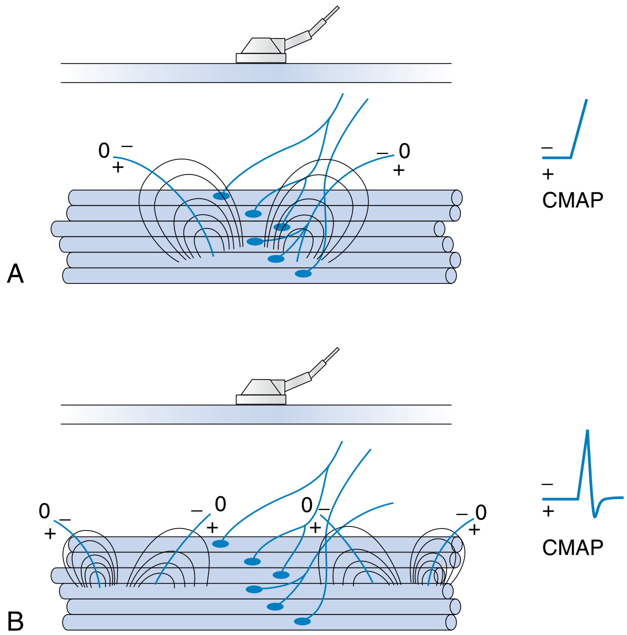
Antidromic and orthodromic bipolar SNAP waveform recordings are typically biphasic rather than triphasic. The reference location has a dramatic effect on SNAP waveforms as demonstrated in eFig. 9A.2.[4] The median SNAP is indeed a triphasic waveform and conforms to the principles of volume conduction but only appears biphasic because of the recording montage used.

To elicit a motor response from a particular muscle, the active electrode is located on the surface of the skin directly over the end-plate region of the muscle, or "motor point."[5] The end-plate region typically lies midway between the origin of the muscle and insertion. The reference electrode is usually placed on or distal to the tendinous insertion of the muscle, so as not to record electrical activity from the activated muscle. Stimulating the peripheral nerve innervating the muscle under investigation will result in a relatively large, biphasic waveform with an initial negative deflection (eFig. 9A.3).

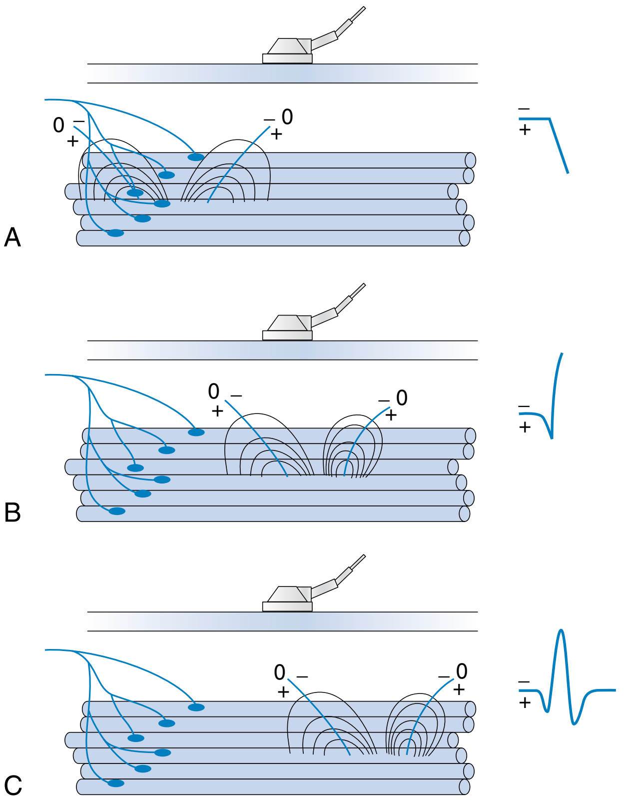
Occasionally, a positive deflection can precede the negative phase of the CMAP. Volume conductor theory can explain this observation (eFig. 9A.4). The active electrode may not be located directly over the motor point but displaced longitudinally away from it. Relocating the active electrode over the anticipated motor point region will usually remedy the situation.[5]



• **eFig. 9A.2** The median nerve is stimulated at the wrist with an antidromic sensory recording montage. Active *(a)* and reference *(r)* are located on fingers as designated in the figure. (A) Bipolar recording with the commonly observed biphasic median nerve sensory nerve action potential (SNAP). (B) The same active electrode location in (A) is referenced to the fifth digit, resulting in a triphasic SNAP. (C) An active electrode placed on the fifth digit but referenced to the third digit permits one to observe what this electrode records when the median nerve is stimulated. An inverted triphasic potential of small magnitude is detected. It is inverted because of its connection to the inverting amplifier port. (D) Electronically summating the potential recorded in (B) and (C) yields that recorded with a bipolar montage in (A).(Redrawn from Dumitru D. Volume conduction: theory and application. In: Dumitru D, ed. *Physical Medicine and Rehabilitation State of the Art Reviews: Clinical Electrophysiology.* Hanley & Belfus; 1989.)



• **eFig. 9A.3** (A) Locating an active recording electrode over the motor point of a muscle places this electrode over the central region of the negative current sink generating all the action potentials of the muscle. An initial negative deflection is recorded because of this location, thereby producing the characteristic morphology of the compound muscle action potential *(CMAP)*. (B) A terminal positive phase is then recorded as the action potential propagates away from the electrode. (Redrawn from Dumitru D, Amato AA, Zwarts MJ. *Electrodiagnostic Medicine.* 2nd ed. Hanley & Belfus; 2002.)



• **eFig. 9A.4** (A) Locating an active electrode off the motor point results in a compound muscle action potential with an initial positive deflection because some of the action potentials of the muscle no longer originate under the electrode but propagate toward it. (B) When the main negative sink reaches the electrode, the main negative spike of the potential is detected. (C) Finally, the terminal positive source currents are recorded, generating the terminal positive phase of the potential. (Redrawn from Dumitru D, Amato AA, Zwarts MJ. *Electrodiagnostic Medicine.* 2nd ed. Hanley & Belfus; 2002.)

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eAppendix 9B

Motor Unit Recruitment and Morphology: Assessing Phases

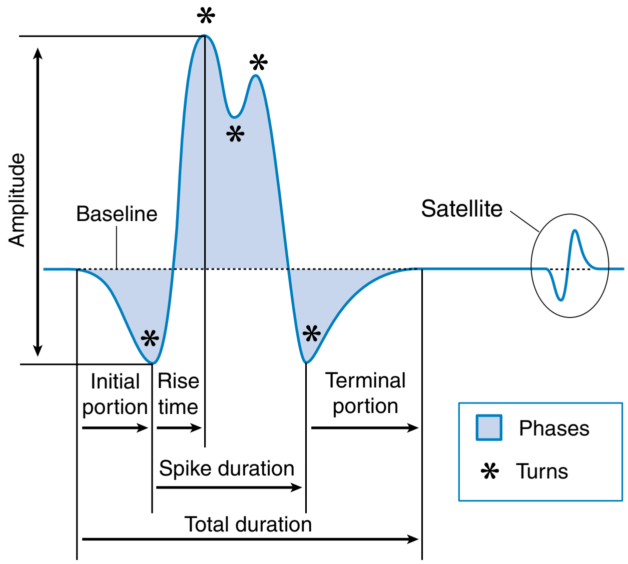
The motor unit is the anterior horn cell and the myocytes innervated by that anterior horn cell. The number of motor units per muscle varies from about 3000 in the extraocular muscles to 100 in the hand intrinsic muscles. Other muscles in the arm, forearm, thigh, and calf are closer to 500 motor units per muscle.

The number of myocytes per motor unit varies somewhat inversely as the number of motor units per muscle. For example, the extraocular muscles have about 10 myocytes per anterior horn cell, while hand and limb muscles have between 100 and 2000 myocytes per motor unit.

The cross-sectional territory of a motor unit is between 5 and 12 mm for most muscles. This covers 20% to 30% of the muscle's total volume and thus is larger in leg muscles than in the arm.[1]

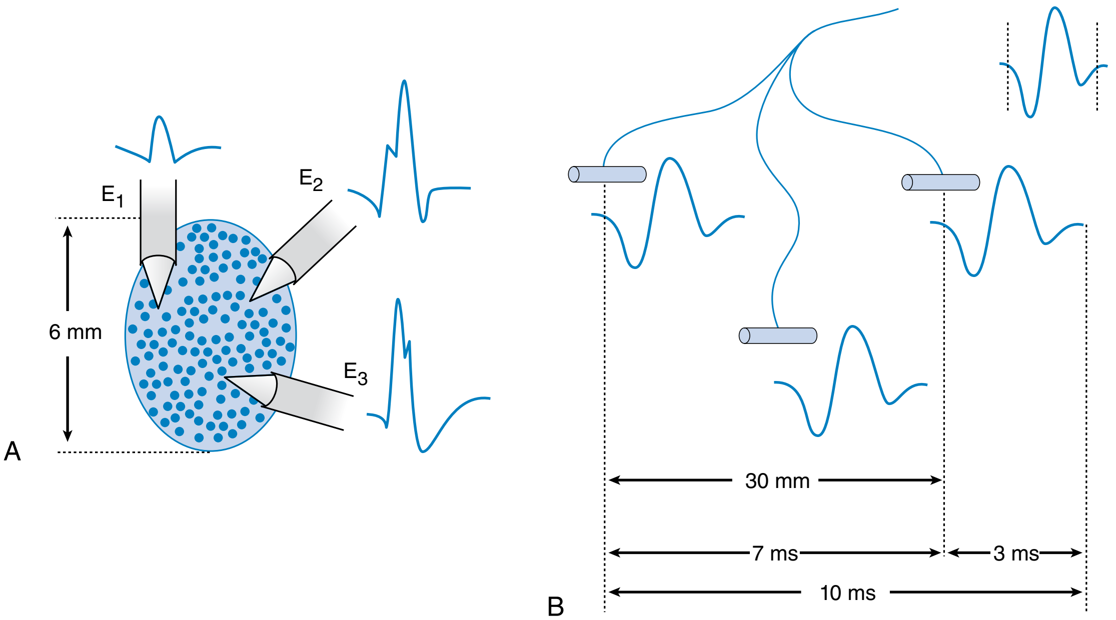
As previously stated, the single muscle fiber action potential usually has a triphasic appearance when recorded outside the endplate zone and away from the tendinous insertion. The voltages from all the single muscle fibers belonging to one motor unit summate to yield a motor unit action potential (MUAP), which is also usually triphasic: positive-negative-positive. This voltage summation does not always produce a smooth result, and small serrations, or turns, can occasionally be seen as part of the major phase of an MUAP (eFig. 9B.1). The number of phases is defined as the number of trace baseline crossings plus one. Normal MUAPs are considered to have four or fewer phases. MUAPs with five or more phases are called polyphasic potentials. Recordings of multiple MUAPs from normal muscle tissue can have between 12% (concentric needle) and 35% (monopolar needle) polyphasic potentials, depending on the type of recording electrode used.[1] Slightly different MUAP configurations can be expected, depending on the exact location of the recording electrode with regard to different single muscle fibers within the motor unit territory (eFig. 9B.2). Satellite potentials are late waves that are linked to the rest of the waveform.

Pathology can also alter the number of phases a MUAP waveform can have. The motor unit can be affected in two general ways after injury or disease: A pathologic condition may affect either the anterior horn cell, the peripheral nerve (including the neuromuscular junction), or the muscle fibers comprising the motor unit. If the neural component of a motor unit is compromised severely enough to experience degeneration, all the muscle fibers innervated by the parent nerve will become denervated. These denervated muscle fibers induce nearby terminal axons of intact nerves to send out neural projections to reinnervate the orphaned muscle fibers most likely through neurotrophic factors. Through collateral sprouting, the total number of muscle fibers belonging to a specific motor unit may increase dramatically (eFig. 9B.3). Neurogenic diseases can lead to larger-amplitude, longer-duration, and highly polyphasic MUAPs.

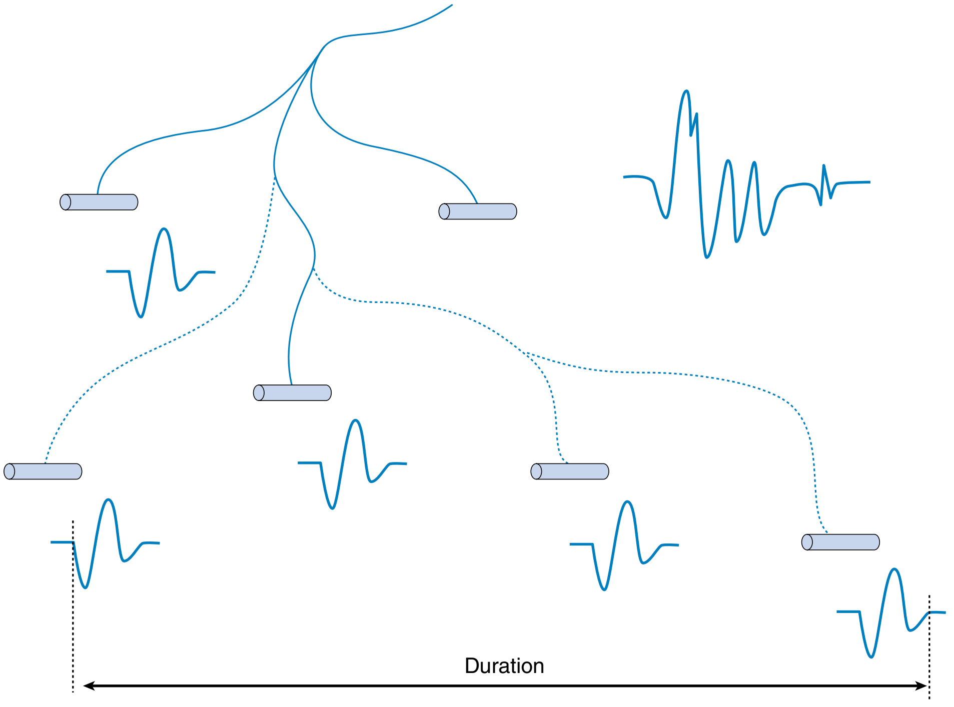


• **eFig. 9B.1** A motor unit action potential is depicted, with various morphologic aspects measured.(Redrawn from Dumitru D, Amato AA, Zwarts MJ. *Electrodiagnostic Medicine.* 2nd ed. Hanley & Belfus; 2002.)

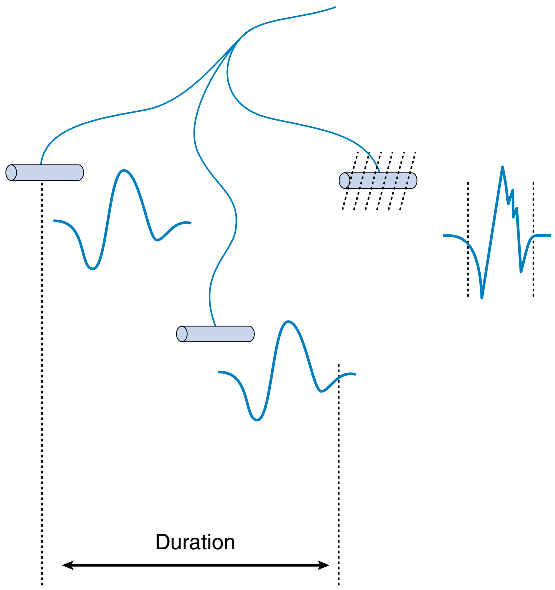
If the muscle fibers comprising a motor unit undergo random degeneration, such as occurs in some myopathies, there is a decrease in the total number of fibers belonging to that motor unit (eFig. 9B.4). A decrease in muscle fiber numbers would result in a reduction of the overall MUAP amplitude, including its initial baseline departure and subsequent return. If the initial baseline departure of the original MUAP and its subsequent return to baseline decline in magnitude, this portion of the waveform can no longer be observed (assuming the gain of the amplifier is not increased). As a result, the total duration of the MUAP would correspondingly appear to decrease. Finally, the dropout of single muscle fiber waveforms would produce less voltage with regard to the spatial summation of single fiber potentials. Fewer muscle fibers could lead to "gaps" in the MUAP waveform, causing an increase in the number of phases. Consequently, a primary myopathic process tends to yield shorter-duration, highly polyphasic, low-amplitude MUAPs.



• **eFig. 9B.2** (A) Muscle fibers belonging to a single motor unit are randomly distributed within an approximately 6-mm oval territory. (B) Three individual motor unit action potential waveforms are recorded from the same motor unit and depend on the location of the recording electrode with regard to different muscle fibers within the spatial territory of that motor unit. (Redrawn from Dumitru D, Amato AA, Zwarts MJ. *Electrodiagnostic Medicine.* 2nd ed. Hanley & Belfus; 2002.)



• **eFig. 9B.3** Several muscle fibers are denervated in the example in the figure. Collateral terminal nerve sprouts from an intact motor unit *(dotted lines)* grow to reinnervate those denervated muscle fibers. The end result is a large-amplitude, long-duration potential with more phases. (Redrawn from Dumitru D, Amato AA, Zwarts MJ. *Electrodiagnostic Medicine.* 2nd ed. Hanley & Belfus; 2002.)



• **eFig. 9B.4** Loss of single muscle fibers from a motor unit results in the generation of motor unit action potentials with smaller amplitude, shorter durations, and possibly more phases. (Redrawn from Dumitru D, Amato AA, Zwarts MJ. *Electrodiagnostic Medicine.* 2nd ed. Hanley & Belfus; 2002.)

Reference (for eAppendix 9B)

1. Dumitru D, Amato AA, Zwarts MJ. *Electrodiagnostic Medicine*. 2nd ed. Hanley & Belfus; 2002.

eAppendix 9C

Instrumentation

An electrodiagnostic instrument actually comprises many separate components, of which the electrodes, amplifier, filters, sound, and stimulator are discussed in this chapter. Practitioners of electrodiagnostic (EDX) medicine are urged to gain a thorough understanding of instrumentation principles and all subcomponents of the instrument (eFig. 9C.1).[2,3,6]

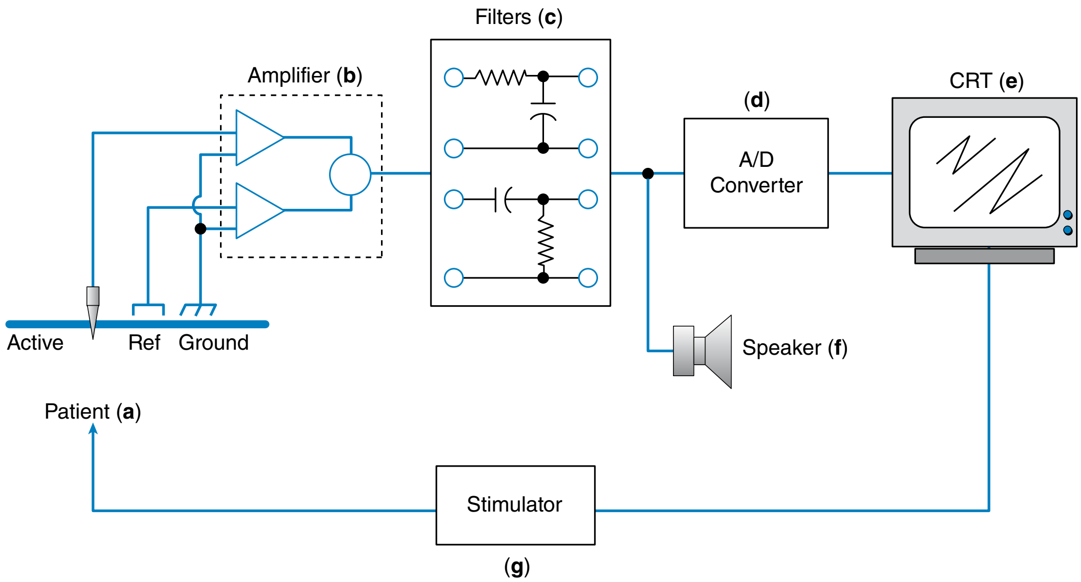
Electrodes

The two types of electrodes are surface and needle. Surface electrodes are manufactured in various sizes and shapes so as to best conform to the body part under investigation. The electrode must be firmly secured to the patient to preclude any movement. Well-secured electrodes minimize movement artifacts that could contaminate the desired signal.

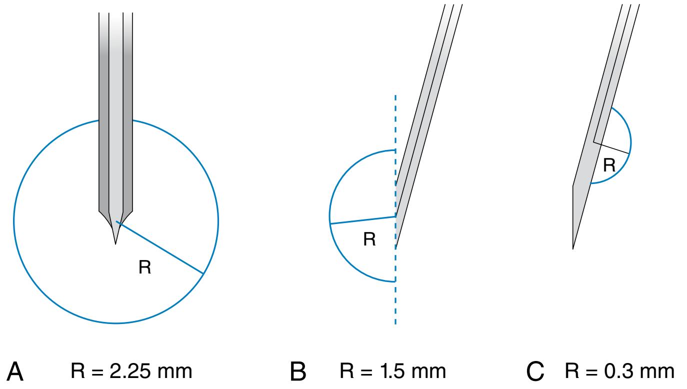
Two commonly used needle recording electrodes are monopolar and concentric (eFig. 9C.2). The monopolar needle is a solid, stainless-steel shaft coated completely with Teflon except for the bare metal tip. It is this bare metal tip that acts as the recording surface. The needle is typically 12 to 75 mm in length and 0.3 to 0.5 mm in diameter, with a recording surface of 0.15 to 0.6 mm2. Separate reference and ground electrodes are required.

The concentric needle electrode is a hollow, stainless steel hypodermic needle with a central platinum or Nichrome silver wire approximately 0.1 mm in diameter, surrounded by epoxy resin acting as an insulating material from the surrounding cannula. The cannula has a similar length and diameter to that of the monopolar needle. A separate ground is required, but the cannula serves as the reference electrode.

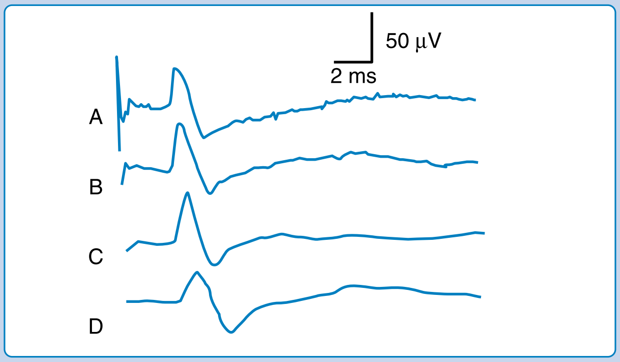
There has been considerable discussion about the merits of each of these electrodes compared with the other. Both electrodes have advantages and disadvantages depending on the clinical circumstances. Monopolar needle electrodes have a wider recording territory and a distant reference, which makes the recording more "noisy" with regard to distant activity and interference. On the other hand, the Teflon coating probably reduces patient discomfort depending on the technique. The concentric needle electrode has the active and reference electrodes close together, making them electrically quieter than monopolar needles. Concentric needle electrodes yield the following as compared with monopolar needle electrodes: smaller potential amplitudes, possibly fewer phases, comparable durations, and less distant motor unit activity. The introduction of commercially available, disposable monopolar and concentric needle electrodes has eliminated concerns such



• **eFig. 9C.1** The subcomponents of the electrophysiologic instrument are depicted. Electrodes on or in the patient *(a)* detect bioelectric changes, which are transmitted to a differential amplifier *(b)*. This signal is filtered *(c)*, undergoes analog to digital (A/D) conversion *(d)*, and is displayed on the cathode ray tube *(CRT) (e)* as well as the sound being presented through a loudspeaker *(f)*. Time-locked evoked potentials can be generated with the stimulator *(g)*. (Redrawn from Dumitru D. Volume conduction: theory and application. In: Dumitru D, ed. *Physical Medicine and Rehabilitation State of the Art Reviews: Clinical Electrophysiology.* Hanley & Belfus; 1989.)



• **eFig. 9C.2** Three needle electrodes and their recording areas. (A) A monopolar needle electrode requires a separate reference and ground electrode. (B) The concentric needle electrode uses the cannula as the reference, while a separate ground is required. (C) The single-fiber electrode is essentially a modified concentric needle electrode. *R*, Radius. (Redrawn from Dumitru D. Volume conduction: theory and application. In: Dumitru D, ed. *Physical Medicine and Rehabilitation State of the Art Reviews: Clinical Electrophysiology.* Hanley & Belfus; 1989.)



Trace	High frequency filter (Hz)	Latency (ms) Onset	Latency (ms) Peak	Amplitude (µV)
A	10,000	2.7	3.3	76
B	2 000	2.8	3.4	76
C	1 000	2.8	3.8	75
D	500	3.0	4.2	64

• **eFig. 9C.3** An antidromic median sensory nerve action potential is recorded from the third digit, while a constant low-frequency filter of 10 Hz but different high-frequency filters are used. Note how the onset and peak latencies are sequentially delayed with decreasing high-frequency filter settings. The amplitude of the potential also decreases.(Redrawn from Dumitru D, Walsh NE. Practical instrumentation and common sources of error. *Am J Phys Med Rehabil.* 1988;67:55-65.)

as Teflon peeling back on the monopolar needles and hook formation on the tip of the concentric needle electrodes. The quality of disposable needle electrodes has improved, eliminating the need to use nondisposable needle electrodes. If electrodes are reused (e.g., single-fiber electrodes), they should be properly sterilized with presoaking in sodium hypochlorite and steam autoclaving.[1] There are concerns about the possibility of spreading prion disease with sterilized, reusable needles.[5] However, documentation of this is lacking.

Single-fiber electrodes are essentially modified concentric needle electrodes. A small, 25-µm recording port is placed opposite the bevel of the electrode and several millimeters from the tip. This makes this special electrode capable of recording the electrical activity from a single muscle fiber. The uptake area for this electrode is approximately 300 µm (eFig. 9C.3). Recent studies show comparable jitter values comparing disposable concentric needles and single-fiber electrodes.4

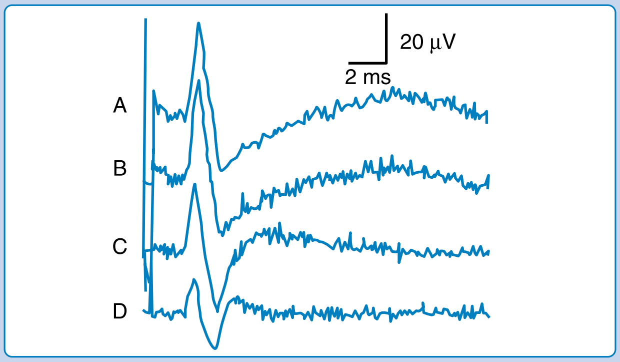
Amplifier

Biologic signals range in size from microvolts to millivolts, thereby requiring significant amplification before being analyzed. An amplifier is simply a device with the ability to magnify the desired signals and minimize unwanted signals or noise. Amplification is expressed as gain or sensitivity. Gain is a ratio of the output of the signal divided by the input. For example, an output of 1 V for an input of 10 µV implies that the amplifier has a gain factor of 100,000 (output/input = 1 V/0.00001 V = 100,000). Sensitivity is the ratio of the input voltage to the size of deflection on the cathode ray tube (CRT) and is usually measured in centimeters. For example, an amplifier that produces a 1-cm deflection for an input of 10 mV has a sensitivity of 10 mV/cm or 10 mV/division. The sensitivity or gain setting used is important because it can influence the onset latency. Increasing the sensitivity for a given waveform results in the instrument displaying the initial departure of the potential from baseline earlier in time because progressively earlier and hence smaller baseline deviations can now be observed.

Understanding amplification is simplified with two concepts: (1) differential amplification in which different signals are amplified and (2) common mode rejection where common signals are eliminated. The standard electromyograph has two amplifiers. The amplifier connected to the active electrode is known as the noninverting amplifier, whereas the reference electrode is connected to the inverting amplifier. The inverting amplifier magnifies the signal presented to it in essentially the same manner as the noninverting amplifier, with the exception of inverting the signal. Both amplified signals (inverted and noninverted) are then electronically summated, and similar signals are canceled. When an identical signal is presented to both amplifiers, theoretically there should be no output from the instrument because there is elimination of the same or so-called common signals. For example, 60-Hz interference recorded by the active and reference electrodes is eliminated as a common mode signal. It is impossible to build two amplifiers with identical properties; thus common mode rejection can never be perfect. The ratio of the differential to common output (common mode rejection ratio) of the instrument should exceed 100,000:1.

Filters

Perhaps the most misunderstood and ignored aspect of the EDX instrument is the filters. The main purpose of filters is to form a window or bandwidth of frequencies containing the desired waveform but exclude those frequencies not comprising the signal of interest ("noise"). Low- and high-frequency filters are used to prevent those frequencies below and above the respective filter



Trace	Low frequency (Hz)	Latency (ms) Onset	Latency (ms) Peak	Duration negative spike (ms)	Amplitude (µV)
A	1	2.6	3.3	1.4	65
B	10	2.6	3.3	1.4	65
C	100	2.6	3.1	1.0	54
D	300	2.6	3.0	0.8	30

• **eFig. 9C.4** An antidromic median sensory nerve action potential is evoked from the third digit, while the low-frequency filter is sequentially elevated with a constant high-frequency filter of 10,000 Hz. Note how the onset latency does not change; however, the peak latency decreases, as does the amplitude of the potential. A third phase is also produced at the low-frequency filter setting of 300 Hz. (Redrawn from Dumitru D, Walsh NE. Practical instrumentation and common sources of error. *Am J Phys Med Rehabil.* 1988;67:55-65.)

settings from being amplified and subsequently interfering with the signal of interest. A high-frequency filter eliminates those frequencies higher than its numeric designation and permits frequencies lower than its numeric designation to pass. Effects of the high-frequency filter are illustrated in eFig. 9C.4. That is why a high-frequency filter is also known as a low-pass filter: It permits low frequencies to pass through and be displayed. Similarly, a low-frequency filter extracts low frequencies from the total recorded signal below its numeric designation. A low-frequency filter permits frequencies greater than its numeric designation to pass. Effects of the low-frequency filter are illustrated in eFig. 9C.4.

Any biologic signal can be conceptualized as a series of sine waves of various frequencies and amplitudes. The combination of "appropriate" sine wave amplitudes and frequencies can result in the formation of essentially any waveform. In this way, the biologic signal recorded by the instrument consists of multiple subcomponent waveforms with specific frequency and amplitude characteristics. Eliminating any of these subcomponent waveforms will result in a distortion of the appearance of the waveform. That is exactly what can happen if the high- or low-frequency filters are set such that the desired biologic waveform has various subcomponent frequencies eliminated.

There are no universally agreed filter settings for any EDX medicine procedure. Arriving at optimal filter settings is highly empirical. The high- and low-frequency filters are lowered and raised, respectively, until waveform distortions are observed. The filters are then expanded until no waveform changes are noted. The goal is to include the major components of the waveforms while eliminating undesired signals or noise. The most important factor is to reproduce all filter settings originally described by those investigators whose reference data are being used.

Sound

After the biologic signal is filtered, it is fed to a loudspeaker. The acoustic analysis of both normal and abnormal potentials is extremely important. It is not uncommon for practitioners to "hear" an abnormality before viewing it on the CRT. The instrument must have a relatively good speaker so as to accurately present the sounds associated with the biologic signals.

Stimulator

Two different types of stimulators are commercially available: constant current and constant voltage. For both types of devices, neural tissue is activated under the cathode (negative pole) while the anode (positive pole) completes the stimulating circuit. A constant current stimulator effectively delivers the desired current output for each stimulus irrespective of the resistance between the skin and cathode or anode. This is accomplished by varying the voltage or current driving force as is necessitated by any alterations in the resistance of the skin-stimulator interface. Similarly, a constant voltage stimulator is designed to deliver the same voltage with each stimulus, even if the resistance between the skin and stimulator changes. Both stimulators are acceptable for most purposes. The constant current stimulator is preferred when the same current must be delivered for each stimulus in clinical situations requiring quantification of current delivery, such as in somatosensory evoked potentials, research, or evaluating side-to-side stimulation thresholds during facial nerve excitability testing.[7]

An important problem associated with stimulators is the stimulus artifact. It is commonly a large potential recorded during the delivery of the stimulus. At times, the magnitude of the potential is large enough to compromise the desired neural or muscular response. In such situations, it is necessary to minimize the shock artifact. An effective method of reducing the shock artifact is to use fast recovery amplifiers that act to suppress this artifact by quickly recovering from the overwhelming voltage delivered. These amplifiers are not available on all instruments, and thus other means must be found to deal with the artifact. The skin surface must be dry, and any perspiration, body lotion, makeup, or other surface conductors should be removed. Wiping a large portion of the body segment under investigation with an alcohol pad usually removes all surface conducting films. A ground electrode is best placed between the stimulus site and the active recording electrode. Wire leads between the patient and stimulator should be separated to avoid any type of capacitive interaction. The stimulator circuit should be isolated from the ground circuit of the instrument, which is true of virtually all commercially manufactured instruments. Perhaps the most effective method of reducing stimulus artifact, once all the above have been addressed, is to rotate the anode about the cathode. This optimizes the voltage output of the stimulator, as recorded by the active and reference electrode, to take advantage of differential amplification and the elimination of the shock artifact as a common mode signal. An attempt is made to have both the active and the reference electrodes record similar voltages, thereby minimizing the signal of the stimulator from being amplified and displayed along with the signal.

Anodal block is an interesting concept that has been widely discussed but has little supporting experimental data. Theoretically, the anode hyperpolarizes the neural tissue in its immediate vicinity and should result in an action potential failing to conduct past the anode. In humans, investigations with bipolar and monopolar anodal current stimulation at the highest current outputs failed to document any type of anodal block. Because the anode is capable of stimulating neural tissue and not blocking it, at this time it appears that anodal block does not occur during the routine EDX medicine consultation.

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